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Development and Implementation of a Permanent Magnet Synchronous Motor Control Platform

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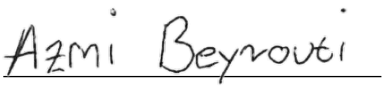
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Überlingen, den 21.09.2026



Azmi Beyrouti

Abstract

Permanent-magnet synchronous motor control strategies differ in their performance, sensing requirements, computational demand, and implementation complexity. However, existing comparisons often use different motors, hardware platforms, feedback configurations, or evaluation procedures, preventing observed differences from being attributed solely to the control strategy. This thesis develops a common motor-control platform and a consistent methodology for comparing six-step control, sinusoidal control, and field-oriented control. An equation-based MATLAB/Simulink framework evaluates the three strategies using identical motor parameters, operating conditions, controller-design objectives, and performance metrics. Custom motor-control hardware is designed and manufactured, while six-step control and field-oriented control are experimentally evaluated on a substitute embedded platform. In simulation, field-oriented control achieved complete current-vector utilization and produced a mean torque of 0.682 N m at the high-current operating point, compared with 0.218 N m for six-step control and 0.220 N m for sinusoidal control. During a low-to-high-speed reference ramp, its average relative tracking error was 0.036 %, compared with 0.645 % and 0.616 %, respectively. The experiments confirmed the relative advantages of field-oriented control in speed smoothness and transient tracking. At the low-speed operating point, the relative speed variation decreased from 12.48 % with six-step control to 0.76 % with field-oriented control, while processor utilization increased from 0.63 % to approximately 38 %. The results demonstrate that improved current utilization and control performance require greater sensing, parameterization, tuning, and computational, but demonstrate significant differences in simulated and experimental performance. Experimental validation of all three strategies and functional validation of the custom hardware remain pending.

Kurzfassung

Regelungsverfahren für permanentmagneterregte Synchronmotoren unterscheiden sich hinsichtlich Regelgüte, Messaufwand, Rechenbedarf und Implementierungskomplexität. Veröffentlichte Vergleiche erlauben jedoch häufig keine eindeutige Zuordnung der beobachteten Unterschiede zum Regelungsverfahren, da Motoren, Hardware, Rückführungsgrößen oder Versuchsbedingungen voneinander abweichen. Ziel dieser Arbeit ist daher die Entwicklung einer einheitlichen Plattform und Methodik zur vergleichenden Untersuchung von Blockkommutierung, sinusförmiger Ansteuerung und feldorientierter Regelung. Hierzu werden die drei Verfahren in einem gemeinsamen, gleichungsbasierten MATLAB/Simulink-Modell unter identischen Motorparametern, Betriebsbedingungen, Auslegungsvorgaben und Bewertungskriterien untersucht. Ergänzend wird eine modulare Motorsteuerungshardware entwickelt und gefertigt. Da deren Inbetriebnahme nicht abgeschlossen werden konnte, erfolgt die experimentelle Untersuchung von Blockkommutierung und feldorientierter Regelung auf einer Ersatzplattform. In der Simulation erreicht die feldorientierte Regelung am höchsten untersuchten Strombetriebspunkt ein mittleres Drehmoment von 0,682 N m, während Blockkommutierung und sinusförmige Ansteuerung 0,218 N m beziehungsweise 0,220 N m erzielen. Bei einer Drehzahlrampe beträgt ihr mittlerer relativer Folgefehler 0,036 %, gegenüber 0,645 % und 0,616 % für die einfacheren Verfahren. Die Messungen bestätigen insbesondere die geringere Drehzahlschwankung der feldorientierten Regelung. Am niedrigen Drehzahlbetriebspunkt sinkt diese von 12,48 % bei der Blockkommutierung auf 0,76 % bei der feldorientierten Regelung, während die Prozessorauslastung von 0,63 % auf etwa 38 % steigt. Die Ergebnisse zeigen, dass eine höhere Stromausnutzung und Regelgüte mit größerem Mess-, Parametrierungs-, Abstimmungs- und Rechenaufwand verbunden sind, obwohl die Ergebnisse zwischen simulation und experiment start unterscheiden. Die experimentelle Bewertung aller drei Verfahren und die Funktionsvalidierung der entwickelten Hardware stehen noch aus.

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1. Introduction

Owing to their high energy-conversion efficiency, high torque and power density, favorable dynamic behavior, and comparatively robust operational characteristics, permanent magnet synchronous motors (PMSMs) have become increasingly relevant in a wide range of industrial, traction and motion-control applications [7, 24]. In general, PMSMs are three-phase alternating current (AC) machines supplied through a power electronic inverter. The inverter is fed by a direct current (DC) link and synthesizes controlled AC phase currents using a transistor-based switching network [7]. This configuration combines several advantages of DC-fed power electronics and AC machine operation, including precise controllability, high efficiency, and flexible control [7].

As a result of this inverter-fed drive structure, PMSMs can be operated from various DC energy sources, including battery systems [7]. This makes them particularly suitable for mobile applications, where they are used as primary or auxiliary propulsion systems in electric vehicles and electric aircraft [10, 5]. Beyond this energy-source flexibility, inverter-fed operation enables precise torque, speed, and position control [7]. Consequently, PMSMs are widely used in precision-oriented applications, including robotic actuators [17], as well as in critical auxiliary systems like electric compressors, water pumps, and power steering units [15].

The wide applicability of PMSM drives is closely linked to the performance of the underlying control system [7, 6]. In high-performance applications, the control task extends beyond the generation of a rotating stator field and includes precise current regulation, torque production, speed control, and the systematic handling of physical constraints imposed by the inverter, the DC-link, the measurement system, and the embedded control hardware and software [7]. Since the electromagnetic torque of a PMSM is directly determined by the stator current components, the quality of the current-control system has a decisive influence on the dynamic response, efficiency, operating smoothness, and torque ripple of the drive [11, 6, 9, 10]. Therefore, the development of a PMSM drive requires

an integrated design approach in which control-method selection, simulation framework, hardware architecture, and software implementation are considered jointly under realistic operating constraints.

Although numerous control approaches for PMSM drives have been proposed, they differ significantly with respect to implementation complexity, required sensing hardware, computational requirements, and achievable control performance [11]. Simpler methods, such as six-step control, can be implemented with limited computational resources but generally provide only coarse current excitation and reduced dynamic performance [11]. More advanced approaches, such as sinusoidal control and field-oriented control FOC, require increasingly sophisticated current generation and control algorithms but can offer smoother operation and improved torque control [11]. Selecting an appropriate control strategy therefore involves balancing performance requirements against implementation effort and hardware constraints [11]. A systematic comparison of these methods on a common simulation environment and embedded hardware can provide insight into their practical trade-offs and suitability for different application scenarios.

1.1. Research Objectives

To investigate the practical trade-offs between different PMSM control approaches under comparable conditions, this thesis aims to develop and implement a motor control platform for permanent magnet synchronous motors (PMSM). In the context of this work, the term *motor control platform* refers to the combined simulation, hardware, and software framework required for the development, implementation, and evaluation of motor control algorithms. The platform serves as a common environment in which selected control strategies can be implemented, tested, and evaluated under defined and reproducible operating conditions. By providing a consistent modelling, simulation, hardware, and software environment, the platform enables a systematic comparison of different control methods while minimizing the influence of external factors.

To ensure that observed differences in performance can be attributed primarily to the control strategies rather than to variations in modelling, simulation environment, hardware, software architecture, or testing conditions, all investigated strategies are implemented within this common framework. Consequently, the motor control platform must enable consistent and reproducible evaluation in both simulation and real-time operation of the

PMSM, provide the necessary interfaces for modelling, sensing, actuation, and control execution, and maintain a modular architecture that facilitates the integration and assessment of control algorithms with varying levels of complexity.

Within this framework, three control strategies of increasing complexity are investigated: six-step control, sinusoidal control, and FOC. The strategies differ with respect to rotor-position requirements, current waveform generation, computational complexity, and the expected quality of current control. Implementing all three approaches on a common motor control platform enables their advantages, limitations, and practical implementation requirements to be evaluated under consistent conditions.

Accordingly, the objectives of this thesis are:

- to develop a modular motor control platform for PMSM applications comprising a simulation framework, embedded control hardware and supporting control software;
- to establish a common modelling, simulation, and experimental environment that supports the implementation, verification, and evaluation of motor control strategies under reproducible conditions;
- to implement six-step control, sinusoidal control, and FOC within the developed motor control platform;
- to compare the implemented control strategies in simulation and experimental investigations under defined operating and measurement conditions.
- to assess their dynamic performance, current quality, implementation characteristics, and practical limitations with respect to PMSM control.

Based on these objectives, the central research question of this thesis is formulated as follows:

> How do different PMSM control strategies compare with respect to dynamic performance, current quality, implementation complexity, and practical limitations when implemented and evaluated on a common motor control platform?

1.2. Delimitations

This thesis is limited to the development and comparative evaluation of six-step control, sinusoidal control, and FOC for a permanent magnet synchronous motor. All three strategies are investigated in simulation, whereas the experimental comparison is limited to six-step control and FOC. The experiments are conducted using a substitute motor-control platform because the developed custom hardware could not be commissioned within the available research period. Consequently, the experimental results do not constitute functional validation of the custom hardware.

The investigated strategies are evaluated during unloaded operation at selected current and speed operating points that are not derived from a specific application. The evaluation focuses on current and torque behavior, speed-control performance, operating limits, and selected implementation-related characteristics. Operation under externally applied load torque, position control, field weakening, regenerative operation, fault behavior, and application-specific motion-control functions are not investigated.

Controller tuning is performed to obtain functional implementations with comparable nominal dynamics. Application-specific optimization, exhaustive tuning across the operating range, and determination of the maximum achievable performance of each strategy are outside the scope of this thesis. The results therefore characterize the relative performance of the implemented strategies under the defined evaluation conditions rather than their absolute performance limits.

1.3. Thesis Structure

This thesis is structured as follows. Chapter 2 introduces the theoretical foundations of PMSM drives, including motor modelling, inverter operation, control strategies, feedback measurement, and operating constraints. Chapter 3 reviews existing simulation frameworks, motor-control hardware, comparative control studies, and relevant development tools, and subsequently identifies the research gap addressed by this work. Chapter 4 describes the methodology applied to the hardware development, modelling and simulation, and experimental evaluation. Chapter 5 presents the design and implementation status of the custom motor-control hardware. Chapter 6 describes the developed simulation models

and presents the simulation-based comparison of six-step control, sinusoidal control, and FOC. Chapter 7 presents the experimental comparison of six-step control and FOC using the substitute motor-control platform. Chapter 8 discusses the results, the agreement between simulation and experiment, the limitations and transferability of the findings, and the contribution of the work. Finally, Chapter 9 summarizes the main conclusions and outlines directions for future work.

2. Theoretical Background

This chapter establishes the theoretical basis for the modelling, control, and implementation of the investigated PMSM drive system. It presents the operating principle and control-oriented mathematical model of the PMSM, including the reference-frame transformations, electrical and mechanical dynamics, torque production, and power relations. The operation of the three-phase voltage-source inverter is also described as the basis for implementing six-step, sinusoidal, and FOC. These control strategies, as well as a generic cascaded speed-control structure are introduced. The chapter further addresses the acquisition of the required feedback quantities and the operating constraints relevant to the control system.

2.1. Permanent Magnet Synchronous Motors

Permanent magnet synchronous motors (PMSMs) are inverter-fed electrical machines in which the rotor excitation is provided by permanent magnets [10, 7]. To illustrate the principal components and operating principle of a PMSM, a simplified machine structure is shown in Figure 2.1.

As shown in Figure 2.1, the machine consists of a stator equipped with three-phase windings and a rotor carrying permanent magnets. The stator windings are spatially displaced by 120 electrical degrees and are supplied by a power electronic inverter. The permanent magnets establish a rotor magnetic field without requiring external excitation.

When appropriately controlled three-phase currents are applied to the stator windings, a rotating magnetic field is generated. Electromagnetic torque is produced through the interaction between the rotating stator field and the rotor flux created by the permanent magnets [10, 7].

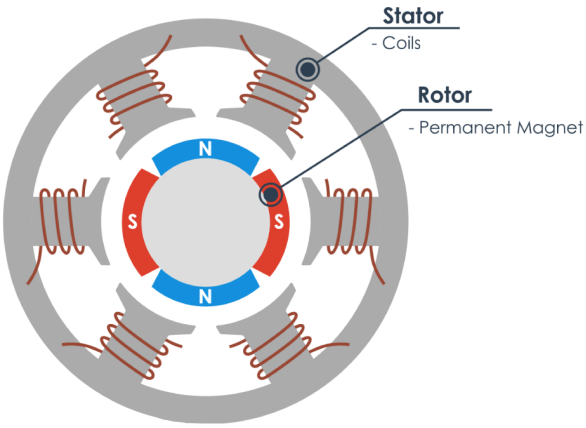


Figure 2.1.: Simplified structure of an inverter-fed permanent magnet synchronous motor with three-phase stator windings and a permanent-magnet rotor. Source: *www.mechatronics.com*, licensed under CC BY-SA 4.0, via Wikimedia Commons.

Permanent-magnet synchronous machines can differ in their rotor construction [10]. In surface-mounted PMSMs, the magnets are mounted on the rotor surface, which typically results in approximately uniform magnetic properties with respect to the rotor position [10]. In interior permanent-magnet machines, the magnets are embedded in the rotor, which can cause the magnetic properties to depend on the direction of the stator magnetic field relative to the rotor [10]. This directional dependence is referred to as magnetic saliency and affects electromagnetic torque production [10].

The three-phase stator windings may be connected in either a wye or delta configuration, which affects the phase voltages and currents [10]. The motor investigated in this thesis and its mathematical model use a wye-connected winding configuration.

2.2. Mathematical Model of the Motor

The control-oriented PMSM model used in this thesis is based on simplifying assumptions commonly adopted for PMSM modelling and controller design [10, 27]. The machine is assumed to be a balanced three-phase system, such that the stator windings are electrically symmetric and no zero-sequence component is considered [10]. The induced back-EMF and the permanent-magnet flux linkage are assumed to be sinusoidal, and the magnetic circuit is considered linear within the investigated operating range [10]. Furthermore, the

machine is modelled as non-salient. Magnetic saturation, iron losses, spatial harmonics, cogging torque, temperature-dependent parameter variations, and inverter nonlinearities are not included in the model [10].

A control-oriented description of a PMSM requires the relation between mechanical and electrical rotor quantities [10]. This relation is determined by the number of pole pairs p, where one pole pair consists of one magnetic north pole and one magnetic south pole on the rotor [10], illustrated in Figure 2.2, which shows three motors with a different number of pole pairs.

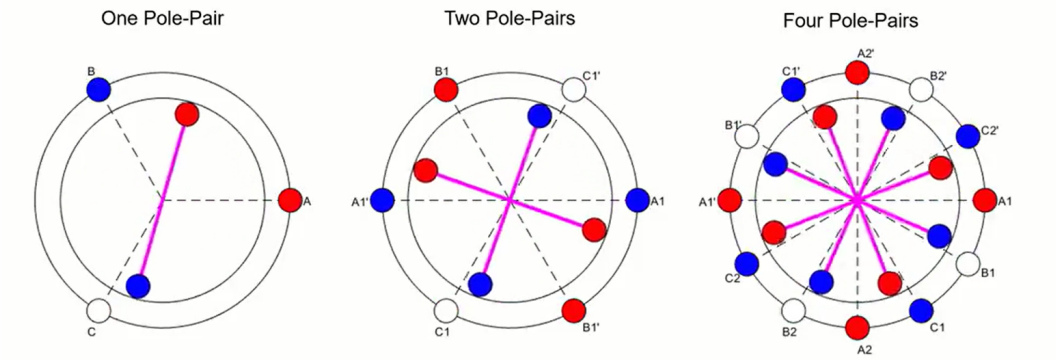


Figure 2.2.: PMSMs with a varying number of pole pairs. Source: *www.mathworks.com*, licensed under CC BY-SA 3.0, via Wikimedia Commons.

As evident through Figure 2.2, with a higher number of pole pairs, the controller has to perform multiple electrical revolutions to achieve a single mechanical revolution [10].

The mechanical rotor angle θ_m and the electrical rotor angle θ_e, and consequently mechanical speed ω_m and electrical speed ω_e are related by

$$\theta_\mathrm{e} = p\theta_\mathrm{m} \tag{2.1}$$

$$\omega_\mathrm{e} = p\omega_\mathrm{m} \tag{2.2}$$

Thus, the electrical angle completes p revolutions during one mechanical revolution of the rotor [10].

The electrical behavior of the motor can be described directly in the three-phase *abc*-stator reference frame, where the phase quantities are represented in their physical form [10]. However, this representation is not well suited for torque-oriented control design

because the phase variables are sinusoidal and inherently coupled [10]. To simplify the control problem, the three-phase quantities are first transformed into a two-axis stationary reference frame and subsequently into a rotor-oriented rotating reference frame, while preserving the information contained in the three phase quantities [10, 27].

The relation between the stationary *abc* frame, the stationary $\alpha\beta$ frame, and the rotating *dq* frame is illustrated in Figure 2.3.

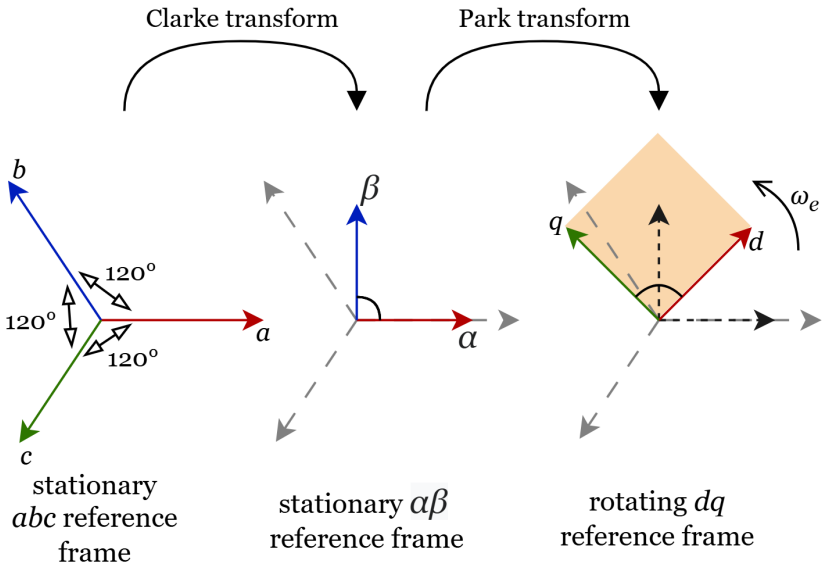


Figure 2.3.: Stationary *abc*, stationary $\alpha\beta$, and rotating *dq* reference frames of a PMSM.

For a balanced three-phase system with no zero-sequence component, the transformation from the three-phase stator system to the stationary $\alpha\beta$ frame is performed using the Clarke transformation, while the subsequent transformation into the rotating *dq* frame is performed using the Park transformation [10, 27]. In the rotor-oriented *dq* frame, which is rotated by the electrical angle θ_e , the *d*-axis is aligned with the rotor permanent-magnet flux, whereas the *q*-axis is electrically orthogonal to the *d*-axis. This alignment enables a control-oriented decomposition of the stator current into the flux-related component i_d and the torque-producing component i_q [10, 27].

The Clarke transformations for the three phase currents i_a, i_b, i_c into the two $\alpha\beta$-currents $i_\alpha, i\beta$ is defined as

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_\mathrm{a} \\ i_\mathrm{b} \\ i_\mathrm{c} \end{bmatrix}. \tag{2.3}$$

The same transformation can be applied to the stator voltage u_{abc} and flux-linkage ψ_{abc} vectors [10].

The inverse Clarke transformation maps the $\alpha\beta$ quantities back to the three-phase *abc* frame [10]. For the stator currents, it is given by

$$\begin{bmatrix} i_{\mathrm{a}} \\ i_{\mathrm{b}} \\ i_{\mathrm{c}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}. \tag{2.4}$$

The subsequent Park transformation for the stator currents is expressed as [10]

$$\begin{bmatrix} i_{\mathrm{d}} \\ i_{\mathrm{q}} \end{bmatrix} = \begin{bmatrix} \cos\theta_{\mathrm{e}} & \sin\theta_{\mathrm{e}} \\ -\sin\theta_{\mathrm{e}} & \cos\theta_{\mathrm{e}} \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}. \tag{2.5}$$

The inverse Park transformation maps rotor-oriented *dq* quantities back to the stationary $\alpha\beta$ frame [10]. For the stator currents, it is given by

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \begin{bmatrix} \cos\theta_{\mathrm{e}} & -\sin\theta_{\mathrm{e}} \\ \sin\theta_{\mathrm{e}} & \cos\theta_{\mathrm{e}} \end{bmatrix} \begin{bmatrix} i_{\mathrm{d}} \\ i_{\mathrm{q}} \end{bmatrix}. \tag{2.6}$$

Again, equivalent transformations can be applied to the voltage and flux-linkage vectors.

In the *dq* reference frame, magnetic saliency is represented by different inductances along the direct and quadrature axes, denoted by L_{d} and L_{q}, respectively [10]. As the machine is modelled as non-salient, the direct- and quadrature-axis inductances are assumed to be equal and equal the stator inductance L_{s}:

$$L_{\mathrm{d}} = L_{\mathrm{q}} = L_{\mathrm{s}}. \tag{2.7}$$

Using the non-salient assumption in Equation 2.7, the electrical dynamics of the PMSM in the rotor-oriented *dq* reference frame are described by

$$u_{\mathrm{d}} = R_{\mathrm{s}} i_{\mathrm{d}} + L_{\mathrm{s}} \frac{di_{\mathrm{d}}}{dt} - \omega_{\mathrm{e}} L_{\mathrm{s}} i_{\mathrm{q}}, \tag{2.8}$$

$$u_{\mathrm{q}} = R_{\mathrm{s}} i_{\mathrm{q}} + L_{\mathrm{s}} \frac{di_{\mathrm{q}}}{dt} + \omega_{\mathrm{e}} \left(L_{\mathrm{s}} i_{\mathrm{d}} + \psi_{\mathrm{PM}} \right), \tag{2.9}$$

where u_{d} and u_{q} are the direct- and quadrature-axis stator voltage components, R_{s} is the stator resistance, and ψ_{PM} is the permanent-magnet flux linkage [10, 27].

The terms containing the electrical angular velocity ω_{e} represent speed-dependent voltage components. The terms $-\omega_{\mathrm{e}} L_{\mathrm{s}} i_{\mathrm{q}}$ and $\omega_{\mathrm{e}} L_{\mathrm{s}} i_{\mathrm{d}}$ describe the cross-coupling between the d- and q-axis dynamics, whereas the term $\omega_{\mathrm{e}} \psi_{\mathrm{PM}}$ represents the back-EMF induced by the permanent-magnet rotor.

The electromagnetic torque of a non-salient PMSM is determined by the interaction between the permanent-magnet flux linkage and the quadrature-axis current [10]. Since $L_{\mathrm{d}} = L_{\mathrm{q}}$, no reluctance torque component is considered. The torque equation is therefore expressed as

$$T_{\mathrm{e}} = K_{\mathrm{t}} i_{\mathrm{q}}, \tag{2.10}$$

$$K_{\mathrm{t}} = \frac{3}{2} p \psi_{\mathrm{PM}}. \tag{2.11}$$

where T_{e} denotes the electromagnetic torque and K_{t} represents the motor torque constant [10, 27]. Accordingly, the torque is directly proportional to the quadrature-axis current i_{q}, while the direct-axis current i_{d} does not contribute to torque production.

The generated electromagnetic torque determines the mechanical behavior of the rotor, which is described by

$$J \frac{d\omega_{\mathrm{m}}}{dt} = T_{\mathrm{e}} - T_{\mathrm{L}} - B\omega_{\mathrm{m}}, \tag{2.12}$$

where J is the moment of inertia, T_{L} is the load torque, and B is the viscous friction coefficient [10, 7]. The equation relates the rotor acceleration to the net torque acting on the shaft.

Based on the electrical and mechanical quantities introduced above, the electrical input power P_{el} is given by [7]

$$P_{\mathrm{el}} = u_{\mathrm{a}} i_{\mathrm{a}} + u_{\mathrm{b}} i_{\mathrm{b}} + u_{\mathrm{c}} i_{\mathrm{c}}, \tag{2.13}$$

while the mechanical output power P_{m} is [7]

$$P_{\mathrm{m}} = T_{\mathrm{e}} \omega_{\mathrm{m}}. \tag{2.14}$$

The dc-link current I_{dc}estimated from the electrical power consumption of the motor-drive system and the dc-link voltage U_{dc} [7]

$$I_{\mathrm{dc}} = \frac{P_{\mathrm{el}}}{U_{\mathrm{dc}}}. \tag{2.15}$$

2.3. Three-Phase Inverter Operation

The controlled operation of a PMSM requires a power electronic converter that generates three-phase stator voltages from a DC supply [10]. In electric drive systems, this function is commonly performed by a two-level three-phase voltage-source inverter [7]. As illustrated in Figure 2.4, the inverter consists of three half-bridges connected to a common DC-link voltage U_{DC}. Each half-bridge contains an upper and a lower switching device. These devices must not conduct simultaneously, since this would short-circuit the DC link. Therefore a short dead time is inserted between switching transitions to prevent shoot-through currents [7].

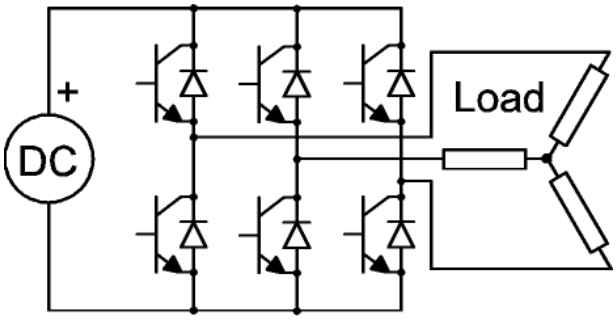


Figure 2.4.: Two-level three-phase voltage-source inverter supplying a PMSM. Source: *C J Cowie at the English-language Wikipedia*, licensed under CC BY-SA 3.0, via Wikimedia Commons.

Each inverter leg connects the corresponding motor phase either to the positive or to the negative DC-link potential, or can, depending on the control strategy, be left floating. The combination of the three leg states defines the instantaneous switching state of the inverter.

For a wye-connected PMSM without an accessible neutral point, the switch-node voltages u_{x0} are generally not identical to the phase-to-neutral voltages u_{x} applied to the stator windings, since the motor neutral-point potential may differ from the inverter reference potential [7]. For the present motor model, the neutral-point potential is determined by the common-mode component of the three switch-node voltages [7]:

$$u_{\mathrm{n0}} = \frac{1}{3}\left(u_{\mathrm{a0}} + u_{\mathrm{b0}} + u_{\mathrm{c0}}\right). \tag{2.16}$$

The phase-to-neutral voltage of each phase is obtained by subtracting this neutral-point potential from the corresponding switch-node voltage [7]

$$u_{\mathrm{x}} = u_{\mathrm{x0}} - u_{\mathrm{n0}}, \tag{2.17}$$

This relation removes the common-mode voltage component and yields a balanced set of phase voltages satisfying [7]

$$u_{\mathrm{a}} + u_{\mathrm{b}} + u_{\mathrm{c}} = 0. \tag{2.18}$$

The resulting phase voltages represent the differential voltage components that drive the stator currents and can be transformed into the rotating dq-reference frame using the transformations introduced in section 2.2.

Since a two-level inverter can apply only a finite number of discrete voltage states instantaneously, pulse-width modulation (PWM) is used to synthesize the required average stator voltage over one switching period [7]. The duty cycle of each inverter leg defines the fraction of the switching period during which the corresponding phase is connected to the positive DC-link potential. The averaged switch-node voltages obtained in this way are converted into phase-to-neutral voltages according to the neutral-point relation

given in Equation 2.17. Consequently, the achievable average stator voltage is limited by the available DC-link voltage U_{DC}. If the requested voltage exceeds this limit, voltage saturation occurs and must be considered in the design and implementation of the control system [7].

The switching frequency determines how often the inverter states are updated within one second. A higher switching frequency can reduce current ripple and improve the approximation of the commanded average voltage, but it also increases switching losses and places higher timing requirements on the embedded control system [7]. Therefore, the selected switching frequency represents a trade-off between current quality, converter losses, thermal loading, and real-time implementation constraints [7].

2.4. Control Strategies

This section presents the three control strategies investigated in this work and compares them with respect to their control principles, feedback requirements, parameter dependency, and implementation complexity.

2.4.1. Six-Step Control

Six-step control is a sector-based control method in which the electrical rotor position is divided into six sectors of 60° electrical angle [10]. In each sector, two inverter legs are energized, connecting one motor phase to the positive and another to the negative DC-link terminal, while the third phase is left electrically floating and is not intentionally excited for torque production [10, 7]. The terminal voltage of the floating phase is therefore not directly imposed by the inverter but is influenced by the motor back-EMF, winding coupling, and freewheeling paths [7]. As the rotor transitions between sectors, the energized phase pair is changed according to a predefined commutation sequence. This results in six discrete voltage vectors per electrical revolution and produces trapezoidal commutation [10]. The corresponding sequence for one electrical revolution is presented in Table 2.1 and illustrated in Figure 2.5.

Table 2.1.: Generic six-step commutation sequence for one electrical revolution.

Sector	Electrical angle range	High-side phase	Low-side phase
1	$0° \leq \theta_e < 60°$	A	B
2	$60° \leq \theta_e < 120°$	A	C
3	$120° \leq \theta_e < 180°$	B	C
4	$180° \leq \theta_e < 240°$	B	A
5	$240° \leq \theta_e < 300°$	C	A
6	$300° \leq \theta_e < 360°$	C	B

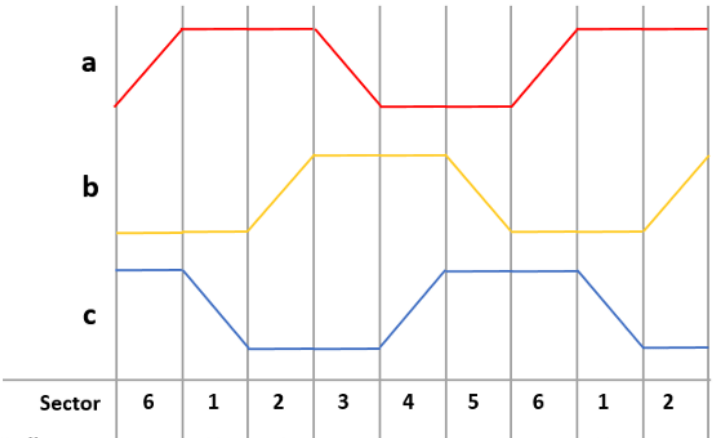


Figure 2.5.: Generic six-step commutation sequence for one electrical revolution. Source: *MathWorks*, licensed under CC BY-SA 4.0, via Wikimedia Commons.

Figure 2.5 shows wether a phase is currently set to high or to low. The floating phase slowly falls or rises in potential. This zero crossing can be recognized using phase voltage sensors [10].

The commutation state can be determined from Hall sensor signals, by mapping a measured rotor angle to the corresponding electrical sector, or by estimating the rotor position from the back-EMF voltage measured at the floating phase [7].

The torque and speed of the motor are influenced by the average voltage applied during each commutation sector. The selected commutation state determines the active phase pair, whereas the PWM duty cycle defines the average line-to-line voltage applied between

these phases over one switching period [7]. For a sector in which phase p is connected to the positive DC-link terminal and phase q is connected to the negative terminal, the average applied line-to-line voltage can be expressed as

$$\overline{u}_{pq} = d^* U_{\mathrm{DC}}, \qquad 0 \leq d^* \leq 1, \tag{2.19}$$

where $\overline{u}_{pq}$ denotes the average voltage applied between the active phases p and q, d^* is the PWM duty cycle, and U_{DC} is the DC-link voltage [7]. Consequently, the duty cycle can be used as the manipulated variable for basic open-loop or closed-loop speed control.

Six-step control is typically presented as the least complex method because it requires only discrete rotor-position information and simple commutation logic [10, 11]. However, the discrete switching between commutation sectors leads to non-sinusoidal phase currents, which increases torque ripple and acoustic noise. These effects are especially relevant at low speed and during commutation transitions [10, 11].

2.4.2. Sinusoidal control

Sinusoidal control improves the phase-current and torque behavior by generating three phase references that vary sinusoidally with the electrical rotor angle. In contrast to six-step control, where only one of six discrete commutation states is selected, sinusoidal control provides balanced continuous phase commands for the phases, satisfying Equation 2.18. These commands are phase-shifted by 120° electrical angle [10] and can be expressed as

$$u_{\mathrm{a}}^* = U^* \sin(\theta_{\mathrm{e}}), \tag{2.20}$$

$$u_{\mathrm{b}}^* = U^* \sin\left(\theta_{\mathrm{e}} - \frac{2\pi}{3}\right), \tag{2.21}$$

$$u_{\mathrm{c}}^* = U^* \sin\left(\theta_{\mathrm{e}} + \frac{2\pi}{3}\right). \tag{2.22}$$

Here, u_{a}^*, u_{b}^*, and u_{c}^* denote the desired sinusoidal phase voltage references. The quantity U^* refers to the peak amplitude of these sinusoidal phase voltage references, i.e., the

amplitude of the fundamental phase-to-neutral voltage reference of the balanced three-phase system [10].

Sinusoidal control can be represented by the control block diagram shown in Figure 2.6, comprising of the sinusoidal voltage reference generator and external blocks like the inverter, or the PMSM.

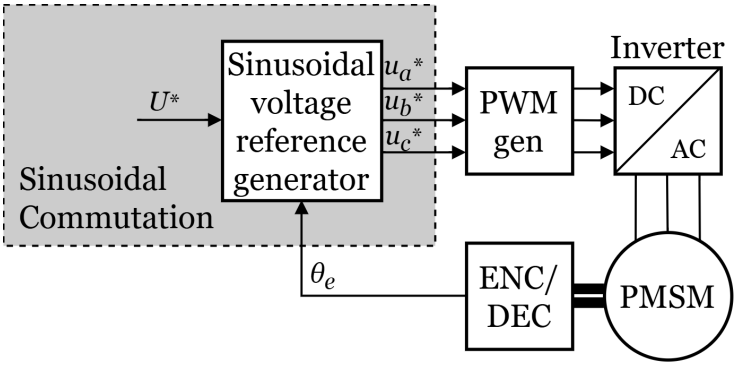


Figure 2.6.: Control block diagram of sinusoidal control.

Figure 2.6 indicates that the PWM generation block is functionally separated from the block that determines the phase voltage references.

A direct implementation of sinusoidal control uses **sinusoidal pulse-width modulation (SPWM)** for PWM generation [7]. In this approach, the PWM duty cycles are computed directly from the three-phase reference voltages:

$$d_{\mathrm{x}} = \frac{1}{2} + \frac{u_{\mathrm{x}}^*}{U_{\mathrm{DC}}}, \qquad x \in \{a, b, c\}. \tag{2.23}$$

where d_{x} denotes the duty cycle of phase x, and u_{x}^* denotes the desired sinusoidal phase voltage reference. The offset term $\frac{1}{2}$ maps the zero-centered sinusoidal voltage references to a duty cycle of 50 %. Positive and negative voltage references therefore correspond to duty cycles above and below 50 %, respectively [7]. In contrast to the six-step duty-cycle formulation introduced in subsection 2.4.1, the duty cycle does not only scale a selected active voltage vector, but also represents the sign and magnitude of a bipolar phase voltage reference.

As the duty cycle cannot exceed 100%, the phase voltage reference must remain within

the available PWM range for SPWM [7]:

$$-\frac{U_{\mathrm{DC}}}{2} \leq u_{\mathrm{x}}^{*} \leq \frac{U_{\mathrm{DC}}}{2}, \qquad x \in \{a, b, c\}. \tag{2.24}$$

This limits the voltage utilization of SPWM to a maximum phase voltage reference amplitude of

$$U_{\mathrm{max,spwm}}^{*} = \frac{U_{\mathrm{DC}}}{2}. \tag{2.25}$$

To improve the utilization of the available DC-link voltage, a common-mode voltage injection can be applied equally to all three phase references before the duty cycles are computed [25]. This approach constitutes a carrier-based implementation of continuous **space-vector PWM (SVPWM)**, where the PWM switching instants of all three phases are synchronized and the shift is determined from the instantaneous maximum and minimum phase references [25, 7]:

$$u_0 = -\frac{1}{2}\left(\max\left(u_{\mathrm{a}}^{*}, u_{\mathrm{b}}^{*}, u_{\mathrm{c}}^{*}\right) + \min\left(u_{\mathrm{a}}^{*}, u_{\mathrm{b}}^{*}, u_{\mathrm{c}}^{*}\right)\right). \tag{2.26}$$

The modified phase voltage references are then given by

$$u_{\mathrm{x,mod}}^{*} = u_{\mathrm{x}}^{*} + u_0, \qquad x \in \{a, b, c\}. \tag{2.27}$$

These modified references are converted into PWM duty cycles according to

$$d_{\mathrm{x}} = \frac{1}{2} + \frac{u_{\mathrm{x,mod}}^{*}}{U_{\mathrm{DC}}}, \qquad x \in \{a, b, c\}. \tag{2.28}$$

The injected common-mode component shifts all three phase references by the same instantaneous value and therefore cancels in the phase-to-phase voltages [7]. As a result, the individual phase commands and duty-cycle waveforms become non-sinusoidal, whereas the torque-relevant line-to-line voltage references remain unchanged, yielding a maximum achievable fundamental phase voltage reference in the linear modulation region of [25]

$$U^*_{\text{max,svpwm}} = \frac{U_{\text{DC}}}{\sqrt{3}}. \tag{2.29}$$

Compared with SPWM, this corresponds to an increase in the achievable fundamental phase voltage amplitude of approximately 15.5 % [25].

The resulting voltage waveforms are shown in Figure 2.7, which plots the phase and line-to-line references of SPWM and SVPWM over time.

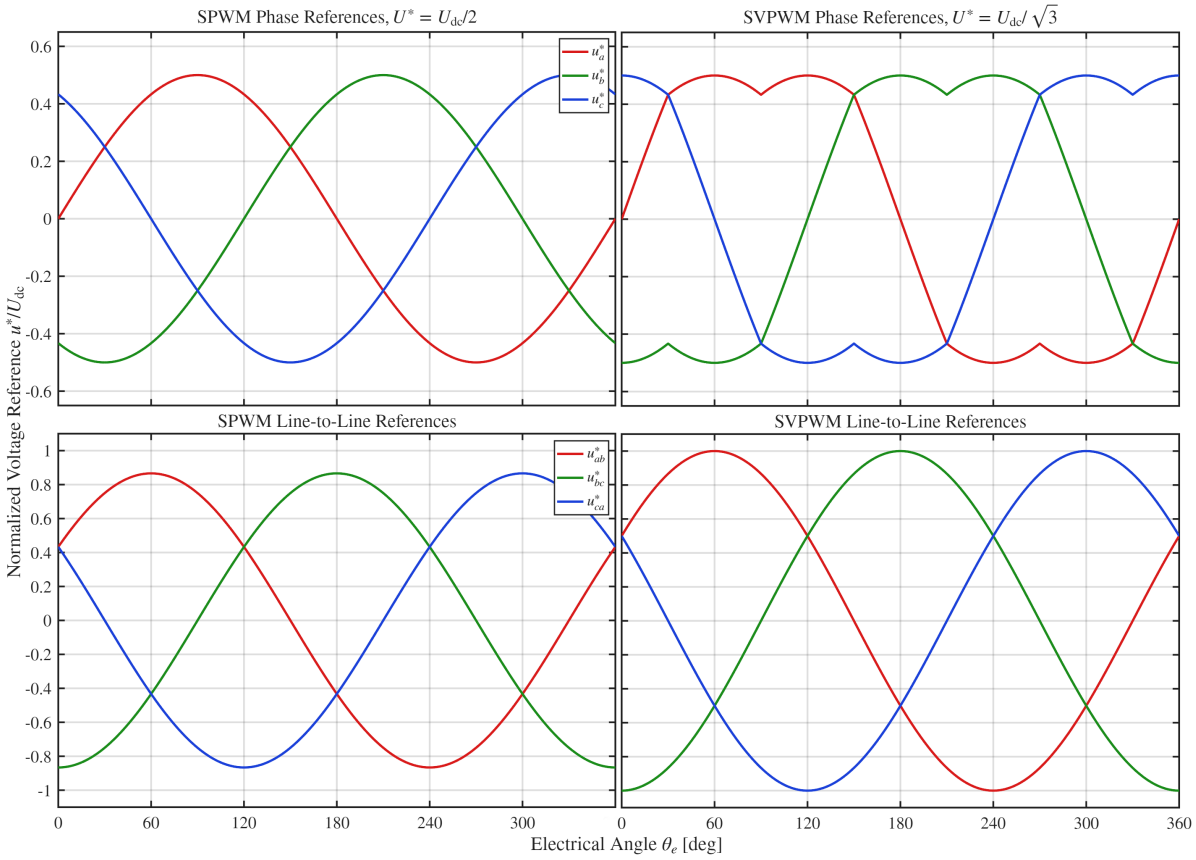


Figure 2.7.: Comparison of SPWM and SVPWM waveforms at maximum modulation.

Figure 2.7 shows that at the maximum linear modulation index, the SPWM phase command voltages remain sinusoidal and reach the limits of the available PWM range. By contrast, the SVPWM phase commands exhibit the characteristic saddle-shaped waveform resulting from the addition of a common-mode voltage component. The line-to-line references achieve higher amplitudes for SVPWM than for SPWM.

Unlike SPWM, in which the phase command voltages are generated from individual sinusoidal reference signals, SVPWM treats the three-phase inverter as a single system. The

switching states of all three inverter legs are coordinated to generate a single resultant voltage space vector in the stationary reference frame. Its magnitude and angular position are determined by the commanded three-phase voltages. As the electrical angle changes, the reference voltage vector passes successively through the six sectors of the space-vector diagram. The corresponding modulation principle is illustrated in Figure 2.8.

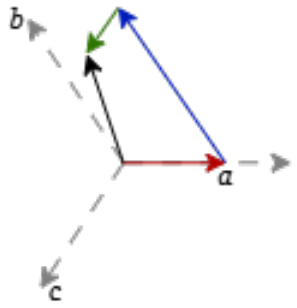


Figure 2.8.: Synthesis of the reference voltage vector from the voltage vectors generated by the available inverter switching states.

Figure 2.8 illustrates how the three phase voltages are added to result in any phase vector in the plane.

Sinusoidal control represents an intermediate level of complexity. The continuously varying phase references improve the current waveform and generally reduce torque ripple and acoustic noise during steady-state operation [11]. However, sinusoidal control requires continuous rotor-angle information and a more complex PWM reference generation scheme.

2.4.3. Field-Oriented Control

Field-oriented control (FOC) is a closed-loop current-vector control method formulated in the rotor-oriented *dq* reference frame [10]. In contrast to six-step and sinusoidal control, the manipulated variables are not determined directly from the rotor angle and a scalar voltage reference. Instead, FOC regulates the flux- and torque-related current components separately [10, 27]. This enables more precise torque control and current limitation. The simplified control structure used for this principle is shown in Figure 2.9.

For the structure shown in Figure 2.9, The measured phase currents are first transformed from the *abc* reference frame into the rotating *dq* reference frame using the reference frame transformations, given in Equation 2.3 and Equation 2.5. The Park transformation

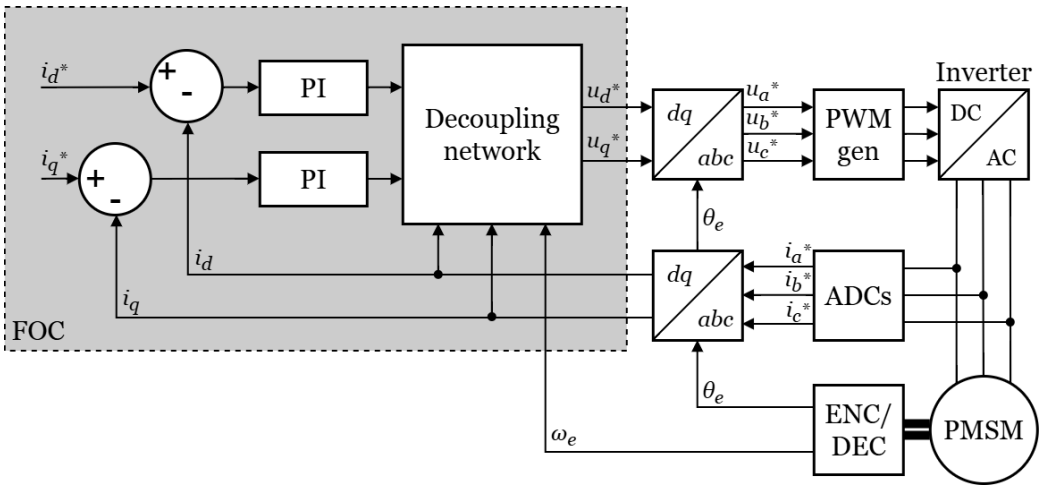


Figure 2.9.: Simplified FOC structure showing the coordinate transformations, current controllers, and PWM generation.

requires the electrical rotor angle θ_e feedback from a sensor or a sensorless estimator [10, 27].

Since the electromagnetic torque is proportional to i_q, as shown in Equation 2.11, the quadrature-axis current reference i_q^* specifies the desired torque-producing current. The direct-axis current reference is commonly set to $i_\mathrm{d}^* = 0$ [10]. For the present PMSM model, the direct-axis current would only increase the stator current magnitude [10]. Separate proportional-integral (PI) current controllers are therefore used to regulate the d- and q-axis current errors.

The speed-dependent coupling terms in Equation 2.8 and Equation 2.9 increase with the electrical angular velocity and limit the validity of treating the two current-control loops as independent single-input single-output systems. Feedforward decoupling is therefore introduced to compensate the rotational coupling and the permanent-magnet back-EMF [10]:

$$u_{\mathrm{d,ff}} = \omega_\mathrm{e} L_\mathrm{s} i_\mathrm{q}, \tag{2.30}$$

$$u_{\mathrm{q,ff}} = -\omega_\mathrm{e} \left(L_\mathrm{s} i_\mathrm{d} + \psi_\mathrm{PM} \right). \tag{2.31}$$

The compensated voltage references are obtained by adding the feedforward terms to the controller outputs:

$$u^*_{\mathrm{d,comp}} = u_{\mathrm{d,ctrl}} + u_{\mathrm{d,ff}} = u_{\mathrm{d,ctrl}} + \omega_{\mathrm{e}} L_{\mathrm{s}} i_{\mathrm{q}}, \tag{2.32}$$

$$u^*_{\mathrm{q,comp}} = u_{\mathrm{q,ctrl}} + u_{\mathrm{q,ff}} = u_{\mathrm{q,ctrl}} - \omega_{\mathrm{e}} \left(L_{\mathrm{s}} i_{\mathrm{d}} + \psi_{\mathrm{PM}} \right). \tag{2.33}$$

With this compensation, the current controllers primarily regulate the resistive and inductive parts of the current dynamics, while the model-based feedforward terms account for the dominant speed-dependent voltage components [10, 27].

The compensated voltage references $u^*_{\mathrm{d,comp}}$ and $u^*_{\mathrm{q,comp}}$ are transformed back into three-phase voltage references by applying the inverse Park and Clarke transformations. These references are supplied to the PWM generation stage using the same voltage synthesis principles as in sinusoidal control, either SPWM or SVPWM [10, 27].

FOC provides the highest level of control authority among the three considered methods [11]. Seperate control of the d- and q-axis current components allows the torque-producing current component to be controlled directly and provides a suitable basis for dynamic torque control, current limitation, and low-ripple operation. These advantages are obtained at the cost of higher implementation complexity, phase-current measurement, coordinate transformations, controller tuning, and increased real-time computational requirements [11].

2.4.4. Strategy Overview

Based on the control principles presented in the preceding subsections, the three strategies differ primarily in their required feedback, dependence on motor parameters, and implementation complexity. Table 2.2 summarizes these characteristics and provides a concise overview of the main distinctions between the investigated strategies.

Table 2.2.: Comparative overview of the investigated PMSM control strategies.

Strategy	**Principle**	**Required feedback**	**Required motor parameters**	**Complexity**
Six-step control	Sector-based switching	Rotor sector	None for basic operation	Low
Sinusoidal control	Sinusoidal phase references	Electrical rotor angle	None for basic operation	Medium
FOC	*dq*-current regulation	Phase currents and rotor angle	R_s, L_s, and ψ_PM	High

2.5. Cascaded Control Structure

PMSM drives are commonly controlled using a cascaded control structure, in which a slower outer control loop regulates the mechanical speed and a faster inner control level determines the electrical actuation [10, 27]. This separation reflects the different time scales of the drive system. The electrical dynamics are typically faster than the mechanical dynamics of the rotor and load [10]. Therefore, the inner current or torque control loop is usually designed with a substantially higher bandwidth than the outer speed loop, commonly by a factor of approximately 5 to 10 [14]. This bandwidth separation helps the inner loop track the current references generated by the speed controller before the outer loop changes significantly, thereby supporting stable and well-damped closed-loop operation [23].

In this thesis, the outer loop is implemented as a speed controller. The measured mechanical speed ω_m is compared with the reference speed ω_m^*, and the resulting controller output y is used as the reference quantity for the inner control level.

Figure 2.10 illustrates the generic cascaded control structure used for the comparison of the investigated control strategies.

Since the outer loop structure remains identical, performance differences can be attributed

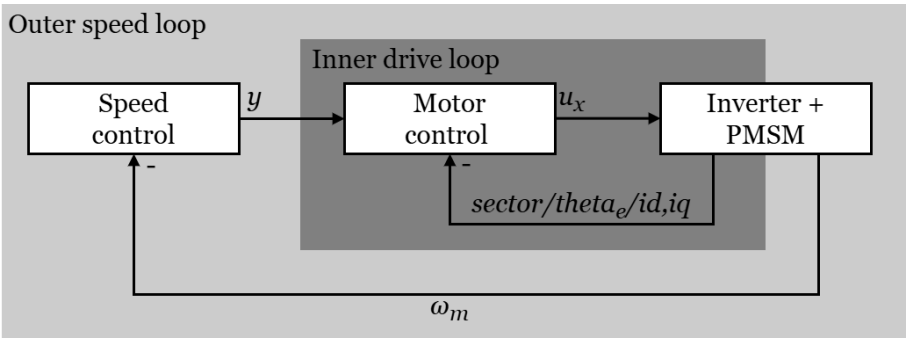


Figure 2.10.: Generic cascaded control structure used for the comparison of the investigated PMSM control strategies.

primarily to the inner control strategy, provided similar controller tuning. The dynamic performance can therefore be evaluated using common time-domain quantities, such as rise time, overshoot, settling time, and steady-state error [3].

2.6. Measurement of Feedback Quantities

Accurate feedback of rotor position, rotational speed, phase currents, and phase voltages is required for the implementation and evaluation of the investigated PMSM control strategies [10]. The required feedback quantities depend on the selected control method and are described in section 2.4. This section briefly introduces the measurement methods considered in the control platform design.

The mechanical rotor angle can be acquired using a magnetic rotary encoder mounted on the motor shaft. Such encoders provide an absolute angular position within one mechanical revolution [17, 1]. The measured mechanical angle θ_m is converted into the electrical rotor angle θ_e using the pole-pair relation introduced in Equation 2.1. The electrical rotor angle can then be mapped to discrete rotor sectors, thereby emulating the sector information typically provided by Hall sensors [7, 10]. The mechanical angular velocity ω_m can be estimated from the time derivative of the measured mechanical angle [17].

In sensorless control schemes, rotor position and speed are not measured directly but are reconstructed from the machine's electrical quantities. Such estimation methods commonly rely on back-EMF information or observer-based algorithms that utilize measured currents and voltages to estimate the rotor state [10]. Since the back-EMF magnitude

is proportional to the rotor speed, it decreases as the rotational speed approaches zero. Consequently, the observability of rotor position and speed is reduced during startup and low-speed operation, which generally limits estimation accuracy in this operating range [10]. Furthermore, the dynamic characteristics of the estimator influence the response to startup, speed changes, and other transient operating conditions. Therefore, in sensorless operation, the reported rotor speed should be interpreted as an estimated quantity provided by the observer or estimator rather than as an independent mechanical shaft-speed measurement.

The phase currents can be measured using inline shunt resistors placed in the motor phase paths. The voltage drop across each shunt resistor is proportional to the corresponding phase current and is processed by bidirectional current-sense amplifiers [13]. Bidirectional current measurement is required because the phase currents of an inverter-fed PMSM change polarity during normal operation. The amplified analogue signals are sampled by an analogue-to-digital converter (ADC) and converted into phase-current values after offset and gain correction. Direct inline phase current sensing generally offers higher accuracy compared to high- and low-side current sensing, where the current is measured between the inverter and the positive and negative DC-link respectively and does not require phase current reconstruction [13].

The phase voltages can be measured using resistive voltage dividers that scale the inverter phase potentials to the input range of the ADC. The measured voltages provide information about the electrical actuation applied by the inverter and can be used for monitoring, validation, diagnostic purposes or for sensorless position estimation [20, 10].

2.7. Operating Constraints

The operation of the investigated control strategies is limited by the physical capabilities of the motor, inverter, and embedded hardware platform. In particular, the generated control references must remain within the available DC-link voltage, the admissible inverter duty-cycle range, and the current limits of the motor and power electronics. A requested value exceeding the corresponding physical limit cannot be realized by the drive system and must therefore be constrained.

For the generic cascaded control structure introduced previously, the limitation of a con-

troller output can be expressed as

$$y_{\text{lim}} = \text{sat}\left(y^*, y_{\text{min}}, y_{\text{max}}\right), \tag{2.34}$$

where y^* denotes the unconstrained controller output, y_{lim} is the constrained reference, and y_{min} and y_{max} are the corresponding lower and upper limits, respectively [3]. The physical interpretation of these quantities depends on the selected control strategy seen in section 2.4.

If a controller with an integral term reaches an output limit, the discrepancy between the requested and realizable control actions can cause integrator windup and thereby impair the transient response [3]. To reduce this effect, the integrator can be corrected using a back-calculation term:

$$\dot{x}_{\text{I}} = e + K_{\text{aw}}\left(y_{\text{lim}} - y^*\right), \tag{2.35}$$

where x_{I} is the integrator state, e is the controller input error, and K_{aw} is the anti-windup gain [3]. This correction counteracts further integration when the requested controller output exceeds the physically achievable range.

A common starting point for determining the anti-windup gain of PI-controllers K_{aw} is defined as [3]

$$K_{\text{aw}} = \frac{K_{\text{i}}}{K_{\text{p}}}, \tag{2.36}$$

where K_{i} and K_{p} are the controller gains for the PI integral and proportional terms, respectively.

A further constraint arises from the discrete-time execution of the controllers on the embedded platform. Controllers designed and evaluated in continuous-time simulation cannot generally be transferred directly to a discrete-time implementation without considering the selected sampling period and discretization method [14]. In particular, the numerical representations of integral and derivative terms depend on the sampling period. Consequently, controller gains derived for a continuous-time model must be converted, recalculated, or retuned according to the employed discrete-time formulation [14]. The resulting controller

must subsequently be verified to ensure that its stability and dynamic behaviour remain consistent with the intended continuous-time design [3].

3. State of the Art

This chapter presents a review of existing approaches, control hardware, and development environments relevant to the implementation and evaluation of PMSM control strategies. Existing hardware architectures for PMSM control are examined with regard to their suitability for experimental investigations, followed by an analysis of studies comparing different control methods for permanent magnet synchronous motors. In addition, development tools and software frameworks that facilitate modelling, implementation, and experimental validation are reviewed. Based on the findings of this review, the existing research gap is identified, and the contribution of this thesis is established within the context of current research.

3.1. Approaches to Motor Drive Simulations

Existing simulation approaches for PMSM drives can generally be divided into two categories. The first comprises physical-modelling frameworks that represent the drive using predefined components for the machine, inverter, load, sensors, and associated physical domains. The second comprises custom equation-based models in which motor dynamics, inverter behavior, control algorithms, and measurement signals are implemented explicitly using mathematical and signal-flow blocks.

3.1.1. Physical-Modelling Frameworks

Physical-modelling environments provide component libraries for constructing motor-drive systems from representations of physical components. A representative example is *Simscape Electrical* [37], which extends MATLAB/Simulink with models of electrical ma-

chines, semiconductor devices, power converters, sensors, mechanical components, and electrical networks [37]. These components can be connected according to their physical relationships, allowing electrical, mechanical, and thermal effects to be simulated within a common environment.

For PMSM drives, Simscape Electrical provides predefined models for surface-mounted and interior permanent-magnet machines. Depending on the selected model and parameterization, effects such as rotor saliency, magnetic saturation, flux-linkage characteristics, inverter switching, and mechanical loading can be represented [37]. Such models are well suited for system-level investigations of interactions between the motor, inverter, load, and other physical domains, including effects such as semiconductor switching losses, magnetic nonlinearities, and thermal behavior [37].

However, the suitability of a predefined model depends on the investigation objectives and available parameter data. Predefined components may limit access to underlying equations and modelling assumptions, reducing flexibility when modifications or consistent model structures are required across multiple control strategies.

3.1.2. Custom Equation-Based Simulation Models

Custom equation-based models represent the electrical and mechanical dynamics explicitly through mathematical relationships rather than predefined physical components. Quantities such as phase currents, back-EMF, electromagnetic torque, rotor position, and rotational speed are computed directly from these equations, typically implemented using general-purpose numerical or signal-flow modelling tools such as MATLAB/Simulink, Python, or Modelica.

Explicit implementation provides direct access to internal states, intermediate variables, and modelling assumptions. This facilitates the consistent realization of the control strategies within a common framework while maintaining identical motor parameters, inverter assumptions, sampling behavior, and evaluation conditions. As a result, differences in simulation results can be attributed more directly to the control strategy itself.

Compared with physical-modelling approaches, custom models require greater implementation and verification effort because all relevant equations must be defined explicitly. Effects not included in the model, are not represented automatically. Consequently, sim-

ulation accuracy depends on the selected model structure, parameter quality, and verification of the implemented equations.

3.2. Existing Motor Control Hardware

Motor control hardware for PMSM applications varies considerably in how much of its internal architecture is accessible to the user. At one end are closed, proprietary, or application-specific systems that reveal little about their internal design. At the other are documented evaluation and reference platforms, including manufacturer development boards, inverter expansion boards, and modular reference designs, which are built for adaptability and transparency.

3.2.1. Control Hardware with Limited Architectural Transparency

Control hardware with limited architectural transparency includes industrial motor controllers and research-specific test benches.

Industrial motor controllers are designed for reliable operation in predefined applications and typically provide integrated power stages, protection functions, standardized communication interfaces, and application-specific configuration options. While these features support efficient commissioning and robust operation, their often proprietary and application-driven design limits flexibility regarding control algorithms, sensing configurations, hardware expansion, and supported motor types. Access to circuit schematics, internal signals, switching behavior, and low-level software is frequently restricted, reducing the ability to modify the control architecture, analyze individual hardware and software components.

Research-specific test benches are usually tailored to a particular motor, inverter, sensor configuration, control strategy, or research objective. Although this specialization enables detailed investigation of the targeted research question, it often limits adaptability to alternative hardware configurations or control approaches. In addition, published work commonly focuses on algorithmic performance rather than complete platform documentation. As a result, implementation details may remain insufficiently documented,

reducing reproducibility, hindering transferability to other platforms, and limiting their transparency and configurability.

3.2.2. Control Hardware with Documented Architectures

Control hardware with documented architectures includes reference designs, evaluation kits, and development boards. These platforms are commonly accompanied by circuit schematics, component specifications, signal assignments, measurement interfaces, and software examples or software development kits. The available documentation provides information about the interaction between the control electronics, power stage, sensing circuitry, communication interfaces, and protection mechanisms.

Documented motor-control platforms can generally be classified as compact all-in-one solutions or modular systems. Compact platforms integrate the microcontroller, inverter, gate driver, and measurement circuitry on a single board. An example is the *STMicroelectronics* `B-G431B-ESC1` [28]. Due to the integrated architecture, the functional blocks are closely interconnected, and the available access points are determined by the board layout and provided interfaces, hindering support for future expandability.

Modular platforms separate the control electronics from the power stage. Examples include the *Texas Instruments* `LAUNCHXL-F280049C` [35] combined with the `BOOSTXL-3PHGANINV` [34] and the *STMicroelectronics* `NUCLEO-G431RB` [30] combined with the `X-NUCLEO-IHM16M1` [33]. In these configurations, a microcontroller development board is connected to a dedicated three-phase inverter module through defined electrical and communication interfaces. The individual modules provide the controller peripherals, power-electronic components, sensing circuits, and connection points required for motor-control implementation.

As modular hardware platforms allow for future expandability, they are of significant relevance for the use as the control hardware. The relevant characteristics of the two representative modular motor-control solutions are extracted from the datasheets [35, 35, 30, 33] and summarized in Table 3.1.

Table 3.1.: Comparison of the investigated motor-control development hardware.

Characteristic	LAUNCHXL-F280049C + BOOSTXL-3PHGANINV	NUCLEO-G431RB + X-NUCLEO-IHM16M1
Microcontroller	TI C2000 F280049C	STM32G431RB
Processor frequency	100 MHz	170 MHz
Timer support	Motor-control timers	Motor-control timers
Position-sensor support	Incremental encoder	Incremental encoder and Hall sensors
Current-sensing topology	In-line phase-current shunts	Low-side shunts
Voltage sensing	Phase-voltage sensing	Phase-voltage sensing
Power-stage technology	GaN-based inverter	MOSFET-based inverter
Continuous phase current	$7\,A_{RMS}$	$1.5\,A_{RMS}$

Table 3.1 shows that, while both solutions generally offer the essential motor-control peripherals and sensing capabilities required for the implementation of the three control strategies, there are technical compromises in each platform. While the STM32G431RB-based platform provides a higher CPU frequency and additional position-sensor support, the TI C2000-based platform offers a significantly higher continuous phase-current capability, a high performance GaN-based power stage and the more-accurate in-line phase current sensing topology.

3.3. Comparative Control Studies

Comparative studies of PMSM control strategies typically assess the trade-offs among implementation complexity, computational demand, required feedback information, current waveform quality, torque ripple, acoustic behavior, dynamic performance, and embedded real-time requirements. However, these criteria are strongly influenced by the test setup, sensor configuration, PWM implementation, controller tuning, sampling conditions, and hardware platform. A meaningful quantitative comparison therefore requires a sufficiently transparent and consistent methodology that allows the effects of the control strategy to be distinguished from those of the experimental implementation.

Despite the limited availability of direct quantitative comparisons, the reviewed literature reveals consistent qualitative trends. These trends correspond to the theoretical characteristics described in section 2.4 and indicate a progression from relatively simple commutation methods toward more advanced current-control approaches. These general trade-offs are consistent with the strategy characteristics summarized in Table 2.2.

Establishing these qualitative tendencies is considerably easier than deriving a reliable quantitative ranking. Most comparative studies examine only two of the three strategies or investigate modified control approaches that extend the underlying methods and therefore lie outside the scope of the present comparison. Furthermore, the evaluated strategies are often implemented using different motors, hardware platforms, sensing configurations, or feedback conditions. Low-level implementation details are also frequently documented with insufficient precision. Consequently, reported performance differences cannot always be attributed unambiguously to the control strategies themselves.

Against this background, two publications are particularly relevant because they are among the reviewed sources that consider the three control strategies within a common comparative context. Neither source provides a fully controlled, like-for-like quantitative benchmark. Nevertheless, their broader coverage makes them suitable for examining both the commonly reported differences among the strategies and the methodological limitations of existing comparisons. The following assessment therefore focuses on the extent to which each publication supports a transparent and reproducible comparison.

Lee et al., A Comparison Study of the Commutation Methods for the Three-Phase Permanent Magnet Brushless DC Motor

Lee et al. [11] discuss all three control strategies in conjunction with motors having assumed trapezoidal and sinusoidal back-electromotive force waveforms. Although the study addresses the full set of strategies conceptually, its simulation comparison includes only selected combinations of motor type and control method. A trapezoidal back-electromotive force motor model is evaluated using six-step control, whereas a sinusoidal back-electromotive force motor is evaluated using trapezoidal and sinusoidal control. FOC is discussed conceptually and examined experimentally, but is not included in the presented simulation comparison.

Experimental validation is performed using a common test platform, but the publication

provides no comprehensive quantitative results [11]. The study therefore does not establish a complete like-for-like comparison in which all three strategies are applied to the same motor model under otherwise identical conditions. As a result, the influence of the control strategy cannot be clearly separated from that of the underlying motor type.

The experimental configurations also differ in their feedback arrangements. Hall sensors are used for trapezoidal control, whereas encoder-based feedback is employed for sinusoidal control and for some FOC configurations. Differences in current waveform quality and control behavior may therefore partly result from variations in sensor resolution and feedback processing rather than from the control strategy alone. In addition, the study primarily presents qualitative waveform observations and does not report comprehensive metrics such as torque-ripple percentages, harmonic analyses, or dynamic-performance measurements.

The architecture of the test platform, the detailed measurement procedure, and the structure of the simulation models are only partially documented. These limitations restrict both the reproducibility of the comparison and the ability to isolate individual influencing factors, including the motor model, sensor type, controller implementation, and measurement method. The study is therefore valuable for illustrating general differences among commutation methods, but provides only limited support for a controlled quantitative comparison.

Barrera, TI Precision Labs: Motor Drivers, Comparison of Commutation Methods

A broader application-oriented comparison is presented in the Texas Instruments Precision Labs material by Barrera [26]. The source discusses six-step, sinusoidal, and FOC from an implementation perspective and summarizes the commonly accepted practical trade-offs among the three methods. In contrast to the study by Lee et al., its primary purpose is educational rather than the presentation of a controlled experimental investigation.

This practical orientation makes the source useful for establishing qualitative expectations regarding implementation complexity, waveform quality, torque behavior, and required feedback. However, the material does not describe the methodology used to obtain the presented results. Relevant information concerning the motor, hardware platform, sensing configuration, controller tuning, operating conditions, and measurement procedure is not provided in sufficient detail. The reported observations therefore cannot be reproduced or

evaluated under comparable conditions.

Consequently, the source cannot support precise quantitative conclusions regarding controller performance, dynamic response, efficiency, robustness, or measurement accuracy. Its contribution to the present comparison is instead to provide an application-oriented reference against which the general qualitative tendencies identified in the academic literature can be assessed. Together, the two sources illustrate both the broadly accepted trade-offs among the three control strategies and the lack of a transparent, methodologically consistent benchmark that isolates the control strategy as the principal experimental variable.

3.4. Research Gap and Contribution

The reviewed literature establishes the general trade-offs among the three control strategies, it however, does not provide a controlled and reproducible quantitative comparison for the reasons stated in section 3.3.

Consequently, the influence of the control strategy cannot always be distinguished from that of the motor, sensing configuration, hardware platform, or measurement method. The resulting research gap is the absence of a transparent and methodologically consistent framework for evaluating six-step control, sinusoidal control, and FOC under comparable simulation and experimental conditions using clearly defined quantitative criteria.

Addressing this research gap requires an experimental platform that supports the consistent implementation and evaluation of all three control strategies. The corresponding hardware requirements were defined based on the research objectives established in section 1.1, the capabilities and limitations of the existing motor-control platforms examined in subsection 3.2.2, and the theoretical requirements for inverter actuation, current and voltage measurement, rotor-position feedback, and embedded controller execution discussed in chapter 2. In addition to the requirements of the present comparative study, the intended use of the platform in future investigations and for testing flight actuators was considered. The resulting system requirements are summarized in Table 3.2.

Table 3.2.: System requirements for the motor-control platform.

ID	Requirement	Description
SR-01	Processing platform and board architecture	The motor-control platform shall be implemented as a combination of a processing board and a motor control extension board.
SR-02	Three-phase current measurement	The platform shall support the simultaneous measurement of all three motor phase currents using in-line shunt resistors.
SR-03	Voltage measurement	The platform shall support the measurement of all three motor phase voltages and the applied DC-link voltage.
SR-04	Operating voltage and current	The platform shall support a DC-link voltage ranging from 5 V to at least 35 V and a continuous DC-link current of at least 15 A.
SR-05	Communication and expansion interfaces	The platform shall provide serial peripheral interface (SPI) and controller area network flexible data-rate (CAN FD) interfaces for communication with compatible peripheral devices and for future system expansion.
SR-06	Rotor-position measurement	The platform shall support rotor-position measurement using a magnetic rotary encoder connected through one of the available communication interfaces.
SR-07	Safe deactivation	The platform shall permit the inverter output stages to be deactivated upon detection of an invalid operating condition.
SR-08	PWM frequency	The platform shall support PWM operation over a range of at least 10 kHz to 30 kHz.

The requirements define the minimum functional and electrical capabilities needed to establish common experimental conditions for the three control strategies. Simultaneous in-line phase-current measurement and access to the phase and DC-link voltages provide a consistent measurement basis across the investigated methods. The rotor-position feedback, switching-frequency range, and safe deactivation capability support their embedded implementation and experimental operation. Furthermore, the communication interfaces and accessible measurement quantities facilitate future investigations of alternative sen-

sored and sensorless control methods, different feedback devices, and different motors.

Although the reviewed modular hardware solution provide several of the fundamental capabilities required for embedded motor control, and would allow for the basic comparison of the three strategies, neither investigated solution satisfies the complete set of requirements, focused on this and future studies, listed in Table 3.2.

Custom motor-control hardware is therefore developed according to the defined system requirements. The platform facilitates the controlled experimental implementation and evaluation of the three strategies under common hardware, sensing, and measurement conditions. At the same time, its measurement capabilities, communication interfaces, and support for different feedback and control methods provide a basis for future studies extending beyond the comparison conducted in this thesis.

This thesis addresses the identified research gap through the following contributions:

- **Consistent simulation framework:** Equation-based models of the three control strategies are developed using common motor parameters, supply conditions, mechanical dynamics, load profiles, and evaluation procedures.
- **Requirements-based custom control platform:** Custom embedded PMSM control hardware is developed and according to the defined system requirements. It enables the three control strategies to be implemented and evaluated using the same processing platform, inverter, sensing interfaces, rotor-position feedback, protection functions, communication interfaces, and data-acquisition infrastructure. The platform also facilitates future investigations of alternative control methods, feedback devices, and motors.
- **Systematic quantitative evaluation:** The simulated and experimental implementations are assessed using defined operating points, test procedures, and performance metrics. The evaluation considers speed response, transient behavior, phase-current quality, torque-related behavior, steady-state deviations, and computational demand.

By combining equation-based simulation, embedded implementation, and experimental validation within a common methodology, this thesis provides a transparent basis for assessing the effects of increasing control complexity. Common simulation conditions and a common experimental platform reduce the influence of differences in motor parameters,

sensing configurations, hardware conditions, and measurement procedures. The observed differences can therefore be related more directly to the sensing requirements, computational demand, implementation effort, and operational characteristics of the three control strategies.

4. Methodology

This chapter describes the methodological framework used to investigate the selected control strategies for a permanent magnet synchronous motor. The approach combines the definition of system requirements, model-based simulation, platform development, and experimental evaluation. The motor, inverter, and control structures are modelled to enable a systematic comparison of six-step control, sinusoidal control, and FOC. The evaluation focuses on relevant electrical and mechanical quantities, including current behavior, torque generation, and speed response.

4.1. Hardware Design Approach

The hardware is developed as a custom extension of the `NUCLEO-G431RB`, which provides the processing unit together with the associated programming and debugging interfaces [30]. Its use also enables integration with the STM32 development environment and the STM32 Motor Control Software Development Kit (MCSDK) [32], thereby reducing the effort required for firmware development and subsequent commissioning.

The development follows an iterative engineering process in which the design is divided into functional subsystems. Their electrical behavior, interfaces, and constraints are first considered individually and subsequently integrated into the complete platform. The process comprises the following principal stages:

- architectural decomposition and definition of subsystem interfaces;
- circuit design, component selection, and electrical dimensioning;
- circuit-board design and verification;
- technical review and identification of design deficiencies; and

- redesign based on the identified deficiencies and risks.

Each iteration is assessed against the hardware requirements defined in Table 3.2. The assessment considers electrical functionality, component ratings, measurement capability, power dissipation, manufacturability, mechanical integration, and microcontroller compatibility. Identified deficiencies and design risks are addressed in subsequent iterations.

Only the final design iteration is presented in chapter 5, as the preceding iterations represent intermediate development states rather than independent research results. The chapter therefore focuses on the architectural, circuit, component-selection, and layout decisions that determine the functionality and suitability of the platform. Minor implementation details that do not require an individual design rationale are documented in the complete schematics provided in Appendix D.

The microcontroller pin and peripheral assignments are defined using STM32CubeMX [31] and transferred to the schematic to maintain consistency between the electrical design and the firmware configuration. Components are selected and dimensioned according to the specified operating conditions, electrical ratings, interface compatibility, measurement ranges, expected power dissipation, availability, and suitability for circuit-board integration. Particular attention is given to the current-carrying components and analogue measurement circuits.

The circuit-board layout is developed by separately considering the high-current paths, high-frequency switching loops, analogue measurement signals, digital interfaces, low-voltage supply rails, thermal dissipation, and mechanical constraints. Component placement and signal routing are evaluated with regard to conductor dimensions, commutation-loop areas, coupling from switching nodes, measurement integrity, decoupling, return-current paths, and accessibility for commissioning and external measurements.

Design verification includes schematic electrical-rule checks, circuit-board design-rule checks, manual verification of component ratings and interfaces, consistency checks of the microcontroller pin assignment, and power-loss estimates for the principal current-carrying components. Before manufacturing, the final schematic and circuit-board layout are additionally reviewed by two electrical engineers. These activities assess the consistency, completeness, and manufacturability of the design but cannot establish the electrical functionality or performance of the assembled hardware.

Functional validation was intended to follow an incremental commissioning procedure, progressing from the low-voltage supply rails to the communication interfaces, analogue measurement channels, inverter-control signals, switching behavior, protection functions, and motor operation. As powered commissioning could not be completed within the available research period, the hardware assessment in section 5.3 is limited to the implemented design, engineering calculations, verification activities, manufacturing data, and visual inspection. Experimental verification of the electrical, thermal, measurement, protection, and motor-control performance therefore remains pending.

4.2. Modelling and Simulation Approach

Modelling and simulation are performed in MATLAB/Simulink R2025b using a custom equation-based framework under consistent and transparent conditions. Although Simscape Electrical is well suited to component-oriented and multidisciplinary simulations [37], the selected approach provides greater access to internal variables and direct control over the assumptions governing the PMSM, inverter, sensing system, and controllers.

A common inverter and motor model, parameter set, mechanical model, supply conditions, unloaded profile, controller tuning procedure and evaluation criteria are applied to all strategies.

4.2.1. Motor Modelling

To facilitate a fair comparison between the motor control strategies, where variations in motor behavior can be fully ascribed to the different motor control strategy, a common PMSM motor model is developed using the electrical and mechanical dynamics equations, reference frame transformations, torque relations and electrical to mechanical quantity relations in section 2.2 and the modelling assumptions defined in section 2.2.

Realistic motor parameterization is performed using analytical and experimental methods using an exemplary motor. This motor is selected such that it approximates the idealized modelling assumptions and is compatible with the boundary conditions defined in Table 3.2.

Table 4.1 lists the required motor parameters, extracted from the equations in section 2.2.

Table 4.1.: Required PMSM parameters for motor modelling.

Parameter	Symbol	Usage
Pole pairs	p	Electrical-to-mechanical speed and position conversion, torque equation
Phase resistance	R_s	Stator voltage equations, copper loss calculation
Permanent-magnet flux linkage	ψ_pm	Back-EMF model, torque equation
Stator inductance	L_s	Current dynamics in stator voltage equations
Rotor inertia	J	Mechanical motion equation, acceleration dynamics
Viscous friction coefficient	B	Mechanical motion equation, load-torque modelling

The PMSM is modeled to include phase voltage inputs and a configurable load torque. Outputs include physically measurable quantities such as, phase currents, mechanical rotary positions and speed, as well as internal signals for simpler analytical evaluation like direct- and quadrate-axis currents or electrical torque and power.

4.2.2. Inverter and Controller Modelling

In the simulation framework, the inverter is represented by an ideal voltage-averaging model that directly applies the commanded three-phase voltages to the motor model. Consequently, explicit PWM switching behavior, dead-time effects, semiconductor voltage drops, and switching and conduction losses are neglected. The investigation of these PWM-related effects is beyond the scope of this thesis and is therefore left for future work. The present comparison instead focuses on the influence of the control strategy on the fundamental motor behavior, excluding inverter-specific switching phenomena.

This common modelling basis prevents differences in switching implementation or semiconductor characteristics from influencing the results and therefore allows the observed

differences in motor behavior to be attributed primarily to the respective control strategy.

For the simplified averaged voltage model adopted in this thesis, the inactive, floating phase voltage of six-step control is approximated by the midpoint of the active line-to-line voltage,

$$u_{\text{float}} = \frac{\overline{u}_{pq}}{2}. \tag{4.1}$$

This approximation provides a common-mode-shifted set of phase voltages that preserves the intended line-to-line voltage vector without explicitly modelling the floating terminal dynamics.

For sinusoidal control and FOC, a common-mode voltage is injected into the commanded phase voltages to reproduce the phase-voltage references associated with SVPWM.

The motor controllers are modelled according to the descriptions in section 2.4.

Voltage, current and as such inverter and controller limitations are modeled to be able to depict signal saturation in compliance with the upper voltage and current limits given in requirement SR-04 (Table 3.2), namely a maximum continuous DC-current of $I_{\text{dc}} = 15\,\text{A}$ and a maximum DC-voltage of $U_{\text{dc}} = 35\,\text{V}$. The PI speed controllers shall provide windup protection as described in Equation 2.35, with a windup gain described in Equation 2.36.

4.2.3. Controller Tuning

The controller gains are tuned individually to achieve approximately identical nominal closed-loop speed dynamics across all implementations, using the Simulink *PID Tuner App* [36]. The tuning procedure imposes consistent requirements regarding bandwidth, stability margin, and step-response overshoot.

In FOC, the current-control loop forms the inner loop of the cascaded control structure. Its achievable bandwidth $f_{\text{BW,i}}$ is constrained by the current-sampling rate and control delay introduced by the PWM process. Based on the minimum supported PWM frequency

$f_{\mathrm{PWM,min}}$ specified by requirement SR-08 in Table 3.2, and a seperation factor of 10 from the separation criterion in section 2.5, the target current-loop bandwidth is defined as

$$f_{\mathrm{BW,i}} = 0.1 f_{\mathrm{PWM,min}} = 1\,\mathrm{kHz}. \tag{4.2}$$

Applying the the same criterion to this loop, yields the following target bandwidth for the outer speed-control loop:

$$f_{\mathrm{BW},\omega} = 0.1 f_{\mathrm{BW,i}} = 100\,\mathrm{Hz}. \tag{4.3}$$

This bandwidth, as well as a maximum speed-step overshoot of 10 % is used for all investigated speed controllers, including strategies without an explicit current-control loop, to maintain comparable speed dynamics and compatibility with the minimum switching-frequency requirement.

In addition, a phase margin of at least 60° is imposed to provide sufficient robustness against parameter deviations, model uncertainties, and unmodelled dynamics. This value is commonly regarded as a suitable compromise between response speed, overshoot, and robustness [14].

The tuning strategy objectives are summarized below

- Speed-loop bandwidth: $f_{\mathrm{BW},\omega} = 100\,\mathrm{Hz}$
- Phase margin: $\varphi_{\mathrm{m}} \geq 60°$
- Speed-step overshoot: $M_{\mathrm{p}} \leq 10\,\%$

Among all solutions satisfying these requirements, a phase margin close to 60° is preferred to avoid unnecessarily reducing the closed-loop response speed.

Tuning with the *PID Tuner App* is based on linearizations [36], in this study, obtained at zero-speed and zero-load operating conditions. For the speed controllers, the operating point is defined by

$$\omega_{m,\mathrm{rpm}} = 0\,\mathrm{rpm}, \qquad T_L = 0,$$

while the FOC current controllers are tuned at

$$i_d = 0\,\mathrm{A}, \qquad i_q = 0\,\mathrm{A}, \qquad \omega_m = 0, \qquad T_L = 0.$$

These operating points correspond to start-up, stopping, and low-speed operation. The resulting gains are optimized only with respect to the local linearized dynamics at the selected operating points. They are therefore not expected to provide optimal closed-loop performance throughout the entire investigated operating range, particularly at higher speeds or under substantially different load conditions. This limitation is accepted consistently for all control strategies rather than compensated for through operating-point-dependent or strategy-specific retuning.

The controller gains are determined individually for each control strategy such that the tuning objectives are satisfied.

4.2.4. Simulation

Prior to conducting the comparative simulation studies, the motor and controller models are validated. The validation is performed by comparing the simulated phase-voltage and phase-current waveforms at arbitrarily selected high controller setpoints with the theoretical waveforms expected for each control strategy, as described in section 2.4. Furthermore, for FOC, the direct- and quadrature-axis currents are expected to converge against the current setpoints. Agreement between the simulated and theoretical waveform characteristics serves as evidence that the implemented models adequately represent the intended control behavior.

The simulations are performed in using the fixed-step `ode4` (Runge–Kutta) solver with a step size of $1 \times 10^{-6}\,\mathrm{s}$. The resulting signal data are logged and evaluated using supplementary MATLAB scripts.

All simulations are conducted under unloaded operating conditions, defined by an external

load torque of $T_{\mathrm{L}} = 0\,\mathrm{N\,m}$, excluding the influence of load-dependent motor behavior. The behavior under loaded operating conditions is outside the scope of the present investigation.

Open-Loop Simulation

The open-loop simulation evaluates the intrinsic steady-state performance of the investigated control strategies. Common current-vector magnitude operating points are used as the basis for the comparison because the primary setpoint of each strategy directly influences the stator current-vector magnitude and, consequently, the electromagnetic torque and speed production.

Using the current-vector magnitude as the comparison basis permits the control strategies to be evaluated at comparable electrical loading conditions. It also reduces the influence of the mechanical dynamics on the definition of the operating points, which would be more pronounced if the strategies were compared at their independently achieved mechanical speeds.

The open-loop test cases summarized in Table 4.2 evaluate the investigated control strategies during steady-state operation with respect to phase-current waveform quality, electromagnetic-torque production and ripple, mechanical-speed variation, electrical input-power demand, and current utilization.

The operating condition and performance indicators associated with each test case are specified in Table 4.2.

Operating-point definition. The low-, medium-, and high-current operating points are defined as approximately 15 %, 67 %, and 100 % of the maximum dc-link current reference of 15 A specified in SR-04 (Table 3.2), corresponding to target current-vector magnitudes of 2.25 A, 10 A, and 15 A, respectively. The dc-link current reference serves only as a numerical basis for defining standardized operating levels and does not imply equality between the dc-link current and the stator current-vector magnitude.

For FOC, the target current-vector magnitude is imposed directly through the current references. For six-step and sinusoidal control, the controller setpoints are selected through

Table 4.2.: Open-loop test cases and associated performance indicators used to compare the investigated control strategies

ID	Name	Test Condition	Performance Indicators
OL1, OL2, OL3	Low-, Medium-, and High-Current Operation	Steady-state performance at low, medium, and high current-vector magnitude operating points	Phase-current total harmonic distortion THD_i; mean electromagnetic torque $\overline{T_\mathrm{e}}$; normalized torque ripple τ_T; mean mechanical angular velocity $\overline{\omega_\mathrm{m}}$; relative mechanical-speed standard deviation $\sigma_{\omega,\mathrm{rel}}$; mean electrical input power $\overline{P_\mathrm{el}}$; mean current-vector utilization u_i; and mean torque per ampere TPA.

a parameter sweep, such that the resulting steady-state mean current-vector magnitude $\bar{i}_\mathrm{mag}$ minimizes the deviation from the target value.

Steady-state detection and evaluation interval. For each open-loop test case, a constant control setpoint is applied until periodic steady state is reached. Periodic steady state is assumed when the relative change in the cycle-averaged mechanical speed between consecutive mechanical periods remains below 2 % for at least three consecutive electrical periods. After steady state has been established, all performance indicators listed in Table 4.2 are evaluated over an interval spanning one complete mechanical period. This interval contains an integer number of electrical periods and captures the electromagnetic quantities over all rotor positions.

Steady-state-indicators. For any sampled quantity x, the arithmetic mean over the evaluation interval containing N_s samples is defined as

$$\overline{x} = \frac{1}{N_\mathrm{s}} \sum_{n=1}^{N_\mathrm{s}} x_n, \tag{4.4}$$

where x_n denotes the n-th sampled value. This definition is used to calculate the mean electromagnetic torque $\overline{T}_\mathrm{e}$, mean mechanical angular velocity $\overline{\omega}_\mathrm{m}$, mean electrical input

power $\overline{P}_{\mathrm{el}}$, mean mechanical output power $\overline{P}_{\mathrm{m}}$, mean current-vector magnitude $\bar{i}_{\mathrm{mag}}$, and mean quadrature-axis current $\bar{i}_q$.

The phase-current total harmonic distortion THD_i quantifies the harmonic content of the stator phase current relative to its fundamental component and is defined as

$$\mathrm{THD}_i = \frac{\sqrt{\sum_{k=2}^{N_{\mathrm{h}}} I_k^2}}{I_1} \cdot 100\,\%, \tag{4.5}$$

where I_1 and I_k denote the magnitudes of the fundamental and k-th harmonic components, respectively. Harmonics up to $N_{\mathrm{h}} = 10$ are considered. For the motor model described in section 2.2, which assumes an ideal sinusoidal back-EMF, THD_i characterizes the deviation of the phase-current waveform from the ideal sinusoidal waveform associated with smooth torque production. The harmonic components are obtained from the discrete Fourier transform of the phase-a current over the evaluation interval. Since the balanced three-phase motor model is symmetric, the phase-a current is representative of the three stator phase currents.

The normalized torque ripple τ_{T} quantifies the peak-to-peak torque variation relative to the mean electromagnetic torque $\overline{T}_{\mathrm{e}}$.

$$\tau_{\mathrm{T}} = \frac{T_{\mathrm{e,max}} - T_{\mathrm{e,min}}}{\left|\overline{T}_{\mathrm{e}}\right|} \cdot 100\,\%, \tag{4.6}$$

where $T_{\mathrm{e,max}}$ and $T_{\mathrm{e,min}}$ denote the maximum and minimum instantaneous electromagnetic torque within the evaluation interval. Normalization facilitates comparison between operating points and control strategies with different average torque levels.

Speed variation is quantified by the relative mechanical speed standard deviation

$$\sigma_{\omega,\mathrm{rel}} = \frac{\sqrt{\frac{1}{N_{\mathrm{s}}} \sum_{n=1}^{N_{\mathrm{s}}} \left(\omega_{\mathrm{m},n} - \overline{\omega}_{\mathrm{m}}\right)^2}}{\left|\overline{\omega}_{\mathrm{m}}\right|} \cdot 100\,\%, \tag{4.7}$$

where $\omega_{\mathrm{m},n}$ denotes the n-th sampled mechanical angular velocity. This indicator quantifies the extent to which torque pulsations affect the mechanical response.

Electrical input power $\overline{P}_{\mathrm{el}}$ is calculated according to Equation 4.4. Because the investigated control strategies may achieve different mechanical speeds and output powers at the same current-vector magnitude, $\overline{P}_{\mathrm{el}}$ is used solely to compare the electrical power processed by the drive and is not interpreted as an efficiency metric.

For the considered non-salient PMSM, the mean current-vector utilization and the mean torque per ampere are defined as

$$u_i = \frac{\bar{i}_q}{\bar{i}_{\mathrm{mag}}} \cdot 100\,\%, \tag{4.8}$$

$$\mathrm{TPA} = \frac{\overline{T}_{\mathrm{e}}}{\bar{i}_{\mathrm{mag}}}, \tag{4.9}$$

respectively, where $u_i = 100\,\%$ corresponds to complete alignment of the mean stator current vector with the positive q-axis.

Closed-Loop Simulation

The closed-loop simulation evaluates the performance of the complete drive system under speed-controlled operating conditions.

Using common mechanical-speed references permits the control strategies to be evaluated under identical commanded operating conditions. In contrast to the open-loop comparison, however, the closed-loop response depends not only on the intrinsic characteristics of each control strategy, but also on its interaction with the speed controller and the electrical and mechanical dynamics of the drive.

The closed-loop test cases summarized in Table 4.3 evaluate the investigated control strategies with respect to steady-state speed-tracking accuracy, mechanical-speed variation, electrical input-power demand, drive efficiency, transient reference-tracking performance, and maximum-speed operation. The operating condition and performance indicators associated with each test case are specified in Table 4.3.

Table 4.3.: Closed-loop test cases and associated performance indicators used to compare the investigated control strategies

ID	Name	Test Condition	Performance Indicators
CL1, CL2	Low- and High-Speed Reference Tracking	Steady-state performance at low- and high-speed reference operating points	Relative speed error $e_{\omega,\mathrm{rel}}$; relative speed standard deviation $\sigma_{\omega,\mathrm{rel}}$; mean electrical input power $\overline{P_{\mathrm{el}}}$; and mean drive efficiency $\overline{\eta}$.
CL3, CL4	Low- and High-Speed Reference Steps	Mechanical-speed reference steps from standstill to the low- and high-speed operating points	Rise time t_{r}; settling time t_{s}; and relative speed overshoot M_{p}.
CL5	Maximum-Speed Operation	Mechanical-speed reference ramp until a limiting condition is reached	Achieved mechanical angular velocity $\omega_{\mathrm{m,max}}$; and limiting quantity x_{lim}.
CL6	Low-to-High-Speed Ramp	Linearly increasing 300 ms reference-speed ramp from the low- to the high-speed operating point	Rise time t_{r}; settling time t_{s}; relative speed overshoot M_{p}; and average relative error during rise $\overline{e}_{\mathrm{rel,\,rise}}$.

Operating-point definition. The low- and high-speed operating points are defined as 8 % and 72 %, respectively, of the smallest approximate angular velocity achieved by any of the investigated control strategies in test case OL3 (Table 4.2).

In test case CL6, the mechanical-speed reference is increased linearly from the low- to the high-speed operating point over 300 ms.

Steady-state detection and evaluation interval. For the closed-loop steady-state test cases CL1 and CL2, a constant mechanical-speed reference is applied until periodic steady state is reached. Periodic steady state is assumed when the relative change in the cycle-averaged mechanical speed between consecutive mechanical periods remains below 2 % for at least three consecutive mechanical periods.

After steady state has been established, the performance indicators associated with CL1 and CL2 in Table 4.3 are evaluated over an interval spanning three complete mechanical periods.

Steady-state indicators. The mean steady-state indicators are calculated according to Equation 4.4. The mean steady-state drive efficiency $\overline{\eta}$ is calculated using the instantaneous efficiency values within the closed-loop steady-state evaluation interval.

$$\eta = \frac{P_{\mathrm{m}}}{P_{\mathrm{el}}}. \tag{4.10}$$

The relative steady-state speed error quantifies the deviation of the achieved mean mechanical angular velocity from the applied reference and is defined as

$$e_{\omega,\mathrm{rel}} = \frac{|\omega_{\mathrm{m}}^{*} - \overline{\omega_{\mathrm{m}}}|}{|\omega_{\mathrm{m}}^{*}|} \cdot 100\,\%, \tag{4.11}$$

where ω_{m}^{*} denotes the mechanical-speed reference. The relative mechanical-speed standard deviation $\sigma_{\omega,\mathrm{rel}}$ is calculated according to Equation 4.7.

Reference-tracking indicators. Test cases CL3, CL4, and CL6 evaluate the reference-tracking performance. The reference-tracking indicators are evaluated after a step or during a ramp in the mechanical-speed reference. Prior to applying the reference change, the system is allowed to reach steady state according to the preceding definition.

Let $\omega_{\mathrm{m},0}$ and ω_{m}^{*} denote the mechanical angular velocity immediately before the reference change and the final mechanical-speed reference, respectively. The commanded speed change is

$$\Delta\omega_{\mathrm{m}}^{*} = \omega_{\mathrm{m}}^{*} - \omega_{\mathrm{m},0}. \tag{4.12}$$

The rise time t_{r}, settling time t_{s}, and relative speed overshoot M_{p} are defined as

$$t_{\mathrm{r}} = t_{\omega,90\%} - t_{\omega,10\%}, \tag{4.13}$$

$$t_{\mathrm{s}} = \inf\left\{t - t_0 \,\middle|\, |\omega_{\mathrm{m}}(\tau) - \omega_{\mathrm{m}}^*| \le 0.02\,|\Delta\omega_{\mathrm{m}}^*|\,,\ \forall\tau \in [t, t_{\mathrm{end}}]\right\}, \tag{4.14}$$

$$M_{\mathrm{p}} = \max\left\{0,\ \frac{\max\limits_{t\in[t_0,t_{\mathrm{end}}]} |\omega_{\mathrm{m}}(t) - \omega_{\mathrm{m},0}| - |\Delta\omega_{\mathrm{m}}^*|}{|\Delta\omega_{\mathrm{m}}^*|}\right\} \cdot 100\,\%, \tag{4.15}$$

where $t_{\omega,10\%}$ and $t_{\omega,90\%}$ are the first instants at which the response reaches 10 % and 90 %, respectively, of the commanded speed change, and t_{end} denotes the end of the considered simulation interval. The settling time uses a tolerance band of 2 % of the commanded speed change.

For test case CL6, the average relative error during rise quantifies the mean absolute deviation of the mechanical angular velocity from the time-varying mechanical-speed reference during the 300 ms reference ramp. It is defined as

$$\overline{e}_{\mathrm{rel,\,rise}} = \frac{1}{N_{\mathrm{r}}} \sum_{n=1}^{N_{\mathrm{r}}} \frac{\left|\omega_{\mathrm{m},n}^* - \omega_{\mathrm{m},n}\right|}{|\Delta\omega_{\mathrm{m}}^*|} \cdot 100\,\%, \tag{4.16}$$

where N_{r} is the number of samples within the reference-ramp interval, and $\omega_{\mathrm{m},n}^*$ and $\omega_{\mathrm{m},n}$ denote the sampled mechanical-speed reference and mechanical angular velocity, respectively. Normalization by the commanded speed change permits comparison of the tracking error between control strategies relative to the magnitude of the imposed reference change.

Maximum-speed indicators. For test case CL5, the mechanical-speed reference is increased until a specified current or voltage limit is reached. The maximum achieved mechanical angular velocity $\omega_{\mathrm{m,max}}$ and the corresponding limiting quantity x_{lim} are recorded for all three control strategies.

The mechanical-speed reference is increased using a ramp with a slope of

$$\frac{\mathrm{d}\omega_{\mathrm{m}}^*}{\mathrm{d}t} = 100\,\mathrm{rad/s^2}.$$

4.3. Experimental Evaluation Approach

Due to timeline constraints and supplier delays, the custom motor-control extension board specified by the hardware design approach in section 4.1 could not be commissioned and tested within the available research period. Therefore, the experimental evaluation is performed using the STMicroelectronics `NUCLEO-G431RB` development board [30] in combination with the `X-NUCLEO-IHM16M1` motor-driver expansion board [28]. This substitute platform was selected because it already utilizes the target processing board defined by section 4.1.

The experimental evaluation aims to assess whether the principal differences identified between the investigated control strategies in the closed loop simulation are also observable on a physical motor-drive system. The experimental campaign is limited to six-step control and FOC, as no qualified implementation of sinusoidal control is available for the selected hardware platform within the MCSDK [32]. Consequently, the complete comparison of six-step, sinusoidal, and FOC is conducted at the simulation level, whereas the experimental evaluation provides a hardware-based comparison of six-step control and FOC.

To ensure a consistent evaluation basis, both experimentally investigated control strategies are tested using the same motor, supply conditions, and measurement procedure, as well as common configuration parameters, including the PWM frequency and current limits are used. Strategy-dependent parameters are configured separately only where required by the respective control algorithm.

4.3.1. Experimental Setup

The experimental setup comprises the `NUCLEO-G431RB` processing board, the `X-NUCLEO-IHM16M1` motor-driver expansion board, the exemplary PMSM characterized according to Table 4.1, a regulated DC power supply, and the measurement equipment required to acquire the investigated quantities.

The motor is operated without an externally applied mechanical load, consistent with the unloaded conditions of the simulation test cases. The mechanical arrangement and experimental configuration are kept constant when changing between six-step control and FOC. Before each measurement, the motor and controller parameters applicable to the

respective control strategy are loaded into the MCSDK configuration and verified.

The angular-velocity setpoints are specified using the STMicroelectronics *Motor Pilot* tool [29], while the controller signals are logged using STM32CubeMonitor [19]. The sound pressure level is measured with a microphone positioned at a fixed distance from the motor. The hardware, software, operating conditions, and data-acquisition parameters are summarized in Table 4.4.

Table 4.4.: Experimental setup and data-acquisition parameters.

Quantity	**Equipment or configuration**	**Value**
Processing platform	`NUCLEO-G431RB`	-
Motor-driver platform	`X-NUCLEO-IHM16M1`	-
Control software	STM32 MCSDK	Version 6.4.2 [32]
DC-link voltage	Regulated DC power supply	12V
Current limit	Regulated DC power supply	2A
PWM frequency	Motor-control firmware configuration	30 kHz
Speed-data acquisition	Cube Monitor	Version 1.12.0 [19]; 1000 Hz
CPU-load acquisition	Cube Monitor	Version 1.12.0; 1000 Hz
Sound pressure measurement	Generic smartphone microphone	5 cm; 100 Hz

Figure 4.1 shows a diagram of the experimental setup, excluding the sound pressure measurement microphone.

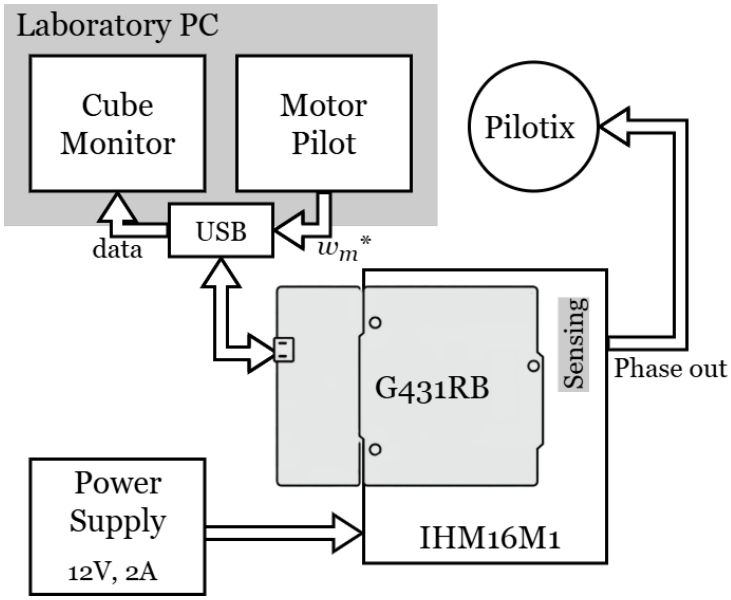


Figure 4.1.: Diagram of the experimental setup used for the experimental evaluation.

The `IHM16M1` control board and as such the motor, are supplied by a standard 12 V, 2 A power supply. As the control board implements a sensorless design, no direct feedback from the Pilotix motor is needed. The sensing circuit is used by the Nucleo `G431RB` processing board to estimate velocity and position. The evaluation data is logged using CubeMonitor using the ST-Link USB connection, while the Motor Pilot tool sends the commanded speed references.

4.3.2. Controller Tuning

For the experimental evaluation, the controllers are configured using the automatic tuning functionality provided by the STMicroelectronics MCSDK Kit. In contrast to the model-based tuning procedure used in the simulation, the experimental controller parameters are therefore directly generated based on the specified motor and control-system parameters.

As in the simulation methodology, the bandwidth of the inner FOC current-control loops is set to

$$f_{\mathrm{BW,i}} = 1\,\mathrm{kHz}.$$

This maintains the previously defined separation between the current-loop dynamics and the outer speed-loop dynamics and provides a consistent basis for comparing the simulated and experimentally implemented FOC systems.

The speed-control loops of both FOC and six-step control are executed with an execution frequency of

$$f_{\text{exec},\omega} = 1\,\text{kHz}.$$

As a practical design guideline, the achievable closed-loop bandwidth is commonly selected approximately one order of magnitude below the controller execution frequency [14]. Applying the same bandwidth-separation ratio used in the simulation gives

$$f_{\text{BW},\omega} \approx 0.1 f_{\text{exec},\omega} = \frac{0.1}{T_{\text{exec},\omega}} = 100\,\text{Hz}. \tag{4.17}$$

The selected execution period therefore places the expected speed-loop bandwidth in the vicinity of the 100 Hz target used for the simulated speed controllers. This provides approximately comparable nominal speed dynamics between the simulation and experimental implementations. However, The exact closed-loop bandwidth, phase margin, and overshoot are not imposed independently in the same manner as in the simulation.

The operating point is again set to

$$\omega_m = 0\,\text{rad/s}.$$

The load or dq-axis current could not be defined like in the simulation.

4.3.3. Experimental Procedure

The experimental procedure is designed to compare the steady-state and transient speed-control behavior of six-step control and FOC, as it was done for the closed-loop simulation test cases.

The experimental test cases are summarized in Table 4.5. Test cases EX1 and EX2 evaluate steady-state operation at the low- and high-speed operating points, and are equivalent to the simulation test cases CL1 and CL2. Test case EX3 evaluates tracking behavior during a low-to-high speed-reference transition and is equivalent to closed-loop simulation test case CL6.

Table 4.5.: Experimental test cases and associated performance indicators used to compare the investigated control strategies.

ID	Name	Test Condition	Performance Indicators
EX1, EX2	Low- and High-Speed Reference Tracking	Constant steady-state low- and high-speed reference speed operation	Relative steady-state speed error $e_{\omega,\mathrm{rel}}$; relative mechanical-speed standard deviation $\sigma_{\omega,\mathrm{rel}}$; average steady-state CPU-load $\overline{\rho}_{\mathrm{CPU}}$; average steady-state sound pressure level $\overline{L}_{\mathrm{p}}$
EX3	Low-to-High Speed Ramp	Linearly increasing 300 ms reference speed from low to high speed levels	rise time t_{r}; settling time t_{s}; relative speed overshoot M_{p}; average relative error during rise $\overline{e}_{\mathrm{rel,\,rise}}$

The low- and high-speed references used in EX1–EX3 correspond to the operating points defined for the closed-loop simulation in subsubsection 4.2.4. The same reference values and the same 300 ms low-to-high-speed ramp are retained to facilitate the result comparison.

For the shared performance indicators, the definitions of the relative steady-state speed error $e_{\omega,\mathrm{rel}}$, relative mechanical-speed standard deviation $\sigma_{\omega,\mathrm{rel}}$, rise time t_{r}, settling time t_{s}, relative speed overshoot M_{p}, and average relative error during rise $\overline{e}_{\mathrm{rel,\,rise}}$ are adopted from subsubsection 4.2.4. The same steady-state detection criterion and evaluation intervals are applied to the experimentally acquired speed signals.

In addition to the speed-related indicators, the experimental evaluation considers the processor load and sound pressure level. The instantaneous processor load is defined as

$$\rho_{\mathrm{CPU},n} = \frac{t_{\mathrm{exec},n}}{T_{\mathrm{ctrl}}} \cdot 100\,\%, \tag{4.18}$$

where $t_{\mathrm{exec},n}$ is the measured execution time of the periodically executed motor-control task and T_{ctrl} is its invocation period. The mean processor load $\overline{\rho}_{\mathrm{CPU}}$ is calculated over the steady-state evaluation interval according to Equation 4.4. Only the execution times of periodically executed motor-control functions are included. Initialization, data transmission, debugging, and non-periodic supervisory functions are excluded. Compiler optimization is disabled.

The mean measured sound pressure level is calculated over the same steady-state evaluation interval accroding to Equation 4.4. Since the smartphone microphone is not calibrated for absolute sound pressure measurements, the recorded values are used only as a relative acoustic indicator for comparing six-step control and FOC. They are not interpreted as absolute acoustic-emission levels.

5. Hardware Design

This chapter presents the final hardware implementation of the PMSM control platform. It describes the system architecture, circuit implementation, circuit-board design, manufactured hardware, and verification activities. The chapter focuses on the principal design decisions that determine the functionality and suitability of the platform. Minor implementation details are documented in the complete schematics provided in Appendix D. Compliance with the hardware requirements defined in Table 3.2 is assessed at the end of the chapter.

The extension and sensor boards were developed, manufactured, and assembled based on the hardware design approach described in section 4.1. However, powered commissioning and motor operation could not be completed within the available research period. The assessment presented in this chapter is therefore based on the implemented circuit design, component ratings, engineering calculations, design reviews, manufacturing data, and visual inspection. Experimental verification of the electrical, thermal, measurement, protection, and motor-control performance remains pending.

5.1. Hardware Implementation

5.1.1. System Architecture and Interfaces

The implemented system consists of the `NUCLEO-G431RB`, a custom extension board, and a separate rotor-position sensor board. The principal subsystems and their interconnections are shown in Figure 5.1, while the complete schematics are provided in Appendix D.

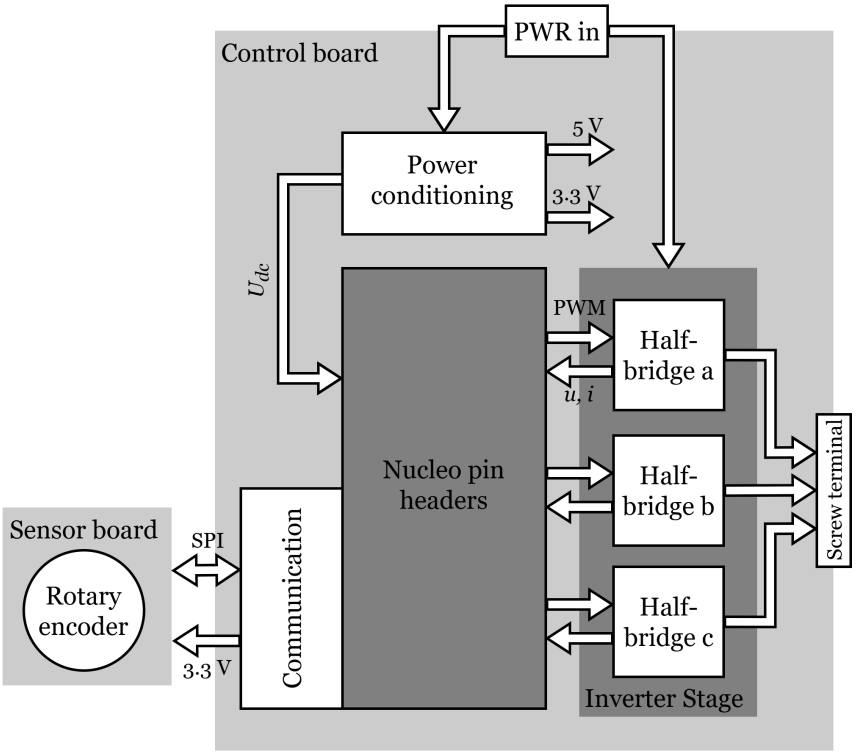


Figure 5.1.: Principal subsystems and interconnections of the implemented motor-control hardware.

The extension board is mechanically and electrically connected to the `NUCLEO-G431RB` through the Nucleo morpho headers. It integrates the DC-link input and conditioning stage, three inverter half-bridges, current- and voltage-measurement circuits, protection interfaces, auxiliary power supplies, and communication interfaces. The motor phases are connected through a dedicated high-current screw terminal. The rotor-position sensor is mounted on the motor and connected to the extension board through a shielded cable.

The microcontroller generates six complementary inverter-control signals and acquires the three phase currents, three switching-node potentials, DC-link voltage, and rotor position. The corresponding pin and peripheral assignments are shown in Figure D.7. The inverter-control signals are assigned to motor-control timer outputs, the electrical measurements to analogue to digital converter (ADC)-capable inputs, and the communication signals to the corresponding digital peripherals.

The available measurement and actuation interfaces support all three control strategies investigated in this thesis without modifications to the principal power-stage hardware. The phase-voltage measurements and expansion interfaces additionally provide a basis for

subsequent investigations of voltage reconstruction, sensorless estimation, and alternative control algorithms.

5.1.2. Power Supply and Three-Phase Inverter

The external DC-link supply is connected through 4 mm laboratory connectors. It directly supplies the three-phase inverter and feeds the converters used to generate the 5 V and 3.3 V auxiliary rails.

Reverse-polarity protection is implemented using an `IPB020N10N5` power MOSFET controlled as an ideal diode with the `LTC4359` ideal diode controller [8, 2]. Compared with a series diode, this arrangement reduces the forward-voltage drop and the associated conduction losses. The MOSFET voltage rating exceeds the specified maximum DC-link voltage of 35 V, while its current rating and thermal connection are selected for the required input-current range.

An `L6982C50DR` synchronous buck converter generates the 5 V rail from the DC-link supply [18]. Its input-voltage range covers the specified operating range of the platform. The 5 V rail supplies the corresponding auxiliary circuits and serves as the input to an `AP2114D` linear regulator, which generates the 3.3 V rail for the digital, measurement, and sensor circuits [4]. The expected loads and the power dissipation of the linear regulator were considered during component selection and layout design.

The inverter consists of three identical half-bridges implemented using `LMG2100R044RARR` integrated GaN power stages [22]. Each device combines a high-side transistor, a low-side transistor, and the associated gate-drive circuitry. This integration reduces the gate-loop length and the number of external gate-drive components. Each half-bridge receives complementary high-side and low-side control signals from a motor-control timer of the `STM32G431RB`. The three resulting switching nodes form the motor-phase outputs.

A hardware-enable connection allows the microcontroller to control the operating state of the power stages. In addition, the timer break input provides a direct path for deactivating the PWM outputs in response to an overcurrent condition. The response and reliability of the complete shutdown path remain to be verified during powered commissioning.

5.1.3. Measurement and Rotor-Position Interfaces

The phase currents are measured using in-line shunt resistors located between the inverter switching nodes and the motor connector. This position permits current measurement independently of which transistor is conducting and supports the simultaneous acquisition of all three phase currents. The resulting shunt voltages are amplified using bidirectional `INA240A1PW` current-sense amplifiers [21]. Their outputs are referenced to $u_{\mathrm{ref}} = 1.65\,\mathrm{V}$, allowing positive and negative phase currents to be represented within the unipolar ADC input range.

The shunt resistance is selected as a compromise between measurement sensitivity and power dissipation. The power loss of the shunt in phase x is given by

$$P_{\mathrm{sh},x} = I_{x,\mathrm{RMS}}^2 R_{\mathrm{sh}}. \tag{5.1}$$

For the conservative current estimate $I_{x,\mathrm{RMS}} = 15\,\mathrm{A}$ and the selected resistance $R_{\mathrm{sh}} = 3.3\,\mathrm{m\Omega}$, the expected dissipation is

$$P_{\mathrm{sh},x} = (15\,\mathrm{A})^2 \cdot 3.3\,\mathrm{m\Omega} = 0.742\,\mathrm{W}. \tag{5.2}$$

A shunt with a rated power dissipation of 2 W is therefore used to provide margin for overload conditions, temperature rise, and the thermal resistance of the PCB assembly.

The three switching-node potentials and the DC-link voltage are measured through resistive dividers connected to ADC-capable microcontroller pins. The switching-node potentials are measured relative to the DC-link ground and therefore include the common-mode voltage introduced by the inverter switching states. The measurement ranges are scaled to remain within the admissible microcontroller input range over the specified DC-link voltage range.

The rotor-position sensor board uses an `AS5047P-ATSM` magnetic rotary encoder to measure the angular position of a diametrically magnetized shaft-mounted magnet [1]. The encoder is supplied from the 3.3 V rail and communicates with the processing board through SPI.

The connection includes MOSI, MISO, serial-clock, and chip-select signals. Series resistors of 33 Ω, placed close to the respective signal sources, are used to limit edge-related ringing.

A shielded RJ-45 cable carries the SPI signals, supply voltage, ground, and auxiliary signals between the sensor and extension boards. This arrangement permits the sensor to be mounted directly on the motor while maintaining a defined electrical connection to the processing hardware. The shielding and grounding concept is intended to reduce interference coupled from the inverter switching nodes and motor conductors. Additional SPI, I^2C, CAN FD, and auxiliary connections provide interfaces for communication and future platform extensions.

5.2. Circuit-Board Implementation

The circuit-board design translates the electrical architecture into a mechanically integrated implementation while accounting for current capacity, switching-loop inductance, measurement integrity, heat dissipation, and compatibility with the `NUCLEO-G431RB`. The resulting extension-board layout is shown in Figure D.8.

5.2.1. Extension-Board Layout

The board geometry is primarily determined by the Nucleo morpho headers, the external power and motor connectors, and the three repeated inverter phases. The `NUCLEO-G431RB` is mounted above the extension board while preserving access to its programming and debugging interfaces. The complete arrangement is illustrated by the three-dimensional model in Figure 5.2.

The DC-link input, bulk capacitance, and inverter stages are grouped within the power-stage region. Local DC-link capacitors are positioned directly adjacent to the supply terminals of the corresponding GaN power stages to reduce the area and inductance of the high-frequency commutation loops. The three half-bridges use a repeated layout to obtain comparable conductor lengths, thermal conditions, and measurement paths. The motor connector is positioned close to the phase outputs.

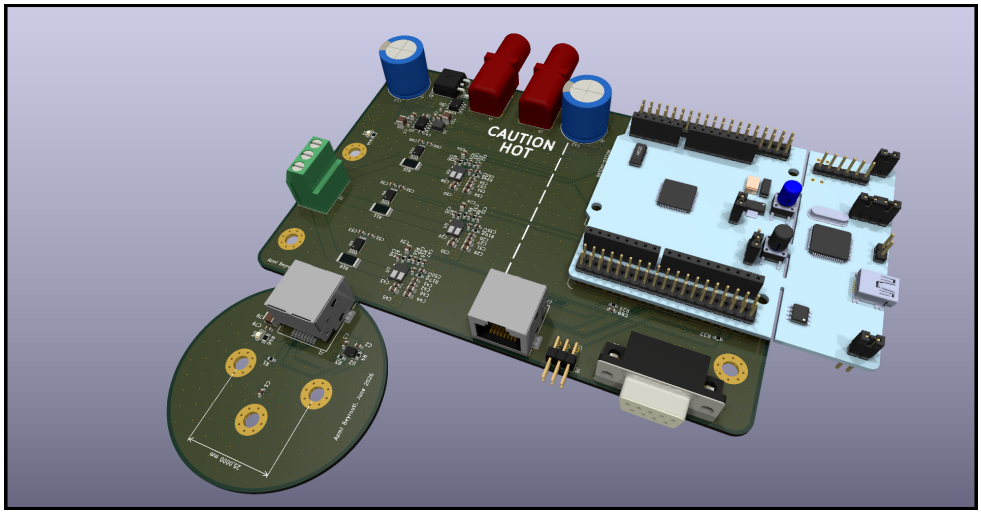


Figure 5.2.: Three-dimensional model of the extension board with the `NUCLEO-G431RB`.

The DC-link and phase-current paths are implemented using enlarged copper regions. Thermal copper areas and interlayer vias distribute the losses generated by the power stages, reverse-polarity protection, voltage regulators, and shunt resistors. The resulting temperature rise additionally depends on the assembled board, airflow, switching conditions, and motor operating point. The thermal performance must therefore be verified experimentally under representative operating conditions.

The switching-node copper areas are restricted to the dimensions required for component connection and current conduction. This limits capacitive coupling from the high-slew-rate nodes to adjacent measurement and communication circuits. The auxiliary power supplies and communication circuits are separated from the inverter region where permitted by the board geometry. Sensitive traces are also prevented from running parallel to the switching nodes over extended distances.

The phase-current shunts are located directly within the phase-output paths. Separate Kelvin connections route the shunt voltages to the current-sense amplifiers, thereby limiting measurement errors caused by voltage drops in the load-current conductors. The amplifiers and associated input-filter components are positioned near the shunts to minimize the measurement-loop area. The analogue measurement traces are routed separately from the PWM signals and high-current conductors, with their return paths arranged to limit voltage errors caused by inverter currents.

5.2.2. Sensor Board and Manufactured Hardware

The sensor-board geometry is determined by the motor mounting interface and the required position of the magnetic encoder relative to the shaft-mounted magnet. The encoder is located on the rotational axis, while the cable connector is positioned away from the rotating components. The mounting holes establish a repeatable mechanical reference relative to the motor housing.

The sensor supply is locally decoupled, and the SPI traces are kept short with continuous ground return paths where permitted by the board geometry. Additional connections allow future sensor extensions without changing the basic encoder interface. Reliable position measurement nevertheless depends on concentric alignment with the shaft-mounted magnet, permissible axial spacing, electrical offset calibration, and robustness against inverter-generated interference. These properties remain to be established during commissioning.

Following completion of the schematic and layout reviews, both boards were manufactured and assembled. Figure 5.3 shows the predominantly assembled extension board.

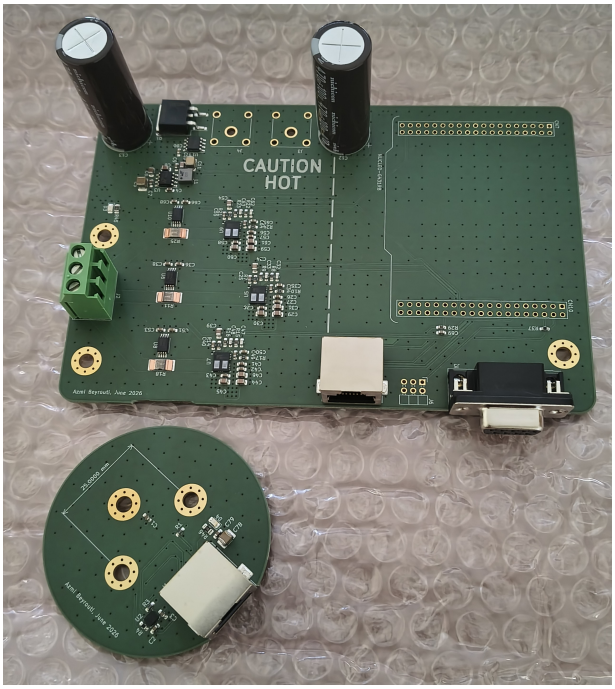


Figure 5.3.: Predominantly assembled motor-control extension board.

Visual inspection was performed to identify incorrect component orientations, incomplete solder joints, unintended solder bridges, and mechanical incompatibilities. No visually

apparent assembly defects or mechanical incompatibilities were identified during this inspection. However, visual inspection does not establish the electrical functionality or performance of the hardware.

5.3. Design Verification and Requirement Assessment

The completed schematic and PCB layout passed the applicable electrical-rule and design-rule checks. The automated checks covered schematic connectivity, footprint assignment, conductor clearances, and compliance with the selected manufacturing constraints.

Manual verification addressed the component voltage and current ratings, auxiliary-supply operating ranges, shunt dissipation, measurement scaling, digital logic levels, microcontroller pin assignments, mechanical interfaces, and separation of power, analogue, and communication paths. Power-loss estimates for the inverter modules, phase-current shunts, reverse-polarity protection MOSFET, and linear regulator informed component selection and copper allocation. The final schematic and PCB layout were additionally reviewed by two electrical engineers before manufacturing.

These activities establish the consistency, completeness, and manufacturability of the design to the extent possible without powered operation. They do not verify the electrical functionality or performance of the assembled boards. Powered commissioning must therefore assess the auxiliary-supply regulation, DC-link current capability, inverter switching behavior, measurement accuracy and noise, ADC synchronization, rotor-position acquisition, communication interfaces, shutdown behavior, and component temperatures during continuous operation.

Table 5.1 summarizes the implementation status of the hardware requirements defined in Table 3.2. The table identifies the corresponding implementation evidence and distinguishes between implementation in the design and experimental validation.

Table 5.1.: Implementation status of the specified hardware requirements.

ID	Implementation evidence	Assessment
SR-01	Nucleo morpho-header connection and verified mechanical board integration.	Implemented in the design.
SR-02	Three in-line shunts, bidirectional current-sense amplifiers, and dedicated ADC connections.	Implemented; calibration and accuracy validation pending.
SR-03	Four resistive voltage-measurement channels with analogue input filtering.	Implemented; measurement validation pending.
SR-04	Component ratings and design calculations cover the specified DC-link voltage and current range.	Fulfilled by design; continuous-operation validation pending.
SR-05	SPI, I^2C, CAN FD, and auxiliary connections are included.	Implemented; communication testing pending.
SR-06	Separate sensor board with magnetic encoder, mechanical mounting, and shielded connection.	Implemented; alignment and calibration pending.
SR-07	Hardware-enable and timer break-input connections are included in the protection path.	Implemented; fault-response testing pending.
SR-08	Three integrated GaN half-bridges are connected to complementary motor-control timer outputs.	Implemented; switching and thermal validation pending.

All specified hardware functions are represented in the schematic and PCB design. The requirements are therefore fulfilled at the implementation level to the extent that can be demonstrated without powered operation. Requirements involving electrical performance, measurement quality, communication reliability, thermal behavior, protection response, and motor operation cannot yet be considered experimentally validated. The manufactured hardware consequently provides the physical basis for the intended control platform, while its functional and performance validation remains subject to subsequent commissioning.

6. Modelling and Simulation

This chapter presents the development and application of the MATLAB/Simulink models used to investigate the selected PMSM control strategies. It describes the motor and controller models, the simulation environment, and the controller-tuning process. The resulting models are subsequently evaluated under the in section 4.2 defined open- and closed-loop test cases, and their suitability for transfer to the embedded control platform is assessed.

6.1. Motor Model Development

The *Pilotix 6210 210KV* [16] was selected as the exemplary motor because its approximately symmetric three-phase windings and surface-mounted permanent magnets support the balanced and approximately non-salient PMSM assumptions adopted in section 2.2. Furthermore, its operating range includes reduced-load operating points compatible with the 15 A current capability specified by SR-04 and the 10–30 kHz PWM-frequency range specified by SR-08. The motor can therefore be operated within the defined system requirements while providing a suitable basis for the development and evaluation of the motor model [16].

The common motor model implemented in Simulink is based on the electrical dynamics of the PMSM given in Equation 2.8 and Equation 2.9, the electromagnetic torque equation given in Equation 2.11, and the mechanical dynamics given in Equation 2.12. The underlying motor equations are formulated in the dq reference frame, as described in section 2.2.

The model interface operates in the abc frame to represent the physically accessible phase quantities. Internally, the phase voltages are transformed into the dq reference frame using

the Clarke and Park transformations defined in Equation 2.3 and Equation 2.5, respectively. After evaluating the electrical and mechanical dynamics, the calculated currents are transformed back into the *abc* frame using the inverse Park and Clarke transformations defined in Equation 2.6 and Equation 2.4, respectively.

In addition to the input and output quantities required for interaction with the surrounding simulation environment, selected internal quantities are routed to the model interface for evaluation. The complete set of motor model signals is listed in Table 6.1.

Table 6.1.: Input and output signals of the motor model.

Signal name	**Label**	**Type**	**Direction**
Phase voltages	$\boldsymbol{u}_x$	Physical interface quantity	Input
Load torque	T_L	Physical interface quantity	Input
Phase currents	$\boldsymbol{i}_x$	Physical interface quantity	Output
Mechanical angular velocity	ω_m	Physical interface quantity	Output
Electrical rotor angle	θ_e	Internal quantity provided for evaluation	Output
Electrical angular velocity	ω_e	Internal quantity provided for evaluation	Output
Electromagnetic torque	T_e	Internal quantity provided for evaluation	Output
Direct-axis current	i_d	Internal quantity provided for evaluation	Output
Quadrature-axis current	i_q	Internal quantity provided for evaluation	Output
Electrical input power	P_el	Internal quantity provided for evaluation	Output
Mechanical output power	P_m	Internal quantity provided for evaluation	Output

The internal calculation follows the signal flow of the motor model. First, the derivatives of the direct- and quadrature-axis currents are calculated from rearranged forms of the

electrical dynamics equations:

$$\frac{\mathrm{d}i_\mathrm{d}}{\mathrm{d}t} = \frac{1}{L_\mathrm{s}} \left(u_\mathrm{d} - R_\mathrm{s} i_\mathrm{d} + \omega_\mathrm{e} L_\mathrm{s} i_\mathrm{q} \right), \tag{6.1}$$

$$\frac{\mathrm{d}i_\mathrm{q}}{\mathrm{d}t} = \frac{1}{L_\mathrm{s}} \left[u_\mathrm{q} - R_\mathrm{s} i_\mathrm{q} - \omega_\mathrm{e} \left(L_\mathrm{s} i_\mathrm{d} + \psi_\mathrm{PM} \right) \right]. \tag{6.2}$$

Integrating Equation 6.1 and Equation 6.2 yields the currents i_d and i_q. The resulting dq-axis currents are subsequently transformed into the phase currents $\boldsymbol{i}_x$, where $x \in \{\mathrm{a}, \mathrm{b}, \mathrm{c}\}$.

The electromagnetic torque T_e is calculated from i_q according to Equation 2.11. The mechanical angular acceleration is then determined from the rearranged mechanical dynamics equation

$$\frac{\mathrm{d}\omega_\mathrm{m}}{\mathrm{d}t} = \frac{T_\mathrm{e} - T_\mathrm{L} - B\omega_\mathrm{m}}{J}. \tag{6.3}$$

Integrating Equation 6.3 yields the mechanical angular velocity ω_m. The corresponding electrical angular velocity ω_e is calculated using Equation 2.2 and subsequently integrated to obtain the electrical rotor angle θ_e. The relationship between the mechanical and electrical rotor angles is given in Equation 2.1.

Finally, the electrical input power P_el and mechanical output power P_m are calculated according to Equation 2.13 and Equation 2.14, respectively. The complete Simulinkmotor model and its subsystems are provided in Appendix C.

An accurate simulation model of the *Pilotix 6210 210KV* requires the identification of the motor parameters needed in the motor model equations. Appendix B shows the complete analytical and experimental parameterization approach performed for the exemplary Pilotix motor. The results of this parameterization, and in turn the adopted motor parameters for the model can be found in Table 6.2

Table 6.2.: Motor parameters adopted for the simulation model.

Parameter	**Value**
R_s	$21.25\,\mathrm{m\Omega}$
L_s	$17.475\,\mu\mathrm{H}$
ψ_PM	$2.165\,\mathrm{mWb}$
J	$1.29 \times 10^{-4}\,\mathrm{kg\,m^2}$
B	$8.61444 \times 10^{-4}\,\mathrm{N\,m\,s/rad}$

6.2. Controller Models

The controller models described in this section use the directly measurable and externally routed internal PMSM signals listed in Table 6.1. Directly measurable signals can be acquired using physical sensors, whereas internal signals generally have to be calculated or estimated from the measured signals and the motor parameters. In the simulation models, these internal signals are routed directly from the PMSM model to the controllers to reduce computational effort and visual complexity. Consequently, the direct availability of these signals is a simulation-specific simplification that must be considered when transferring the controller models to a practical implementation.

Depending on the control strategy and control stage, a operation limit of $I_{\mathrm{dc}} = 15\,\mathrm{A}$ or $U_{\mathrm{dc}} = 35\,\mathrm{V}$ is imposed.

6.2.1. Six-Step Control Model

The six-step control model uses the duty cycle d as its control quantity and is organized into three subsystems: the *Sensor*, *Commutation Logic*, and *Trapezoidal gen* subsystems. The overall model is shown in Figure 6.1, while the internal structures of the subsystems are provided in Appendix C.

The electrical rotor angle θ_e, provided as an output of the PMSM model, is supplied to the *Sensor* subsystem. The subsystem determines the corresponding commutation sector according to the mapping defined in Table 2.1. Accordingly, the model represents a six-step control scheme in which the mechanical rotor angle θ_m, obtained from a continuous position measurement, is first converted into the electrical rotor angle θ_e using the mechanical-to-electrical relationship given in Equation 2.1. The electrical angle is subsequently discretized into sector information. This approach differs from the use of Hall-effect sensors, which provide discrete rotor-position information corresponding to the commutation sectors directly.

The sector information is supplied to the *Commutation Logic* subsystem. Based on the current sector, the subsystem selects the corresponding switching state defined in Table 2.1 and outputs it as a six-element switching vector.

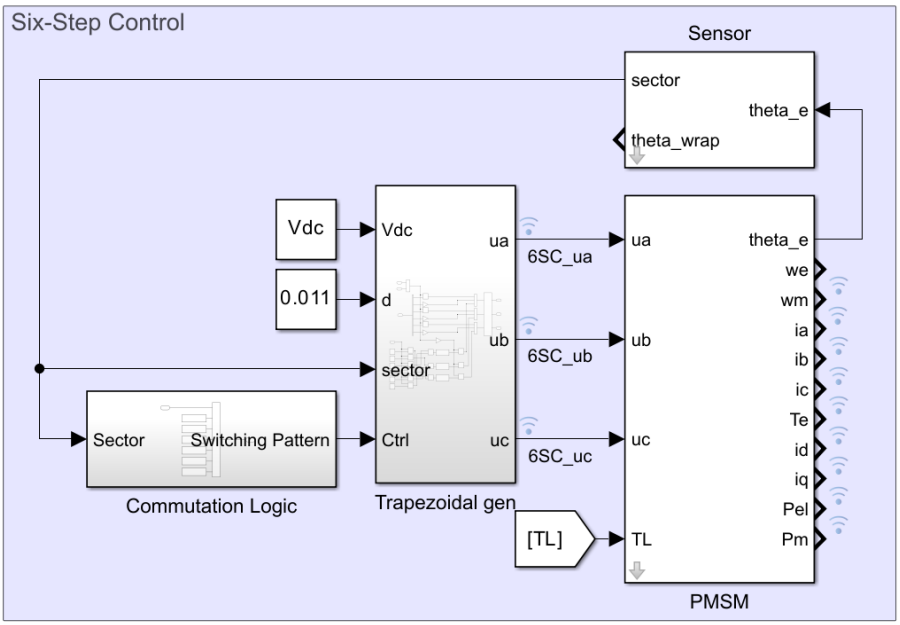


Figure 6.1.: Simulink model of an open-loop six-step-controlled PMSM.

Although the *Trapezoidal gen* subsystem is not strictly part of the controller, it generates the trapezoidal phase voltages and represents the voltage-averaged inverter model. The switch-node voltage of the high-side-driven phase is determined by the dc-link voltage U_{dc} and the duty cycle d. Therefore, $U_{\mathrm{dc}} = 35\,\mathrm{V}$ defines the upper limit of the corresponding switch-node voltage. The switch-node voltage of the low-side-driven phase is set to zero, while the switch-node voltage of the floating phase is set to half the voltage of the high-side-driven phase, as defined in the floating phase voltage approximation in Equation 4.1. The sector information is supplied to the *Trapezoidal gen* subsystem to identify the phase that remains floating in each commutation sector.

Because these voltages are defined relative to the negative dc-link terminal, the neutral-point voltage is calculated using the neutral point voltage equation in Equation 2.16. It is then subtracted from the switch-node voltages according to the neutral point relation equation in Equation 2.17. As such, the resulting phase-to-neutral voltages satisfy the voltage-balance condition given in Equation 2.18.

6.2.2. Sinusoidal Control Model

The sinusoidal control model uses the sinusoidal voltage-reference amplitude U^* as its control quantity and is organized into two subsystems: the *Sinusoidal voltage reference*

generator and the *Common Mode Injection* subsystems. The model is based on the control structure shown in Figure 2.6 and is presented in Figure 6.2. The internal structures of the subsystems are provided in Appendix C.

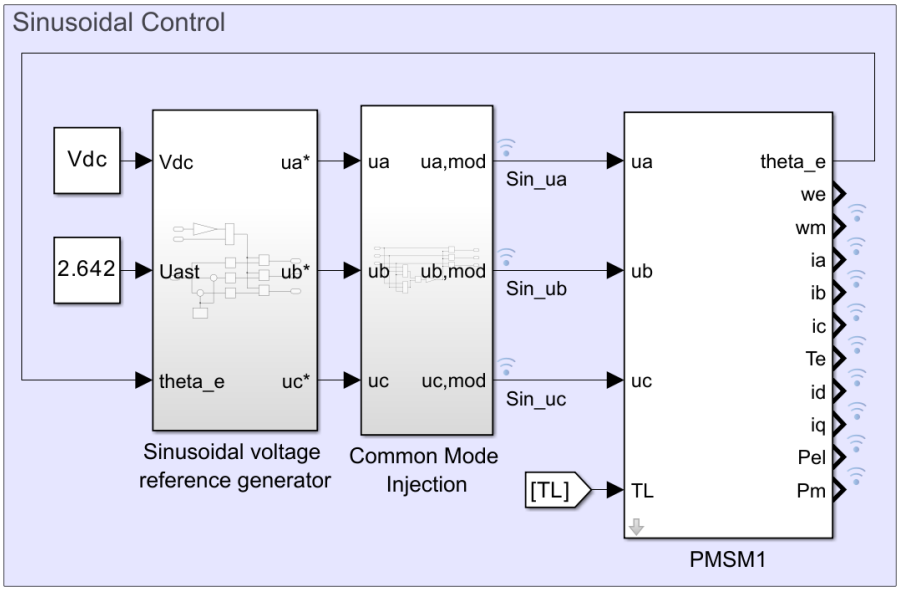


Figure 6.2.: Simulink model of an open-loop sinusoidally controlled PMSM.

The electrical rotor angle θ_e is supplied to the *Sinusoidal voltage reference generator*, which calculates the three reference phase voltages according to equations 2.20, 2.21, and 2.22. The maximum admissible voltage-reference amplitude is determined by the available dc-link voltage of $U_{\mathrm{dc}} = 35\,\mathrm{V}$ and the maximum reference-voltage magnitude for SVPWM defined in Equation 2.29.

The *Common Mode Injection* subsystem represents the modulation effect of SVPWM through the injection of a common-mode voltage. It implements Equation 2.26 and Equation 2.27 by adding a common-mode offset to the three reference phase voltages. The resulting modulated phase voltages are subsequently supplied to the motor model.

6.2.3. Field-Oriented Control Model

The FOC model uses the quadrature-axis current reference i_q^* as its control quantity. It comprises the *Decoupling Network*, and *Common Mode Injection* subsystems, two PI controllers, and the required reference-frame transformations. The model is based on

the control structure shown in Figure 2.9 and is presented in Figure 6.3. The PMSM subsystem is omitted from the figure to improve readability. The internal structures of the subsystems are provided in Appendix C.

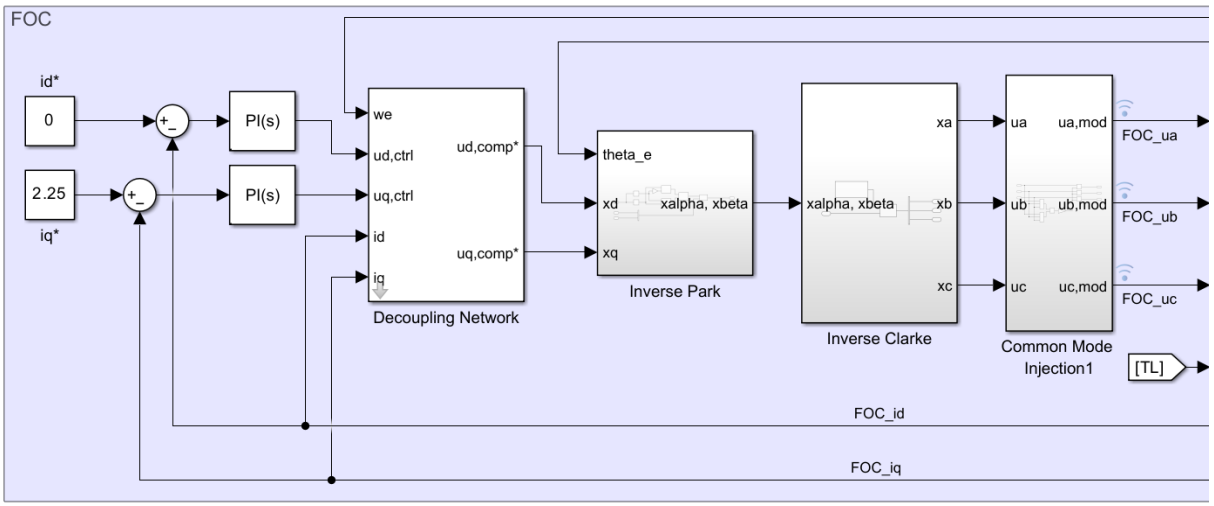


Figure 6.3.: Simulink model of open-loop FOC.

The measured direct- and quadrature-axis currents i_d and i_q are subtracted from their respective reference values i_d^* and i_q^* to obtain the current-control errors. These errors are supplied to two dedicated PI controllers. The simulation model directly uses the externally routed i_d and i_q signals from the PMSM model. In a practical implementation, the measured phase currents must instead be transformed from the *abc* reference frame into the *dq* reference frame using the Clarke and Park transformations.

The PI controllers generate the direct- and quadrature-axis voltage commands u_d and u_q. These voltage commands contain the cross-coupling components of the *dq*-axis motor equations. The *Decoupling Network* compensates for these components by implementing the feed-forward and decoupling equations 2.31 and 2.33. The compensation terms are calculated using the electrical angular velocity ω_e and the *dq*-axis currents from the motor.

The the decoupled voltage references are transformed into the *abc* reference frame using the inverse Park and Clarke transformations. The resulting three-phase voltage references are subsequently supplied to the same *Common Mode Injection* subsystem used in the sinusoidal control model. The modulated phase voltages are then applied to the motor model.

Current saturation for the quadrature-axis controller is not imposed at this stage, as limiting current at this stage would impair the comparability between the control strategies, as the other control strategies do not impose a current limit for the low-level motor control itself.

6.2.4. Speed Control

The speed-control architecture used for the closed-loop investigations employs a standard PI controller. Its implementation for six-step control is shown in Figure 6.4.

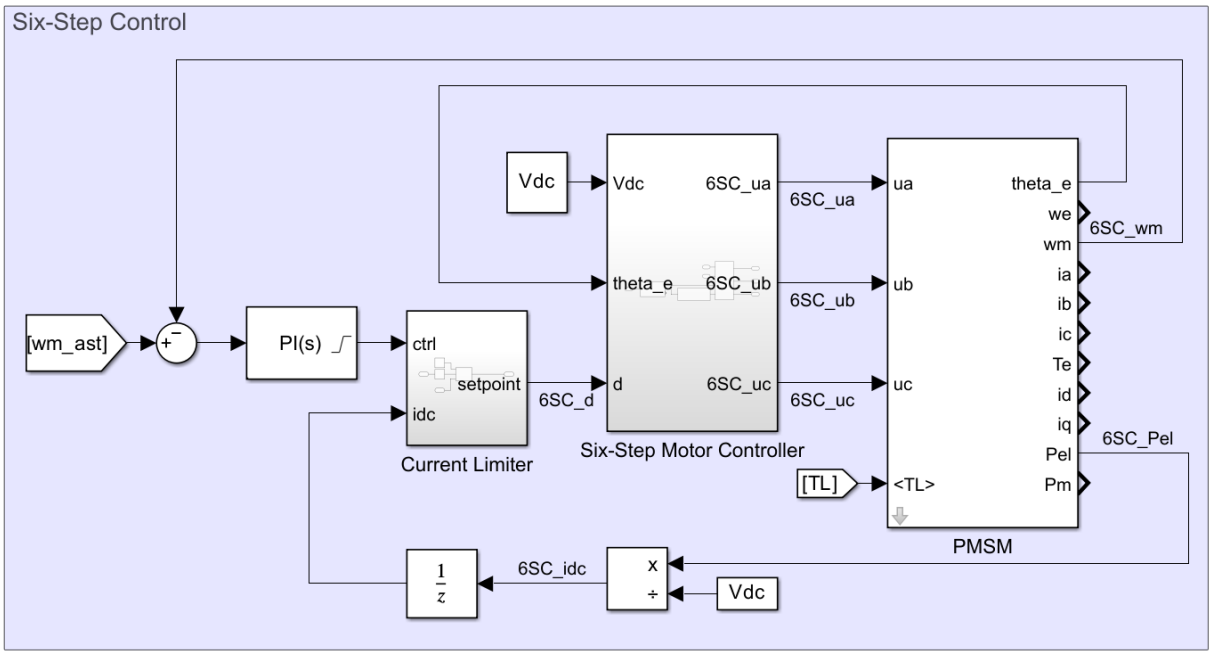


Figure 6.4.: PMSM model with six-step control and closed-loop speed control.

The speed-control error is calculated as the difference between the mechanical angular velocity reference ω^* and the measured mechanical angular velocity. The PI controller converts this error into the control quantity required by the respective motor controller. For the six-step control model shown in Figure 6.4, this control quantity is the duty cycle d. The controller output is constrained to its admissible range, and an anti-windup mechanism is included to prevent excessive accumulation of the integral term during output saturation.

The *Current Limiter* subsystem implements on-off current control with hysteresis, limiting

peak DC-current [12]. This modeling reflects a possible implementation of current limiting in the *Nucleo G431RB*, where PWM gets turned off once a defined current limit is reached [33]. DC-current is calculated using Equation 2.15.

6.3. Simulative Controller Configuration

Following the tuning methodology described previously, the controller gains are determined individually using the Simulink *Simulink PID Tuner App* and plant linearizations obtained at the defined zero-speed and zero-load operating points. As specified in the tuning objectives, a bandwidth of 100 Hz is applied to all speed controllers, while the inner FOC current controllers are tuned to 1 kHz. Furthermore, the controllers are required to achieve a phase margin of at least 60° and a maximum overshoot of 10 %. The resulting PI controller gains and achieved closed-loop dynamics are summarized in Table 6.3.

Table 6.3.: System dynamics and controller gains.

PI-controller	f_{BW}	φ_m	M_p	K_p	K_i
FOC i_d current controller	$1000\,Hz$	75°	9.75 %	0.1006	307.5
FOC i_q current controller	$1000\,Hz$	70°	9.07 %	0.0927	354.1
Six-step control speed controller	$100\,Hz$	60°	7.58 %	0.001356	0.8919
Sinusoidal control speed controller	$100\,Hz$	60°	7.58 %	0.02739	18.02
FOC speed controller	$100\,Hz$	80°	9.25 %	1.842	181.6

6.4. Observations and Results

The implemented motor and controller models are validated as described in subsection 4.2.4. Figure 6.5 presents the simulated phase-voltage and phase-current waveforms obtained at $d = 1$ for six-step and $U^* = 35\,\mathrm{V}$ for sinusoidal control.

The waveforms in Figure 6.5, as well as the FOC waveforms at a setpoint of $i_q^* = 15\,\mathrm{A}$ are consistent with the theoretical characteristics described in section 2.4. Under six-

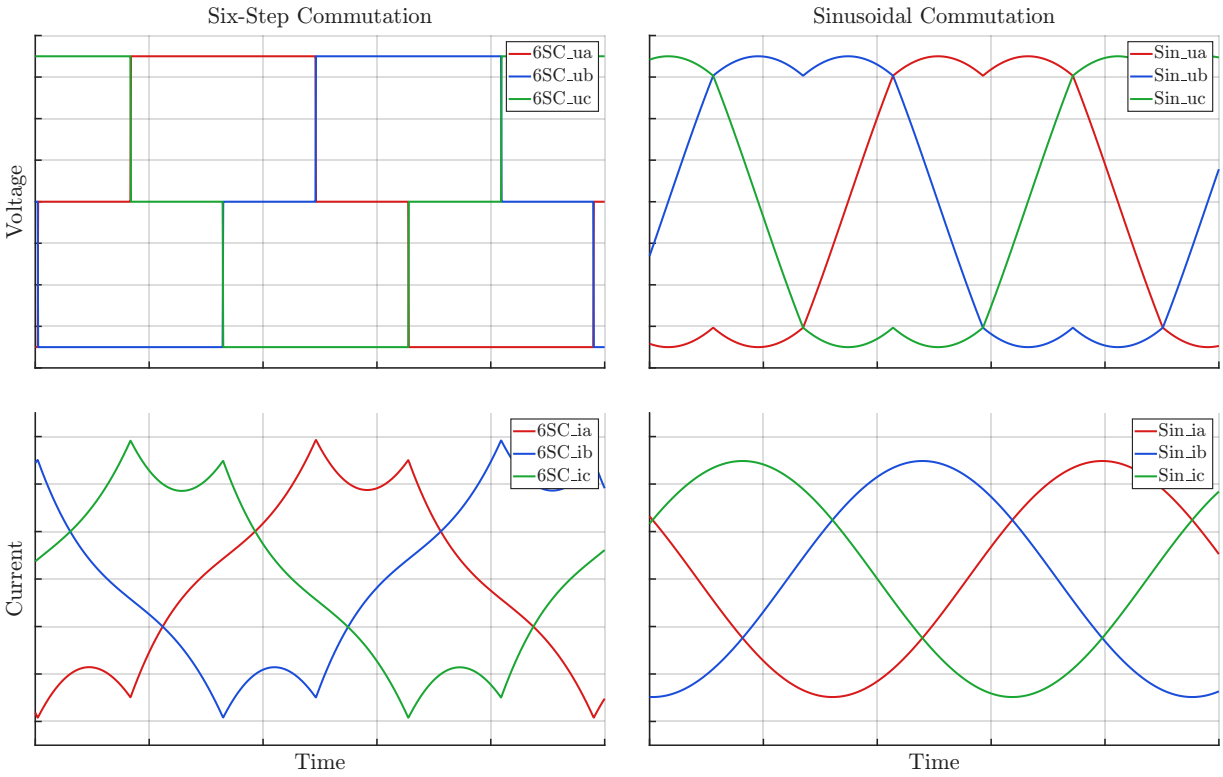


Figure 6.5.: Phase-voltage and phase-current waveforms obtained using six-step and sinusoidal control at selected high controller setpoints.

step control, the phase voltages exhibit the expected block-shaped commutation pattern. Within each commutation sector, one phase is connected to the positive DC-link potential, another phase is connected to the negative DC-link potential, and the remaining phase is left floating at a potential close to the active phase midpoint. By contrast, sinusoidal control and FOC produce saddle-shaped phase-voltage waveforms as a result of the injected common-mode voltage.

The phase currents generated by six-step control are approximately trapezoidal, whereas sinusoidal control and FOC produce approximately sinusoidal phase currents. For FOC, i_d and i_q converge to the specified setpoints. This qualitative agreement indicates that the implemented models reproduce the expected operating characteristics and are therefore considered adequate for the subsequent comparative analyses.

6.4.1. Open-Loop Tests

The setpoint mapping established in subsection 4.2.4 correspond to the control-strategy-specific setpoints, d for six-step control, U^* for sinusoidal control and i_d^* for FOC listed in Table 6.4.

Table 6.4.: Current setpoints and corresponding control references.

Level	$I/I_{\max}$	I	d	U^*	i_q^*
Low	15.0 %	2.25 A	1.1 %	2.642 V	2.25 A
Medium	66.7 %	10.0 A	34.1 %	6.776 V	10.0 A
High	100.0 %	15.0 A	44.8 %	8.728 V	15.0 A

OL1 – Low-Current Operation

During OL1 low-current steady-state operation, six-step control produces the distorted phase-current waveform shown in Figure 6.6. Instead of approaching the trapezoidal profile typically associated with six-step operation, the phase currents exhibit ramp-shaped conduction intervals and sharp transitions at the commutation instants. By contrast, sinusoidal control and FOC generate approximately sinusoidal phase currents under the same operating condition.

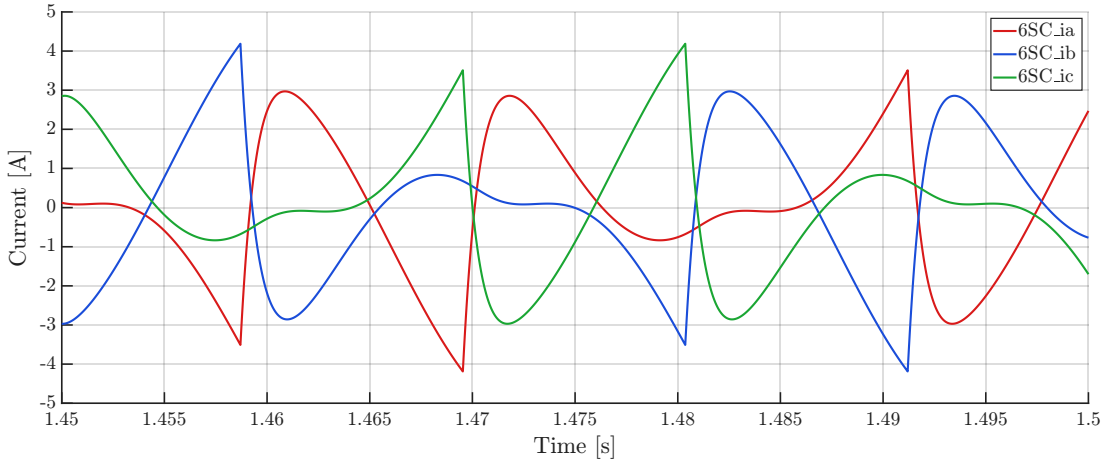


Figure 6.6.: Phase-current waveforms obtained using six-step control under low-current operating conditions.

The ramp-shaped current waveform is likely associated with the low effective voltage applied under the OL1 operating condition. Although the low mechanical speed provides a

comparatively long commutation interval, the six-step duty-cycle command of only 1.1 % produces limited voltage excitation. Consequently, the phase currents remain transient within each commutation sector instead of approaching the relatively constant conduction levels. The commutation sequence and switching instants remain consistent with the intended control principle, thereby enabling rotation. However, the resulting phase-current waveform is discontinuous and deviates substantially from an idealized six-step waveform.

The corresponding mechanical angular velocities are shown in Figure 6.7. Six-step control produces pronounced periodic speed variations and the lowest mean mechanical angular velocity. Sinusoidal control reaches a higher and approximately constant steady-state velocity, while FOC achieves the highest mean velocity.

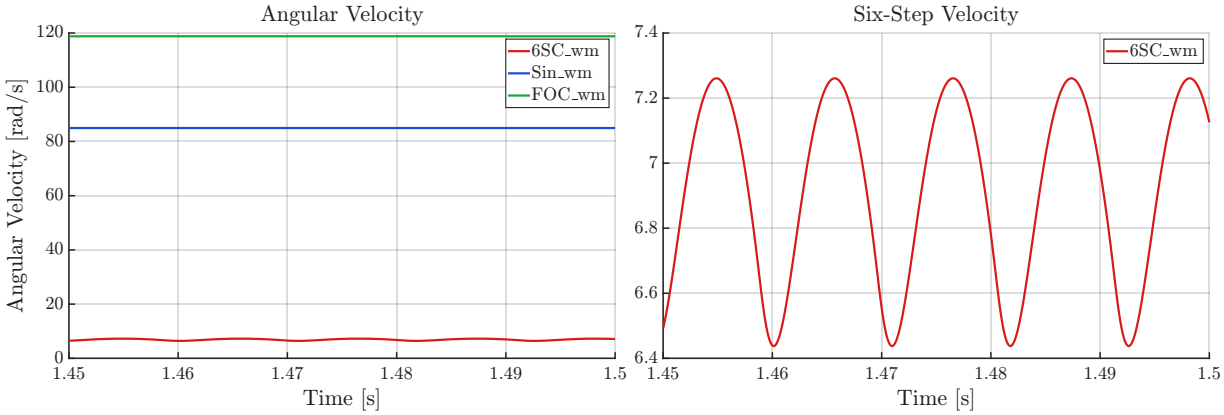


Figure 6.7.: Mechanical angular velocities obtained under low-current operating conditions.

The electromagnetic torque and electrical input power shown in Figure 6.8 exhibit corresponding steady-state behavior. The distorted and poorly utilized phase currents under six-step control result in substantial torque and power pulsations, whereas sinusoidal control and FOC provide considerably smoother responses.

During six-step commutation, the instantaneous electrical input power briefly becomes negative. According to the adopted sign convention, these intervals indicate a temporary return of stored magnetic energy from the motor phases to the DC link while the current is redistributed between the outgoing and incoming phases. Since the mean electrical input power remains positive, the negative instantaneous values represent short-duration energy exchange rather than sustained regenerative operation.

The steady-state performance indicators are summarized in Table 6.5. At the common

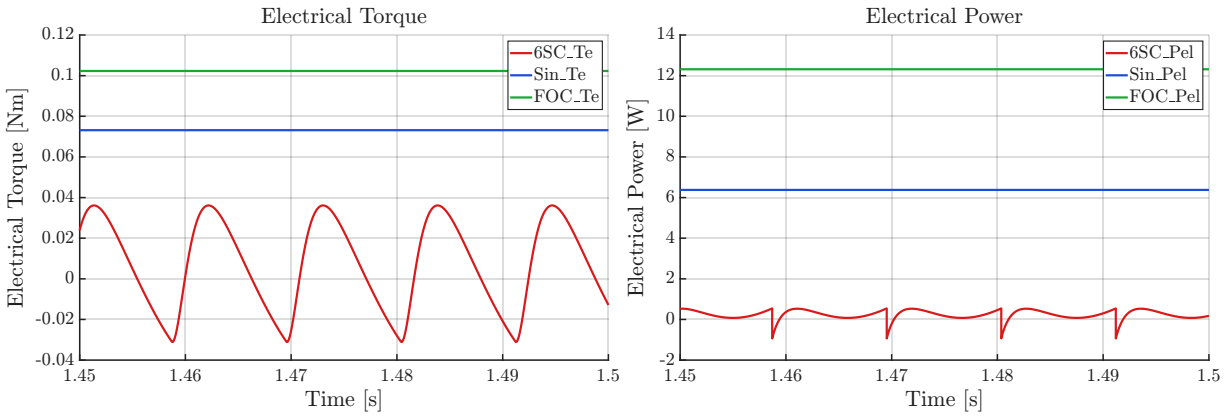


Figure 6.8.: Electromagnetic torque and electrical input power obtained under low-current operating conditions.

mean current-vector magnitude of 2.25 A, FOC provides the highest current-vector utilization and consequently achieves the highest mean torque, torque per ampere, and mechanical angular velocity. Sinusoidal control exhibits proportionally lower current-vector utilization, with corresponding reductions in mean torque and mechanical speed.

By contrast, six-step control directs only a small fraction of the current vector into the positive torque-producing q-axis component. Its poor current-vector utilization therefore substantially limits the mean torque and mechanical speed achieved under the OL1 operating condition.

The exceptionally high normalized torque ripple obtained with six-step control must be interpreted in relation to its very small mean torque. Because the mean torque appears in the denominator of the normalized torque-ripple metric, moderate absolute torque variations result in a disproportionately large relative value. The normalized metric is therefore poorly conditioned at this operating point and should not be considered independently of the absolute mean torque. In the adopted model, sinusoidal control and FOC exhibit negligible current distortion and torque ripple.

Table 6.5.: Performance indicators for the OL1 low-current operating condition.

Metric	Symbol	Six-step	Sinusoidal	FOC	Unit
Mean current-vector utilization	u_i	5.89	71.6	100	%
Current total harmonic distortion	THD_i	55.7	≈ 0	≈ 0	%
Mean torque	$\overline{T}_\mathrm{e}$	0.0059	0.073	0.102	Nm
Normalized torque ripple	τ_T	565.141	≈ 0	≈ 0	%
Mean torque per ampere	$\overline{\mathrm{TPA}}$	0.0027	0.0325	0.0455	Nm/A
Mean mechanical angular velocity	$\overline{\omega}_\mathrm{m}$	6.9	84.96	118.8	rad/s
Relative mechanical-speed standard deviation	$\sigma_{\omega,\mathrm{rel}}$	4.14	≈ 0	≈ 0	%
Mean electrical input power	$\overline{P}_\mathrm{el}$	0.24	6.38	12.31	W

OL2 – Medium-Current Operation

During OL2 medium-current steady-state operation, the six-step phase currents shown in Figure 6.9 approach a more trapezoidal waveform than those obtained under OL1. This behavior indicates more continuous conduction at the higher effective phase voltage, which is associated with the increased six-step duty-cycle command of 34.1 %. Nevertheless, pronounced curved depressions remain visible during the conduction and commutation intervals. The waveform therefore remains considerably distorted relative to both an idealized trapezoidal waveform and the approximately sinusoidal currents generated by sinusoidal control and FOC.

The mechanical angular velocities shown in Figure 6.10 demonstrate that six-step and sinusoidal control achieve similar mean performance under OL2. Both strategies produce comparable mean mechanical angular velocities, whereas FOC reaches a substantially

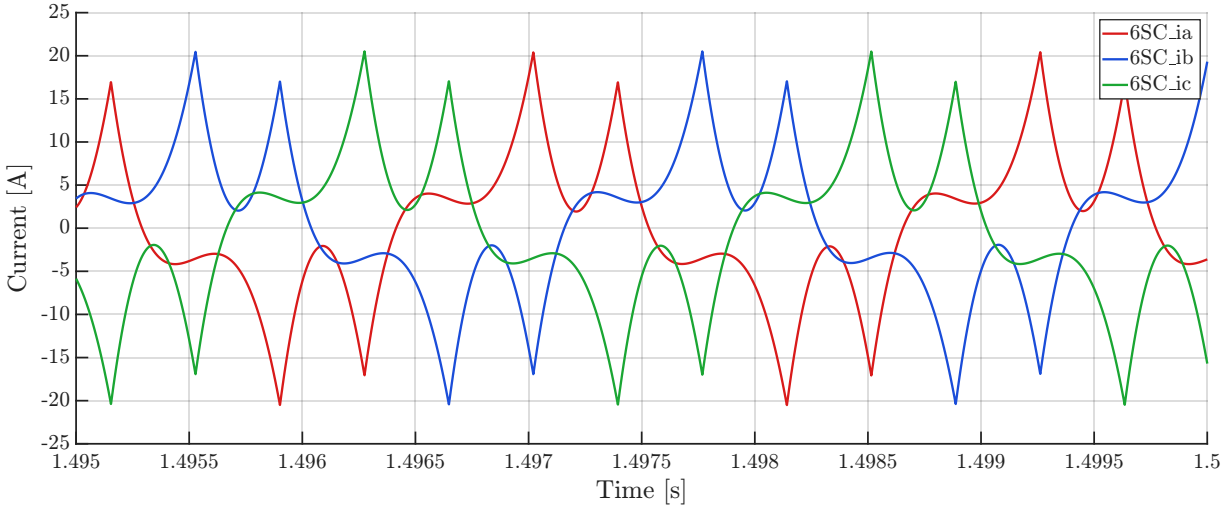


Figure 6.9.: Phase-current waveforms obtained using six-step control under medium-current operating conditions.

higher value. Furthermore, the speed variation produced by six-step control is considerably smaller than under OL1, indicating smoother mechanical operation despite the remaining current distortion.

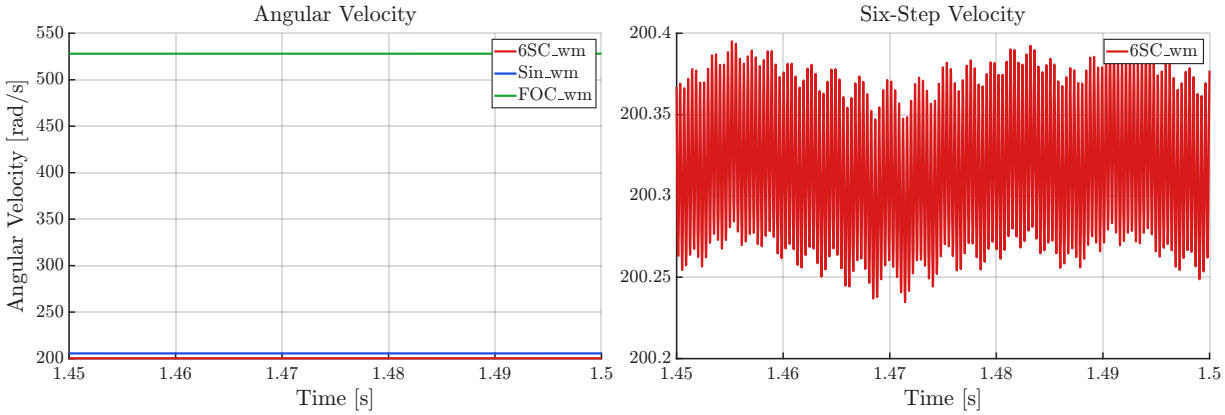


Figure 6.10.: Mechanical angular velocities obtained under medium-current operating conditions.

The steady-state performance indicators are summarized in Table 6.6. Six-step and sinusoidal control exhibit similar current-vector utilization and consequently achieve comparable mean torque, torque per ampere, mechanical angular velocity, and electrical input power. However, this agreement in mean performance does not extend to waveform quality. Six-step control retains substantial current distortion and torque ripple, whereas both quantities remain negligible for sinusoidal control and FOC.

Among the three strategies, FOC achieves complete current-vector utilization and there-

fore provides the highest mean torque, torque per ampere, and mechanical angular velocity. The lower current-vector utilization of six-step and sinusoidal control limits their torque production and mechanical performance under the OL2 operating condition.

Table 6.6.: Performance indicators for the OL2 medium-current operating condition.

Metric	Symbol	Six-step	Sinusoidal	FOC	Unit
Mean current-vector utilization	u_i	37.856	38.925	100	%
Current total harmonic distortion	THD_i	64.99	≈ 0	≈ 0	%
Mean torque	$\overline{T}_\mathrm{e}$	0.17245	0.17697	0.45465	Nm
Normalized torque ripple	τ_T	70.934	≈ 0	≈ 0	%
Mean torque per ampere	$\overline{\mathrm{TPA}}$	0.017211	0.017697	0.045465	Nm/A
Mean mechanical angular velocity	$\overline{\omega}_\mathrm{m}$	200.31	205.55	528.01	rad/s
Relative mechanical-speed standard deviation	$\sigma_{\omega,\mathrm{rel}}$	0.020133	≈ 0	≈ 0	%
Mean electrical input power	$\overline{P}_\mathrm{el}$	38.669	39.564	243.24	W

OL3 – High-Current Operation

OL3 high-current steady-state operation, largely follows the trends established under OL2. The waveform retains the characteristic distortion introduced by the commutation process, but the relative influence of the commutation-related depressions is reduced at the higher current level, indicating improved conduction despite the remaining deviation from an ideal six-step waveform.

The steady-state performance indicators are summarized in Table 6.7. As observed for OL2, six-step and sinusoidal control achieve nearly identical current-vector utilization and therefore produce comparable mean torque, torque per ampere, mechanical angular

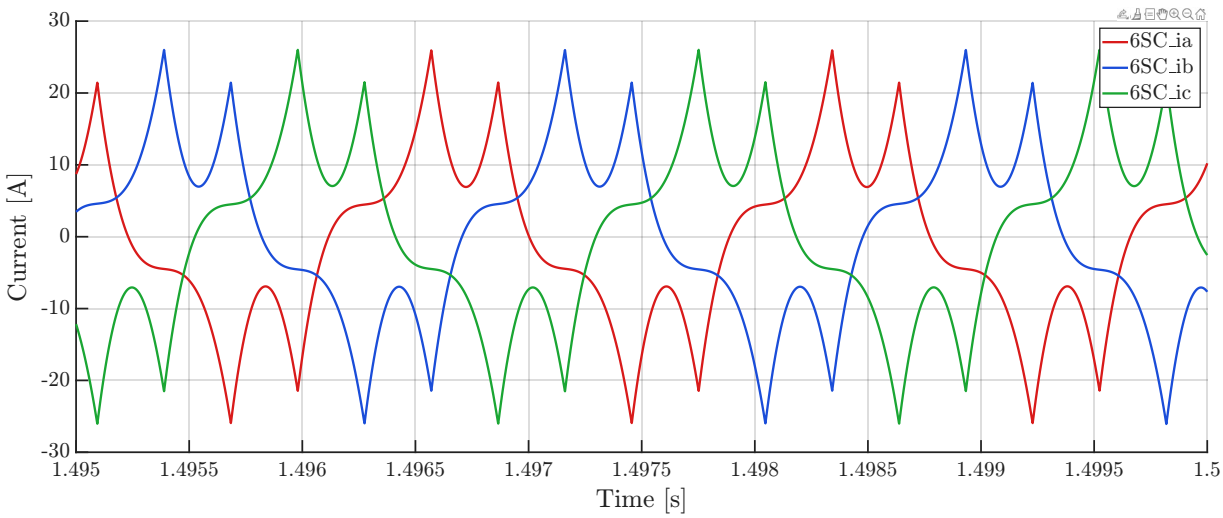


Figure 6.11.: Phase-current waveforms obtained using six-step control under high-current operating conditions.

velocity, and electrical input power. In contrast, FOC maintains complete current-vector utilization and consequently achieves substantially higher mean torque, torque per ampere, and mechanical angular velocity.

Relative to OL2, six-step control exhibits a reduction in current distortion, normalized torque ripple, and relative mechanical-speed variation. These improvements indicate smoother operation at the higher current level, although significant current distortion and torque ripple remain present. As in OL1 and OL2, current distortion and torque ripple remain negligible for sinusoidal control and FOC.

Table 6.7.: Performance indicators for the OL3 high-current operating condition.

Metric	Symbol	Six-step	Sinusoidal	FOC	Unit
Mean current-vector utilization	u_i	31.882	32.210	100.000	%
Current total harmonic distortion	THD_i	43.487	≈ 0	≈ 0	%
Mean torque	$\overline{T}_\mathrm{e}$	0.218	0.220	0.682	Nm
Normalized torque ripple	τ_T	58.093	≈ 0	≈ 0	%
Mean torque per ampere	$\overline{\mathrm{TPA}}$	0.014	0.015	0.045	Nm/A
Mean mechanical angular velocity	$\overline{\omega}_\mathrm{m}$	253.260	255.290	792.010	rad/s
Relative mechanical-speed standard deviation	$\sigma_{\omega,\mathrm{rel}}$	0.014	≈ 0	≈ 0	%
Mean electrical input power	$\overline{P}_\mathrm{el}$	63.521	63.287	547.300	W

6.4.2. Closed-Loop Tests

Based on the lowest mean mechanical angular velocity achieved at the OL3 operating condition of 250 rad/s, the low and high angular reference velocities are set to 20 rad/s and 180 rad/s, as defined in subsection 4.2.4.

CL1 – Low-Speed Reference Tracking

At the CL1 low-speed operating point, all three control strategies track the 20 rad/s reference with negligible relative steady-state speed error. The steady-state mechanical angular velocity remains approximately constant under sinusoidal control and FOC, whereas six-step control exhibits a measurable variation around the reference. The steady-state response for the angular velocities and the electrical powers can be seen in Figure 6.12.

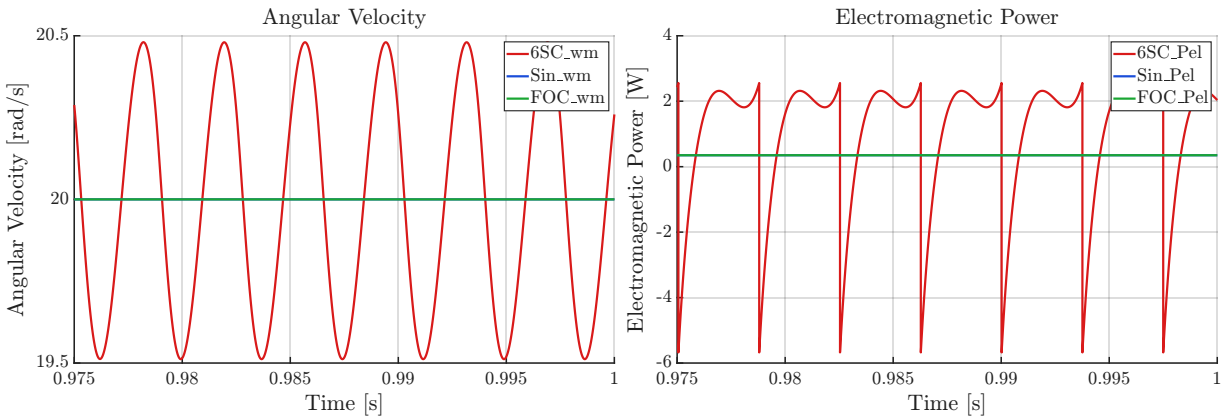


Figure 6.12.: Mechanical angular-velocity and electrical input-power responses obtained during CL1 low-speed reference tracking.

Consistent with the open-loop simulation, the electrical power for six-step commutation briefly falls below 0 W during the commutation instants.

The corresponding performance indicators are summarized in Table 6.8. Sinusoidal control and FOC exhibit nearly identical electrical input power and drive efficiency. By contrast, six-step control requires substantially more electrical input power and achieves considerably lower drive efficiency. Thus, although all three strategies provide accurate mean steady-state tracking, their energetic performance and steady-state speed variation differ substantially.

Table 6.8.: Steady-state performance indicators for the CL1 low-speed operating condition.

Metric	Symbol	Six-step	Sinusoidal	FOC	Unit
Relative steady-state speed error	$e_{\omega,\mathrm{rel}}$	≈ 0	≈ 0	≈ 0	%
Relative steady-state speed standard deviation	$\sigma_{\omega,\mathrm{rel}}$	1.7229	≈ 0	≈ 0	%
Mean steady-state electrical input power	$\overline{P}_{\mathrm{el}}$	1.047	0.3492	0.3490	W
Mean steady-state drive efficiency	$\overline{\eta}$	33.02	98.62	98.69	%

CL2 – High-Speed Reference Tracking

The principal steady-state observations from CL1 are also evident at the CL2 high-speed operating point. All three control strategies track the 180 rad/s reference with negligible steady-state error. As seen in Figure 6.13 Six-step control again exhibits greater speed variation than sinusoidal control and FOC, although this variation is substantially less pronounced than in CL1.

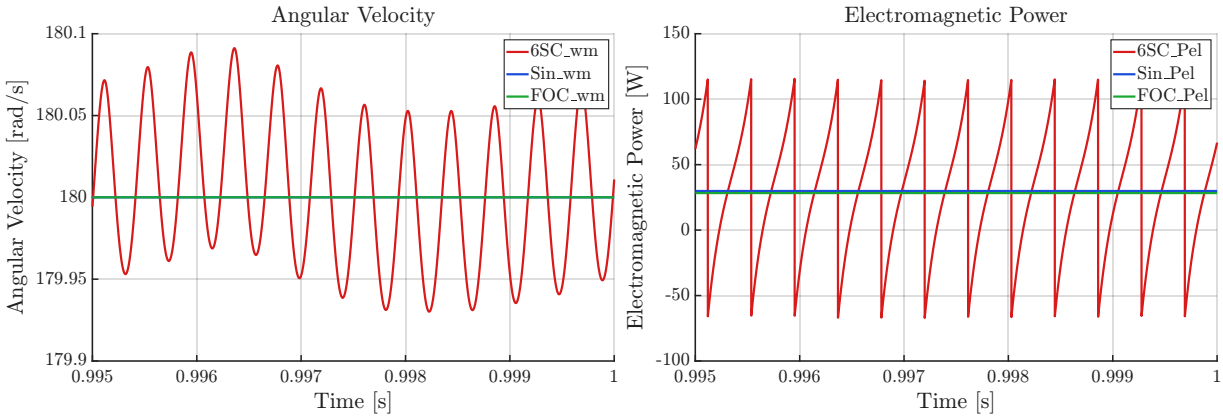


Figure 6.13.: Mechanical angular-velocity and electrical input-power obtained during CL2 high-speed reference tracking.

The corresponding performance indicators are summarized in Table 6.9. The energetic ordering observed in CL1 is retained: six-step control exhibits the highest electrical input power and lowest drive efficiency, FOC exhibits the lowest input power and highest efficiency, and sinusoidal control lies between them. However, the differences between the strategies are less pronounced at the higher operating speed.

Table 6.9.: Steady-state performance indicators for the CL2 high-speed operating condition.

Metric	Symbol	Six-step	Sinusoidal	FOC	Unit
Relative steady-state speed error	$e_{\omega,\mathrm{rel}}$	≈ 0	≈ 0	≈ 0	%
Relative steady-state speed standard deviation	$\sigma_{\omega,\mathrm{rel}}$	0.024	≈ 0	≈ 0	%
Mean steady-state electrical input power	$\overline{P}_{\mathrm{el}}$	31.06	29.86	28.27	W
Mean steady-state drive efficiency	$\overline{\eta}$	89.82	93.43	98.69	%

CL3 – Low-Speed Reference Step

The mechanical angular-velocity and current responses obtained for the CL3 low-speed reference step are shown in Figure 6.14.

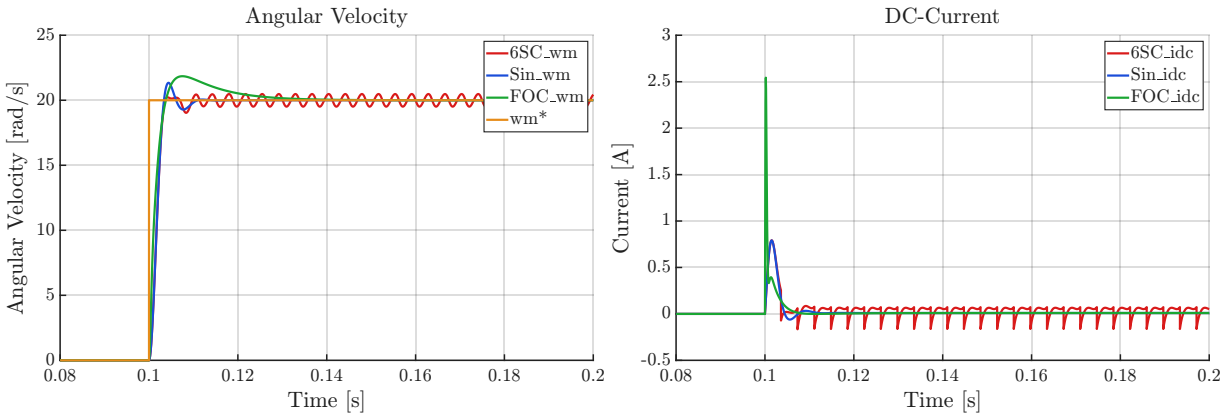


Figure 6.14.: Mechanical angular-velocity and current responses obtained for the CL3 low-speed reference step.

As illustrated in Figure 6.14, all three control strategies exhibit comparable rise times and similar initial overshoots. The most noticeable differences emerge as the rotor approaches the reference speed and transitions into steady-state operation. Six-step commutation results in persistent speed oscillations, sinusoidal control exhibits only minor oscillations before converging to an approximately constant speed, and FOC achieves non-oscillatory

operation at the expense of the largest overshoot and the slowest settling response.

It can be seen that the DC-current spikes for all three control strategies, but especially for FOC at the step instant.

The quantitative results in Table 6.10 confirm that the choice of control strategy has only a minor influence on rise time, whereas overshoot and settling time differ considerably. The comparatively large overshoot observed with FOC indicates that positive torque is maintained for longer as the rotor approaches the reference speed. Since all strategies employ the same speed-controller bandwidth, this behavior is likely attributable to differences in torque production and current-control dynamics, which allow more acceleration energy to accumulate before the reference speed is reached.

Among the control strategies that satisfy the settling criterion, sinusoidal control achieves both the smallest overshoot and the shortest settling time, indicating a well-damped transient response. Although six-step commutation produces the lowest overshoot, the associated torque ripple causes periodic speed fluctuations that prevent convergence within the specified settling band. Nevertheless, the onset of steady-state operation occurs at approximately the same time as for sinusoidal control and earlier than for FOC.

Table 6.10.: Step-response performance indicators for the CL3 low-speed reference step.

Metric	Symbol	Six-step	Sinusoidal	FOC	Unit
Rise time	t_r	0.00225	0.00221	0.00252	s
Settling time	t_s	Did not settle	0.0092	0.0225	s
Relative speed overshoot	M_p	2.40	6.71	9.25	%

CL4 – High-Speed Reference Step

The mechanical angular-velocity responses obtained for the CL4 high-speed reference step are shown in Figure 6.15.

As shown in Figure 6.15, the higher-speed transition exhibits behavior similar to that

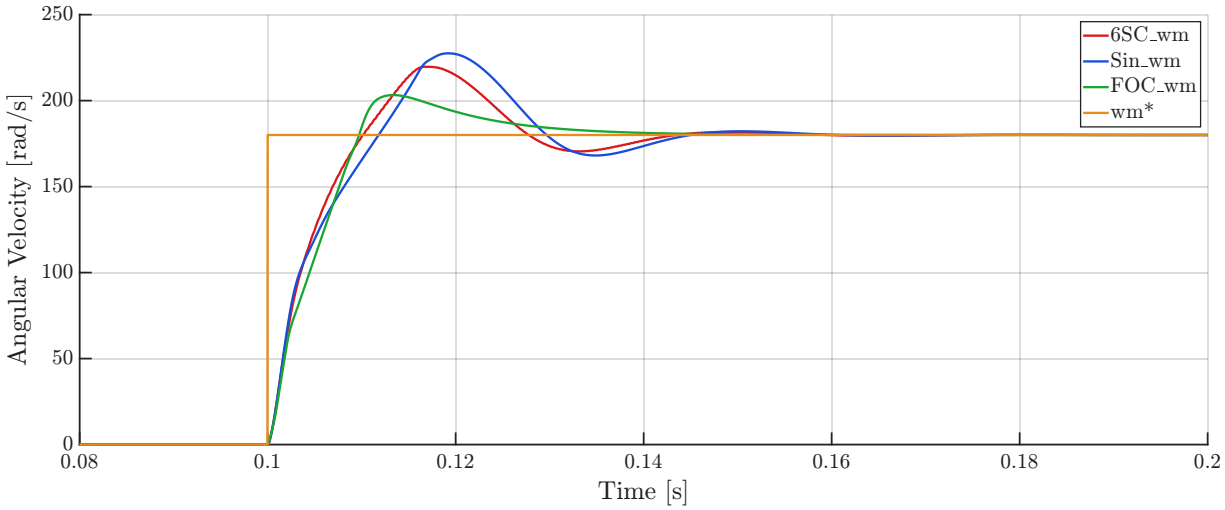


Figure 6.15.: Mechanical angular-velocity responses obtained for the CL4 high-speed reference step.

observed in CL3. The main differences between the control strategies again occur after the initial overshoot. Six-step and sinusoidal control both undershoot the reference only once before converging to the setpoint, whereas FOC again remains non-oscillatory.

The same current spikes as in CL3 are recorded.

The performance indicators show that six-step commutation and FOC achieve comparable rise times, whereas sinusoidal control exhibits a slightly slower initial response. In contrast to the CL3 low-speed reference step, FOC produces the smallest overshoot and reaches steady-state first. This result suggests that, at higher operating speeds, the torque production and current-control dynamics of FOC permit a more accurate regulation of the electromagnetic torque as the rotor approaches the reference speed. Consequently, less excess acceleration energy is accumulated, resulting in a smaller overshoot and faster convergence to steady-state operation.

While there are still some minor oscillations visible during steady-state, six-step control manages to fulfill the settling criterion for the high-speed reference step. Consequently, unlike the CL3 low-speed reference step, all control strategies achieve stable steady-state operation according to the defined settling criterion.

Table 6.11.: Step-response performance indicators for the CL4 high-speed reference step.

Metric	Symbol	Six-step	Sinusoidal	FOC	Unit
Rise time	t_r	0.0073	0.0088	0.0075	s
Settling time	t_s	0.0397	0.0419	0.0311	s
Relative speed overshoot	M_p	22.075	26.381	12.891	%

CL5 – Maximum-Speed Operation

With a steadily increasing speed-reference ramp, the maximum mechanical angular velocities achieved by the three control strategies are summarized in Table 6.12. Six-step and sinusoidal control attain comparable maximum speeds, with six-step control achieving a marginally higher value. In contrast, FOC reaches a substantially higher maximum mechanical angular velocity than both alternative strategies.

For all three control approaches, the maximum-speed operating point is reached when the 35 V DC-link voltage limit becomes active. Consequently, the observed differences in maximum achievable speed indicate how effectively each strategy utilizes the available inverter voltage under the considered operating conditions. The superior maximum-speed performance of FOC therefore suggests a more effective utilization of the available voltage compared with six-step and sinusoidal control.

Table 6.12.: Maximum mechanical angular velocities obtained during the CL5 maximum-speed test.

Metric	Symbol	Six-step	Sinusoidal	FOC	Unit
Maximum mechanical angular velocity	$\omega_{m,max}$	482.24	473.53	758.06	rad/s

CL6 – Low-to-High Speed Ramp

The mechanical angular-velocity responses obtained during the CL6 low-to-high reference-speed ramp are presented in Figure 6.16. All three control strategies closely track the increasing reference speed and successfully reach the high-speed operating point.

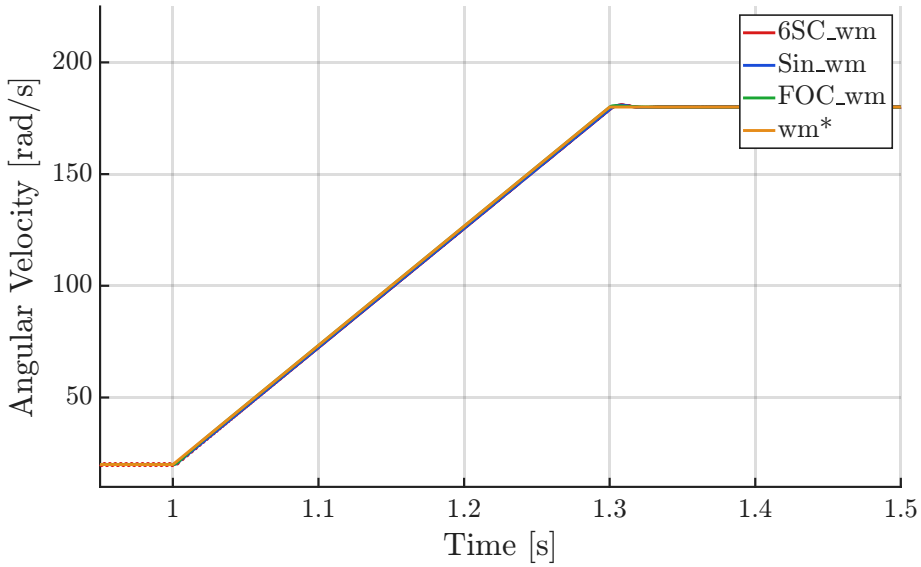


Figure 6.16.: Mechanical angular-velocity responses obtained during the CL6 low-to-high reference-speed ramp.

The corresponding performance indicators are summarized in Table 6.13. All three controllers exhibit nearly identical rise and settling times, while the relative speed overshoot remains below 1 % in each case. The most notable, although still very small difference is observed during the transient ramp phase. Six-step and sinusoidal control achieve comparable average relative tracking errors, whereas FOC yields a substantially lower value. Consequently, although all three strategies demonstrate similar endpoint performance, FOC tracks the continuously varying reference more accurately throughout the speed-ramp interval.

Table 6.13.: Performance indicators for the CL6 low-to-high reference-speed ramp.

Metric	Symbol	Six-step	Sinusoidal	FOC	Unit
Rise time	t_r	0.24029	0.24028	0.23994	s
Settling time	t_s	0.29614	0.29605	0.29410	s
Relative speed overshoot	M_p	0.52358	0.50779	0.37408	%
Average relative error during rise	$\overline{e}_\mathrm{rel,rise}$	0.64539	0.61603	0.03567	%

6.4.3. Summary of Simulation Results

The open-loop tests reveal clear differences in current-vector utilization and waveform quality among the three control strategies. FOC directs approximately 100 % of the current vector into the torque-producing q-axis component and consequently achieves the highest mean torque, torque per ampere, and mechanical angular velocity at all investigated current levels. Sinusoidal control exhibits lower current-vector utilization but maintains approximately sinusoidal phase currents and negligible torque ripple.

Six-step control exhibits strongly operating-point-dependent behavior. Under the OL1 low-current condition, its phase currents remain predominantly transient within each commutation sector, resulting in poor current-vector utilization, low mean torque, and pronounced torque and speed variations. Under OL2 and OL3, the phase currents approach a trapezoidal profile, and the mean torque and mechanical angular velocity become comparable to those obtained with sinusoidal control. However, substantial current distortion and torque ripple remain present. Thus, convergence in mean performance does not imply convergence in waveform quality.

The closed-loop results are consistent with these open-loop observations. Speed feedback compensates for the differences in mean torque production and enables all three strategies to track the investigated steady-state speed references with negligible mean error. However, it does not eliminate the strategy-specific waveform characteristics identified in the open-loop tests. In particular, the poor low-current behavior and pronounced torque pulsations of six-step control reappear under CL1 as increased steady-state speed variation, electrical input power, and drive losses. At the higher CL2 operating point, these differ-

ences become less pronounced, consistent with the improved conduction continuity and lower relative speed variation observed for six-step control at the higher open-loop operating levels. Sinusoidal control and FOC, which produce smooth torque in the open-loop tests, correspondingly maintain approximately constant steady-state speeds in closed-loop operation.

The transient results further show that accurate steady-state tracking does not imply identical dynamic behavior. Although the common speed-controller tuning produces broadly comparable rise times, the overshoot and settling behavior depend on the operating point and the underlying torque-production and current-control dynamics. Sinusoidal control provides the shortest settling time among the strategies satisfying the CL3 settling criterion, whereas FOC achieves the lowest overshoot and shortest settling time for the CL4 high-speed step. During the low-to-high speed ramp, all three strategies exhibit similar rise and settling times, while FOC produces the lowest average tracking error.

Finally, six-step and sinusoidal control attain comparable voltage-limited maximum speeds, whereas FOC reaches a substantially higher maximum mechanical angular velocity within the adopted simulation model.

7. Experimental Evaluation

This chapter presents the experimental evaluation of six-step control and FOC using the substitute motor-drive platform introduced in section 4.3. The investigation covers steady-state speed tracking, transient reference tracking, processor load, and sound pressure under the conditions defined in subsection 4.3.1 and subsection 4.3.3. The implemented controller configuration is presented first, followed by the observations and performance indicators obtained in test cases EX1–EX3.

7.1. Experimental Controller Configuration

Following the tuning procedure described in section 4.3, the experimental controller parameters are obtained using the automatic tuning functionality of the STMicroelectronics Motor Control Software Development Kit. The FOC current controllers are configured with a bandwidth of 1 kHz, and the speed controllers of both control strategies are executed with a period of 1 ms. The resulting PI controller gains are listed in Table 7.1.

Table 7.1.: Experimental controller configuration and corresponding controller gains.

PI controller	**Tuning parameter**	**Specified value**	$\boldsymbol{K}_\mathbf{p}$	$\boldsymbol{K}_\mathbf{i}$
FOC i_d current controller	$f_{\mathrm{BW,i}}$	1000 Hz	0.00513	0.00226
FOC i_q current controller	$f_{\mathrm{BW,i}}$	1000 Hz	0.00513	0.00226
FOC speed controller	$T_{\mathrm{exec},\omega}$	1 ms	168.75	0.17395
Six-step speed controller	$T_{\mathrm{exec},\omega}$	1 ms	24.76563	0.02551

7.2. Observations and Results

As described in subsection 4.3.3, the experiments utilize the same 20 rad/s low speed and 180 rad/s high speed references as the closed loop tests in subsection 6.4.2.

7.2.1. Experimental Tests

EX1 – Low-Speed Reference Tracking

The measured mechanical-speed responses at the nominal reference of 20 rad/s are shown in Figure 7.1. Both control strategies operate close to the applied reference. However, the six-step response exhibits considerably larger speed variations than the FOC response.

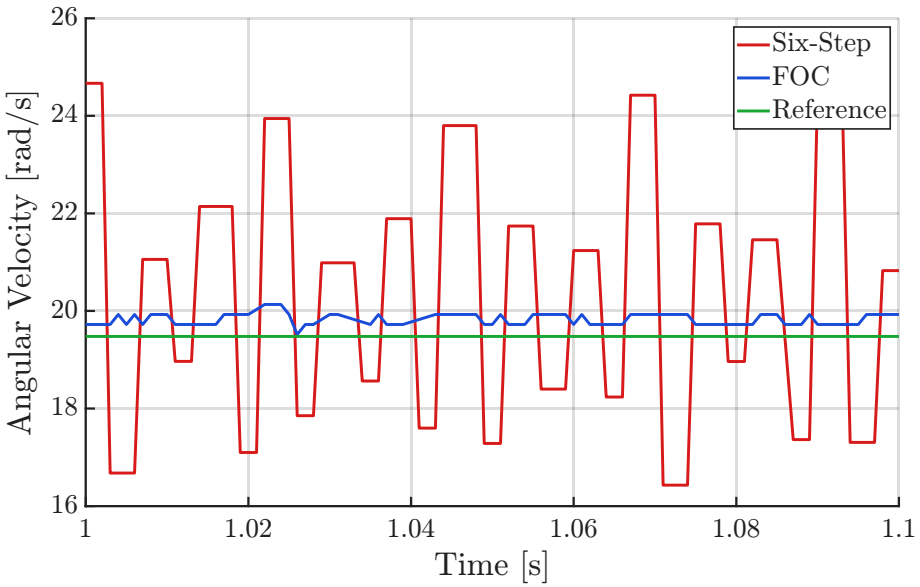


Figure 7.1.: Measured mechanical-speed responses at the EX1 reference of 20 rad/s.

As indicated by the green reference line in Figure 7.1, the reference applied during the experiment is 19.5 rad/s, which is slightly lower than the nominal command of 20 rad/s. Since the speed errors and performance metrics are calculated relative to the actual applied reference of 19.5 rad/s, this deviation slightly shifts the experimental operating point but does not directly affect the calculated metrics.

Both six-step control and FOC exhibit irregular oscillations about mean values that are slightly above the applied reference of 19.5 rad/s. The oscillation amplitude is substantially

larger under six-step control. Furthermore, the measured-speed signals, particularly under six-step control, contain intervals during which the reported value remains unchanged. These intervals indicate that the measured speed is updated comparatively infrequently at this low-speed operating point.

The quantitative results presented in Table 7.2 are consistent with the differences observed in the measured responses. Compared with six-step control, FOC achieves a smaller steady-state speed error and substantially lower speed variation. In contrast, six-step control produces a substantially lower processor load. The measured sound pressure levels are similar for both strategies, although a slightly lower value is obtained with FOC.

Table 7.2.: Performance indicators for the EX1 low-speed reference-tracking operating condition.

Metric	Symbol	Six-step	FOC	Unit
Relative steady-state speed error	$e_{\omega,\mathrm{rel}}$	4.4785	1.7968	%
Relative steady-state mechanical-speed standard deviation	$\sigma_{\omega,\mathrm{rel}}$	12.475	0.76063	%
Average steady-state processor load	$\overline{\rho}_{\mathrm{CPU}}$	0.63034	37.967	%
Average steady-state sound pressure level	$\overline{L}_{\mathrm{p}}$	29.1	27.7	dB

EX2 – High-Speed Reference Tracking

The measured mechanical-speed responses at the nominal reference of 180 rad/s are shown in Figure 7.2. The same trend as observed in EX1 is evident at higher speed. Both control strategies operate close to the applied reference, whereas the six-step response exhibits substantially larger speed variations than the FOC response.

As indicated by the reference line in Figure 7.2, the reference applied during the experiment is 179.7 rad/s, which is lower than the nominal command of 180 rad/s. The speed errors and performance metrics are calculated relative to this actual applied reference. Compared with EX1, the mean responses of both control strategies follow the applied reference more closely, which is also reflected by their smaller relative steady-state speed errors.

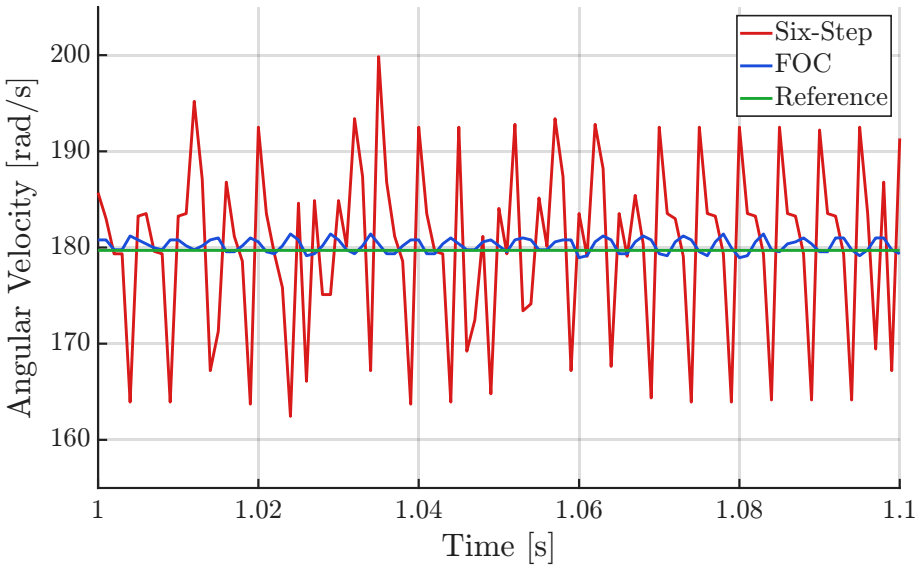


Figure 7.2.: Measured mechanical-speed responses at the EX2 reference of 180 rad/s.

The measured-speed update frequency increases from EX1 to EX2 for both six-step control and FOC, likely reflecting the improved availability of back-EMF-related information at higher rotational speeds. The increase is more pronounced for six-step control, whereas the implemented FOC scheme exhibits a comparatively smaller change due to its estimator-based speed calculation.

The speed oscillations occur at a higher frequency in EX2 than in EX1 for both control strategies. This increase is consistent with the higher rotational and electrical frequencies at the EX2 operating point. The oscillation amplitude remains substantially larger under six-step control, whereas the FOC response remains comparatively smooth.

The quantitative results presented in Table 7.3 confirm that the principal differences identified in EX1 persist at the higher reference speed. Compared with six-step control, FOC achieves a smaller steady-state speed error and substantially lower speed variation. However, the difference between the strategies in steady-state speed error is smaller than in EX1, whereas their difference in speed variation remains pronounced.

As in EX1, six-step control produces a substantially lower processor load than FOC. The measured sound pressure levels also remain similar for both strategies. In contrast to EX1, however, six-step control produces the slightly lower sound pressure level under the EX2 operating condition.

Table 7.3.: Performance indicators for the EX2 high-speed reference-tracking operating condition.

Metric	Symbol	Six-step	FOC	Unit
Relative steady-state speed error	$e_{\omega,\mathrm{rel}}$	0.37453	0.20092	%
Relative steady-state mechanical-speed standard deviation	$\sigma_{\omega,\mathrm{rel}}$	4.6671	0.41343	%
Average steady-state processor load	$\overline{\rho}_{\mathrm{CPU}}$	2.2758	38.063	%
Average steady-state sound pressure level	$\overline{L}_{\mathrm{p}}$	49.3	50.9	dB

EX3 – Low-to-High Speed Ramp

The measured mechanical-speed responses during the 300 ms reference ramp from 20 rad/s to 180 rad/s are shown in Figure 7.3. Both control strategies increase the measured mechanical speed toward the high-speed operating point. However, the six-step response exhibits a larger deviation from the reference during the ramp and substantially greater speed variation after the transition than the FOC response.

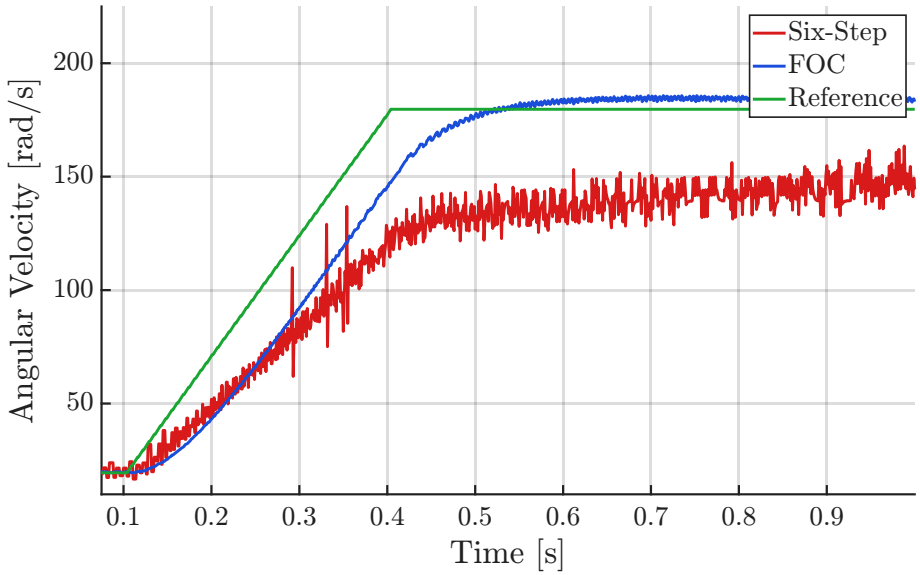


Figure 7.3.: Measured mechanical-speed responses during the EX3 reference ramp from 20 rad/s to 180 rad/s over 300 ms.

During the reference ramp, both measured-speed responses initially remain below the increasing reference. The deviation is more pronounced under six-step control, whereas the FOC response follows the reference more closely throughout the transition. After the reference reaches the high-speed operating point, both responses approach the final reference. However, the six-step response exhibits a larger overshoot and continued speed oscillations, whereas the FOC response approaches the final operating point with comparatively small overshoot and lower speed variation.

The quantitative results presented in Table 7.4 are consistent with the differences observed in the measured responses. Compared with six-step control, FOC achieves a shorter rise time, substantially lower relative speed overshoot, and a smaller average relative tracking error during the ramp. Both responses satisfy the specified settling criterion. However FOC reaches the specified settling range substantially sooner than six-step control.

Table 7.4.: Performance indicators for the EX3 low-to-high speed-reference ramp.

Metric	**Symbol**	**Six-step**	**FOC**	**Unit**
Rise time	t_{r}	0.31294	0.27126	s
Settling time	t_{s}	7.195	0.444	s
Relative speed overshoot	M_{p}	1.85	0.55	%
Average relative tracking error during the ramp	$\overline{e}_{\mathrm{rel,\,rise}}$	30.694	17.598	%

7.2.2. Summary of Experimental Findings

The steady-state experiments show that both six-step control and FOC track the investigated speed references with small mean errors. However, FOC consistently produces substantially lower mechanical-speed variation. This difference is particularly pronounced at low speed, where the sensorless speed estimate is updated less frequently. At the higher operating point, the mean tracking of both strategies improves, but the greater speed variation under six-step control remains evident.

The transient results reinforce the steady-state findings. During the low-to-high speed ramp, FOC follows the reference more closely and reaches the final operating point with a shorter rise time, smaller overshoot, and substantially faster settling. Six-step control exhibits larger tracking deviations during the ramp and persistent oscillations after the

transition. Closed-loop speed feedback therefore compensates for much of the difference in mean behavior but does not eliminate the control-strategy-specific speed variations.

A more qualitative observation is that during all test cases, six-step commutation appeared to cause increased motor vibrations on and on the the table it was resting on. This is likely caused by the torque variations in six-step control, although this effect could not be quantified.

The mean sound-pressure measurements did not reveal a consistent difference between six-step control and FOC. During operation, however, FOC was subjectively perceived as less prominent and less disturbing. Since the mean sound-pressure level does not represent the spectral composition of the emitted sound, the recorded signals were additionally examined in the frequency domain.

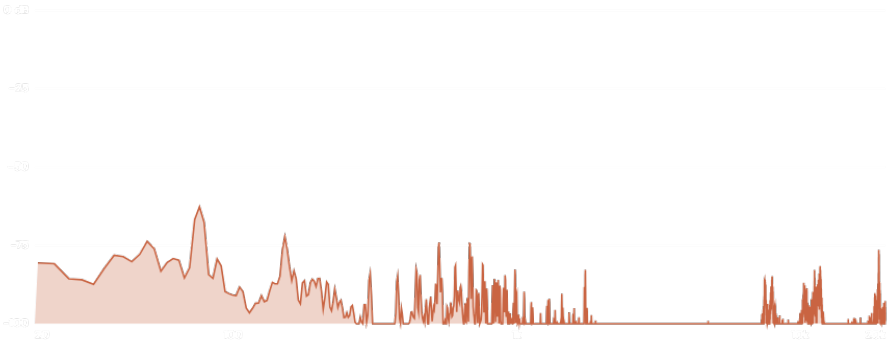


Figure 7.4.: Frequency spectrum of the measured sound-pressure signal for six-step control at the low-speed operating point.

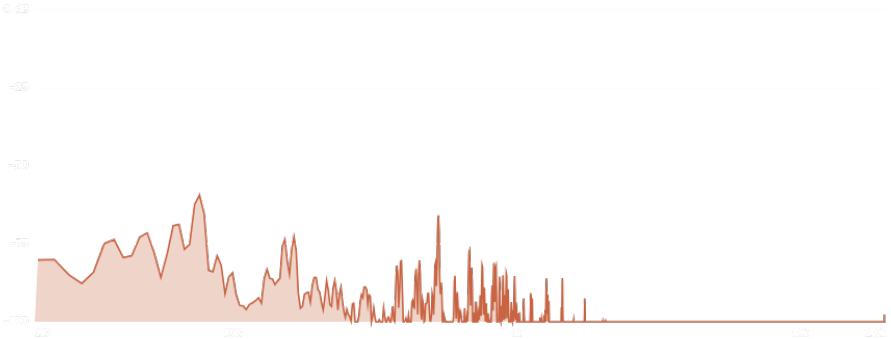


Figure 7.5.: Frequency spectrum of the measured sound-pressure signal for FOC at the low-speed operating point.

As shown in Figures 7.4–7.7, the sound-pressure spectrum of six-step control contains more pronounced discrete frequency components, particularly in the higher-frequency range. The spectral energy under FOC is distributed more broadly and exhibits fewer dominant

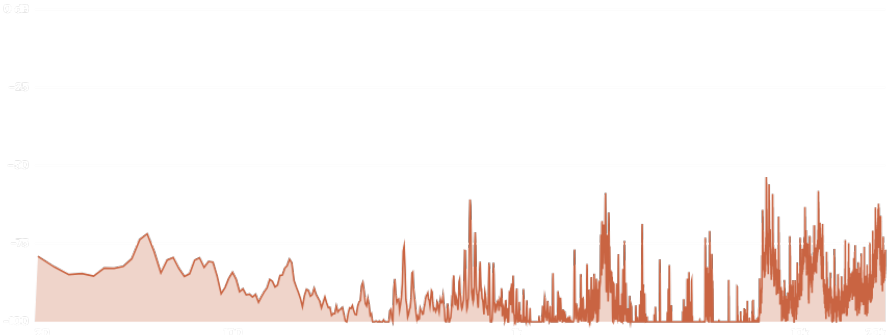


Figure 7.6.: Frequency spectrum of the measured sound-pressure signal for six-step control at the high-speed operating point.

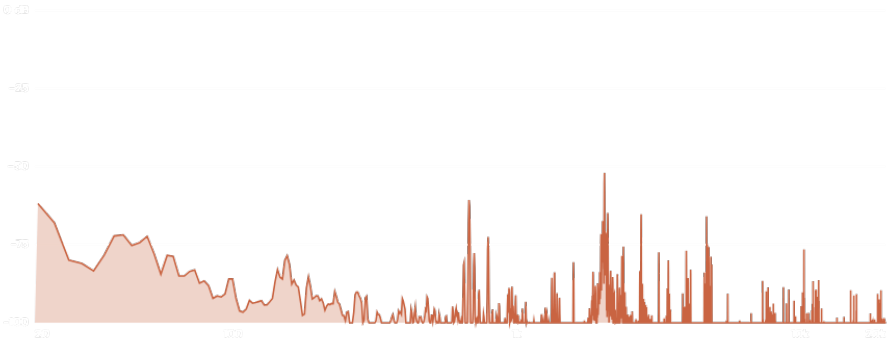


Figure 7.7.: Frequency spectrum of the measured sound-pressure signal for FOC at the high-speed operating point.

peaks. This difference provides a possible explanation for the subjective perception of FOC as less disturbing despite the similar mean sound-pressure levels, and is especially pronounced at lower speeds.

Within the investigated motor, hardware platform, controller configurations, and operating range, FOC provides the strongest overall speed-control performance, while six-step control offers substantially lower processor utilization. The acoustic measurements do not establish a consistent ranking because their ordering changes with the operating point. Since sinusoidal control was not included in the experimental campaign, no experimentally supported comparison with this strategy can be made.

8. Discussion

This chapter examines the validity and transferability of the simulation and experimental investigations. It focuses on the extent to which both investigations can be related, the methodological factors affecting their interpretation, and the limitations of the resulting comparison. The final assessment of the research objectives and contributions is reserved for chapter 9.

8.1. Scope and Validity of the Comparison

The simulation study provided a consistent basis for comparing six-step control, sinusoidal control, and FOC. All strategies were evaluated using the same motor model, inverter representation, parameter set, operating conditions, tuning objectives, and performance indicators. Differences among the simulation results can therefore be attributed primarily to the investigated controller implementations. However, they remain dependent on the adopted models, controller configurations, and operating points and do not represent general performance limits of the underlying control principles.

The experimental investigation had a narrower scope. Because the custom motor-control hardware could not be commissioned within the available research period, six-step control and FOC were evaluated using a substitute motor-drive platform. Sinusoidal control was excluded because no suitable implementation was available within the employed MCSDK. Although this software limitation would also have affected the custom extension board, the intended sensored platform would have provided a more suitable basis for developing an independent implementation. The experiments therefore provide a physical comparison of six-step control and FOC, but neither validate the custom hardware nor provide experimental evidence for sinusoidal control.

The experimental comparison remained internally consistent because both strategies were operated using the same motor-drive system, supply conditions, software environment, operating points, and data-acquisition procedure. However, the experimental responses characterize the complete implementations, including their respective sensorless estimators, rather than the control algorithms in isolation. This distinction is particularly relevant at low speed, where the reduced back-EMF limits the information available for sensorless estimation [10].

The principal differences between the simulation and experimental investigations are summarized in Table 8.1.

Table 8.1.: Methodological differences between the simulation and experimental investigations.

Aspect	**Simulation**	**Experiment**
Feedback	Ideal rotor position and speed	Strategy-specific sensorless estimation
Controller tuning	Common bandwidth, phase-margin, and overshoot objectives	Parameters generated using the MCSDK
Electrical constraints	Limits selected for the intended custom hardware	Lower limits imposed by the substitute supply and motor-driver platform
Inverter representation	Continuous voltage-averaging model	Switched inverter with sampling, delays, and protection functions
Control strategies	Six-step, sinusoidal, and FOC	Six-step and FOC

These differences prevent a direct quantitative validation of the simulation using the experimental results. The comparison between both investigations is therefore restricted to whether the same relative tendencies remain identifiable under physical embedded operation.

8.2. Relationship Between Simulation and Experiment

The simulation and experimental investigations showed the same relative tendencies in speed smoothness and transient reference tracking. Their quantitative results differed, however, because the experiment introduced estimator dynamics, discrete implementation effects, and more restrictive electrical operating limits that were absent or represented differently in the simulation. The simulation should therefore be regarded as a comparative model of the investigated controller implementations rather than as a quantitatively predictive model of the substitute drive.

The use of sensorless feedback is particularly relevant to the interpretation of the experimental speed responses. Because the estimated speed served as both controller feedback and evaluation quantity, physical shaft-speed variation cannot be separated from estimation and reporting effects. The comparatively infrequent speed updates observed at low speed indicate that this dependency contributed to the measured responses. The experimental speed variation can therefore not be interpreted exclusively as mechanical speed ripple.

The more restrictive experimental operating range also prevented the complete simulation test campaign from being reproduced. Protection events under demanding transient conditions restricted the achievable acceleration and excluded an experimental maximum-speed comparison. Consequently, the simulated and experimental operating limits were reached for different reasons. The experimental transient responses characterize the substitute drive within its available electrical limits, whereas the simulated maximum-speed results characterize the interaction of the implemented controllers with the modelled voltage and saturation constraints.

The agreement between both investigations is therefore limited to the relative behavior of six-step control and FOC. It does not establish the validity of the simulated magnitudes of speed variation, transient response, torque ripple, efficiency, or maximum speed.

8.3. Model and Measurement Limitations

8.3.1. Simulation Model

The simulation framework was restricted to the electrical and mechanical behavior required for the comparative investigation. The voltage-averaging inverter omitted PWM switching, dead time, semiconductor voltage drops, switching and conduction losses, current-sampling effects, and measurement noise. The negligible simulated current distortion and torque ripple obtained for sinusoidal control and FOC are therefore partly consequences of the continuous voltage representation and do not predict the harmonic behavior of a physical switched inverter.

The efficiency calculation was subject to similar restrictions. Winding resistance and viscous mechanical friction were represented, whereas semiconductor losses, iron losses, magnetic saturation, temperature-dependent parameters, and nonlinear mechanical losses were omitted. The resulting values are suitable for comparisons within the common model but not as predictions of physical drive efficiency.

The motor model further assumed sinusoidal back-EMF, linear magnetic behavior, and a single stator inductance. However, the rotor-position-dependent inductance observed during parameter identification in Appendix B suggests that the physical motor may exhibit magnetic saliency. Cogging torque, nonlinear friction, and thermal parameter variation were likewise not represented. These simplifications restrict the quantitative fidelity of the simulated current, torque, and speed responses.

The assumed back-EMF waveform also affects the transferability of the six-step results. The investigated model represents a motor with sinusoidal back-EMF, whereas block commutation may be better matched to motors with an approximately trapezoidal waveform [10]. The results therefore describe the interaction between the implemented commutation strategy and the selected motor model rather than the general performance of six-step control.

8.3.2. Experimental Measurements

The experimental evaluation was limited by the available measurement equipment and the use of internal controller signals. In particular:

- the sensorless speed estimate was not verified using an independent shaft-speed measurement;
- the processor-load measurements depended on the employed microcontroller, MCSDK implementation, compiler configuration, and measurement method; and
- the acoustic measurements were obtained using an uncalibrated microphone in a non-standardized acoustic environment.

The processor-load measurements consequently indicate only the relative computational demand of the investigated implementations on the substitute platform. They cannot be transferred directly to other processors, software architectures, compiler settings, or optimized production firmware.

The microphone used for the sound preassure measurements was not calibrated, background noise and structural transmission paths were not controlled, and no psychoacoustic quantities such as loudness, sharpness, tonality, or roughness were evaluated. The spectra therefore reveal differences in the frequency distribution of the recorded signals but do not establish a general acoustic or perceptual ranking of the control strategies.

8.4. Limitations and Transferability

The principal limitation of the work is the incomplete realization of the intended experimental platform. The custom hardware was designed, reviewed, manufactured, and assembled, but its power supplies, inverter, measurement channels, communication interfaces, protection functions, thermal behavior, and motor-control operation were not functionally validated. The experimental findings therefore apply exclusively to the substitute platform.

Further limitations affecting the transferability of the results are:

- **Incomplete strategy coverage:** Sinusoidal control was investigated only in simulation. The corresponding findings apply specifically to the implemented voltage-based controller and adopted simplifications and assumptions.
- **Unloaded operation:** No external load torque was applied. The investigation therefore does not represent load disturbances, rated continuous operation, application-specific inertia, regeneration, field weakening, or thermally relevant duty cycles.
- **No common mechanical output requirement:** The operating points were not defined by an equal useful mechanical output. Electrical power and model-level efficiency can therefore be compared only within the selected test conditions and not as application-level efficiency indicators.
- **Controller-specific tuning:** The controllers were tuned using common nominal objectives but were not optimized individually for each operating point or application-specific requirement. The dynamic results therefore characterize the investigated configurations rather than the maximum achievable performance of each strategy.
- **System dependency:** The findings depend on the motor parameters, back-EMF waveform, saliency, rotor inertia, inverter, feedback method, switching frequency, supply capability, protection behavior, and processor architecture of the investigated systems.

Quantities such as harmonic distortion, efficiency, acoustic behavior, processor utilization, maximum speed, and settling time are particularly dependent on these conditions and should not be transferred directly to other motor-drive systems. Application-specific evaluation would require representative hardware, independent mechanical measurements, an appropriate load profile, and controller tuning aligned with the intended operating range and performance requirements.

9. Conclusion and Outlook

This chapter evaluates the extent to which the research objectives were achieved, the research question was answered, and the identified research gap was addressed. The remaining limitations provide the basis for the subsequent outlook.

9.1. Conclusion

The work was motivated by the absence of a consistent quantitative comparison of six-step control, sinusoidal control, and FOC that considers both control performance and implementation requirements under common evaluation conditions. To address this gap, a common simulation framework, a custom motor-control platform, and an experimental investigation were pursued. These objectives were achieved to different extents.

9.1.1. Achievement of the Research Objectives

The objective of developing a common simulation framework was achieved. Six-step control, sinusoidal control, and FOC were implemented using the same equation-based motor and inverter models, parameter set, operating conditions, tuning objectives, and evaluation metrics. This enabled the investigated implementations to be compared without introducing strategy-specific simulation environments or hardware constraints.

The framework distinguished the strategies according to current utilization, speed variation, dynamic tracking, torque variation, and operating range. The relevance of current-vector utilization was particularly evident at the high-current operating point, where FOC produced approximately 0.682 N m, compared with approximately 0.218 N m for six-step

control and 0.220 N m for sinusoidal control at a comparable current-vector magnitude. During the simulated speed ramp, the mean tracking error was approximately 0.036 % for FOC, compared with 0.645 % and 0.616 % for six-step and sinusoidal control, respectively. The developed framework therefore provides a quantitative basis for identifying the principal performance trade-offs between the investigated implementations.

The objective of experimentally examining the simulation findings was partially achieved. Six-step control and FOC were evaluated on a common substitute motor-drive platform, and their relative behavior agreed with the principal simulation tendencies. At the low-speed operating point, the measured relative speed variation was approximately 12.48 % for six-step control and 0.76 % for FOC. At the high-speed operating point, it was approximately 4.67 % and 0.41 %, respectively. During the experimental speed ramp, the settling times were approximately 7.20 s for six-step control and 0.444 s for FOC, with the difference being dominated by the persistent variation of the six-step response.

These measurements support the relative distinction identified in simulation, but they do not constitute a quantitative validation of the simulation models. The experimental system used sensorless feedback, different controller tuning, lower electrical limits, and a switched embedded implementation. Furthermore, the estimated speed served as both controller feedback and evaluation quantity. The measured differences therefore characterize the complete experimental implementations and cannot be attributed exclusively to the underlying control algorithms or physical shaft-speed variation.

The intended three-strategy experimental comparison was not achieved. Sinusoidal control could not be evaluated because no suitable implementation was available within the employed MCSDK. Statements regarding sinusoidal control therefore remain limited to the investigated voltage-based simulation implementation and cannot be transferred directly to alternative sinusoidal controllers employing phase advance, current feedback, or operating-point-dependent compensation.

The objective of assessing implementation effort was achieved for the embedded implementations of six-step control and FOC. The measured processor utilization was approximately 38 % for FOC at both operating points, compared with 0.63 % at low speed and 2.28 % at high speed for six-step control. These values quantify the substantial difference in computational demand on the substitute platform, although their absolute magnitudes remain specific to the employed processor, software implementation, compiler configuration, and measurement method.

The acoustic investigation did not establish a consistent difference in mean sound-pressure level. However, the frequency-domain analysis indicated different spectral compositions. Six-step control produced more pronounced discrete components, particularly in the higher-frequency range, whereas the spectral energy under FOC was distributed more broadly. This provides a possible explanation for the subjective perception of FOC as less disturbing despite similar mean levels. Because the microphone was not calibrated and no psychoacoustic metrics were evaluated, the observation remains indicative and does not establish an acoustic ranking.

The objective of developing the custom motor-control hardware was partially achieved. The extension and sensor boards were designed, reviewed, manufactured, and assembled according to the derived requirements. However, their power supplies, measurement channels, inverter, communication interfaces, protection functions, thermal behavior, and motor-control operation were not commissioned. The resulting hardware therefore constitutes an engineering outcome of the thesis, but its functionality and suitability as a common experimental platform have not yet been demonstrated.

9.1.2. Answer to the Research Question

For the investigated implementations and operating conditions, the comparison shows that six-step control, voltage-based sinusoidal control, and FOC primarily differ in the balance between control capability and implementation requirements.

Six-step control imposed the lowest computational demand but retained commutation-related variation, particularly at low speed. The investigated sinusoidal controller provided smooth excitation within the averaged simulation model, but its lack of current-vector regulation limited its torque utilization. FOC achieved the highest current utilization and the most consistent dynamic behavior, while requiring phase-current measurement, motor parameterization, current regulation, coordinate transformations, modulation, and substantially greater processing effort. The practical significance of the additional software complexity may nevertheless decrease as established implementation frameworks such as the MCSDK become more widely available.

All three strategies achieved the commanded mean speeds in the closed-loop simulations. Mean speed tracking alone is therefore insufficient for selecting a control strategy. A meaningful comparison must additionally consider current utilization, instantaneous speed

variation, transient tracking, operating range, sensing requirements, and computational demand.

No strategy can consequently be identified as universally preferable. Six-step control remains suitable where implementation simplicity and low processing demand have priority over smooth operation. The investigated sinusoidal approach represents an intermediate level of control and implementation complexity, but did not achieve the current utilization of FOC. Field-oriented control is appropriate where accurate torque regulation, smooth speed behavior, and dynamic tracking justify the additional sensing, parameterization, tuning, and processing requirements.

This answer is limited to the investigated configurations. The tests were conducted without externally applied load torque, the simulation used an idealized averaged inverter and simplified motor model, and the experimental comparison included only six-step control and FOC. The quantitative results therefore characterize the implemented motor-drive systems rather than universal limits of the control principles.

9.1.3. Closure of the Research Gap

The identified research gap was partially closed. The principal contribution of the thesis is a common quantitative methodology for comparing the three investigated control strategies under consistent models, operating points, tuning objectives, and performance indicators. By jointly considering electrical utilization, mechanical response, computational demand, sensing, and implementation requirements, the methodology provides a broader basis for strategy selection than assessments based on individual criteria such as mean speed tracking or current-waveform quality.

Experimental evidence was obtained for the principal comparative tendencies between six-step control and FOC. However, the absence of an experimental sinusoidal controller prevented the complete three-strategy comparison from being verified on a common physical platform. Similarly, although the custom hardware required for such a comparison was developed, its uncommissioned state prevented its use for validating the methodology under controlled sensing, processing, and modulation conditions.

The methodological part of the research gap was therefore addressed more comprehensively than its experimental and hardware-related parts. The thesis establishes a consistent com-

parative basis and quantifies the principal trade-offs of the investigated configurations, but it does not provide a universally transferable ranking, a fully validated custom platform, or a complete experimental comparison of all three strategies.

9.2. Outlook

Future work should primarily address the objectives that were achieved only partially. The following activities are ordered according to their relevance for completing and extending the investigation.

1. **Commission the custom hardware.**
   The individual power supplies, communication interfaces, measurement channels, gate-driver outputs, inverter stages, and protection functions should first be validated using current-limited laboratory supplies and non-inductive test loads. The complete inverter should then be commissioned under gradually increasing voltage and current. Thermal measurements are required to establish the continuous operating range and verify the assumptions made during the hardware design.
2. **Complete the experimental comparison.**
   Six-step control, sinusoidal control, and FOC should be implemented on the custom platform using a common processing, sensing, and modulation environment. In particular, a sensored sinusoidal controller should be developed independently of the MCSDK. This would complete the intended three-strategy comparison and reduce differences caused by strategy-specific library implementations.
3. **Introduce independent mechanical measurements.**
   An encoder or comparable external position sensor should provide a common shaft-position and speed reference for all strategies. This would separate physical speed variation from sensorless-estimator behavior and permit the estimators and controllers to be evaluated independently.
4. **Extend the investigation to loaded operation.**
   A dynamometer or controllable load machine should be used to define operating points by torque, mechanical output power, and representative duty cycle. This would enable the investigation of disturbance rejection, torque tracking, loaded efficiency, continuous thermal operation, regeneration, and startup under load. A common mechanical output requirement would also provide a more meaningful ba-

sis for comparing electrical input power and efficiency.

5. **Increase the fidelity of the simulation model.**
   A switched-inverter model should incorporate PWM, dead time, semiconductor voltage drops, switching and conduction losses, current-sampling timing, measurement noise, and discrete controller execution. This would permit a more realistic evaluation of harmonic distortion, torque ripple, current reconstruction, and operation near the voltage limit. An implementation in Simscape and comparison with real performance would be a way to achieve this.
6. **Refine the motor model and parameter identification.**
   The observed rotor-position dependency of the measured inductance should be investigated to determine whether separate direct- and quadrature-axis inductances are required. The physical back-EMF waveform should also be measured to assess the validity of the sinusoidal assumption and its influence on the six-step comparison. Temperature-dependent resistance, magnetic saturation, cogging torque, iron losses, and nonlinear friction should be incorporated where experimentally relevant.
7. **Separate controller structure from controller optimization.**
   The strategies should first be compared using common design objectives and subsequently optimized individually for selected application requirements. This would distinguish performance under comparable tuning effort from the maximum achievable performance of each implementation. Saturation, anti-windup behavior, update frequency, current limiting, and phase advance should be included in this investigation.
8. **Perform a controlled acoustic assessment.**
   Calibrated sound-pressure and vibration measurements should be conducted under standardized boundary conditions. In addition to mean level and spectral content, psychoacoustic metrics such as loudness, sharpness, tonality, and roughness should be considered. These quantities could determine whether the observed spectral differences explain the subjective perception of the strategies.
9. **Investigate extended operating functions.**
   Following validation of the platform, the comparison could be extended to field weakening, regenerative operation, low-speed sensorless control, startup under load, and transitions between control strategies. These extensions would support the application of the comparative framework to practical motor-drive requirements.

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List of Abbreviations

AC	alternating current
ADC	analog-to-digital converter
CAN FD	controller area network flexible data-rate
DC	direct current
EMF	electromotive force
FOC	field-oriented control
MCSDK	Motor Control Software Development Kit
MISO	master in, slave out
MOSI	master out, slave in
PI	proportional-integral
PMSM	permanent-magnet synchronous motor
PWM	pulse-width modulation
SPI	serial peripheral interface
SPWM	sinusoidal pulse-width modulation
SVPWM	space-vector pulse-width modulation

Nomenclature

a ..... Phase-a index

α ..... First axis of the stationary two-axis reference frame

b ..... Phase-b index

β ..... Second axis of the stationary two-axis reference frame

B ..... Viscous friction coefficient

c ..... Phase-c index

d ..... PWM duty cycle or direct-axis index, depending on context

$\Delta\omega_m^*$ ..... Commanded change in mechanical angular velocity

d^* ..... PWM duty-cycle reference

d_x ..... PWM duty cycle of phase x

e ..... Controller input error

$e_{\text{rel,rise}}$ ..... Average relative tracking error during the reference rise

η ..... Instantaneous or mean drive efficiency

$e_{\omega,\text{rel}}$ ..... Relative steady-state speed error

$f_{\text{BW},i}$ ..... Current-loop bandwidth

$f_{\text{BW},\omega}$ ..... Speed-loop bandwidth

$f_{\text{exec},\omega}$ ..... Speed-controller execution frequency

f_{PWM} ..... PWM switching frequency

$f_{\text{PWM,min}}$ ..... Minimum PWM frequency

i_a ..... Phase-a current

$\mathbf{i}_{abc}$ ..... Three-phase stator-current vector

i_α ..... Stationary α-axis stator-current component

i_b ..... Phase-b current

i_β ..... Stationary β-axis stator-current component

i_c ..... Phase-c current

I ..... Current magnitude

i_d ..... Direct-axis stator-current component

I_{DC} ..... DC-link current

i_d^* ..... Direct-axis current reference

I_k ..... Magnitude of the k-th current harmonic

i_{mag} ..... Stator-current-vector magnitude

I_{max} ..... Maximum current magnitude

$\inf(\cdot)$ ..... Infimum operator

I_1 ..... Magnitude of the fundamental current component

i_q ..... Quadrature-axis stator-current component

i_q^* ..... Quadrature-axis current reference

$I_{x,\mathrm{RMS}}$ ..... RMS current of phase x

J ..... Rotor moment of inertia

k ..... Harmonic-order, switching-sector, or sample index

K_{aw} ..... Anti-windup gain

K_e ..... Back-EMF constant

K_i ..... Integral controller gain

K_p ..... Proportional controller gain

K_t ..... Motor torque constant

K_V ..... Motor speed constant

L_d ..... Direct-axis stator inductance

L_{LL} ..... Mean line-to-line inductance

$L_{\mathrm{LL},ab}$ ..... Line-to-line inductance between phases a and b

$L_{\mathrm{LL},ac}$ ..... Line-to-line inductance between phases a and c

$L_{\mathrm{LL},bc}$ ..... Line-to-line inductance between phases b and c

L_p ..... Measured sound-pressure level

L_q ..... Quadrature-axis stator inductance

L_s ..... Stator phase inductance

m ..... Estimated rotor mass

$\max(\cdot)$ ..... Maximum-value operator

$\min(\cdot)$ ..... Minimum-value operator

m_{motor} ..... Total motor mass

M_p ..... Relative speed overshoot

n ..... Sample index

N_h ..... Highest harmonic order considered in the THD calculation

N_r ..... Number of samples within the reference-ramp interval

N_s ..... Number of samples within a simulation evaluation interval

ω_e ..... Electrical angular velocity

ω_m ..... Mechanical angular velocity

$\omega_{m,\max}$ ..... Maximum achieved mechanical angular velocity

$\omega_{m,n}$ ..... Sampled mechanical angular velocity

ω_m^* ..... Mechanical angular-velocity reference

$\omega_{m,n}^*$ ..... Sampled mechanical angular-velocity reference

$\omega_{m,0}$ ..... Mechanical angular velocity immediately before a reference change

$\omega(t)$ ..... Time-dependent mechanical angular velocity

ω_0 ..... Initial mechanical angular velocity

p ..... Number of motor pole pairs

P_{el} ..... Electrical input power

φ_m ..... Controller phase margin

P_m ..... Mechanical output power

$P_{\text{sh},x}$ ..... Power dissipation of the shunt resistor in phase x

$\boldsymbol{\psi}_{abc}$ ..... Three-phase stator flux-linkage vector

ψ_{PM} ..... Permanent-magnet flux linkage

q ..... Quadrature-axis or phase index, depending on context

ρ_{CPU} ..... Processor load

$\rho_{\text{CPU},n}$ ..... Processor load during control cycle n

R_{LL} ..... Line-to-line winding resistance

r_1 ..... Inner rotor radius or stator radius

R_s ..... Stator phase resistance

R_{sh} ..... Shunt resistance

r_2 ..... Outer rotor radius or motor radius

s ..... Slope of the fitted logarithmic angular-velocity curve

$\mathrm{sat}(\cdot)$ ..... Saturation function

$\sigma_{\omega,\mathrm{rel}}$ ..... Relative mechanical-speed standard deviation

t ..... Time

τ ..... Dummy time variable used in the settling-time definition

τ_T ..... Normalized electromagnetic torque ripple

T_{ctrl} ..... Invocation period of the motor-control task

T_e ..... Electromagnetic torque

$T_{e,\max}$ ..... Maximum instantaneous electromagnetic torque

$T_{e,\min}$ ..... Minimum instantaneous electromagnetic torque

t_{end} ..... End of the evaluated time interval

$t_{\mathrm{exec},n}$ ..... Motor-control task execution time during control cycle n

$T_{\mathrm{exec},\omega}$ ..... Speed-controller execution period

THD_i ..... Phase-current total harmonic distortion

θ_e ..... Electrical rotor angle

θ_m ..... Mechanical rotor angle

T_L ..... Externally applied load torque

TPA ..... Mean electromagnetic torque per ampere

t_r ..... Rise time

t_s ..... Settling time

$t_{\omega,90\%}$ ..... First instant at which the speed reaches 90% of the commanded change

$t_{\omega,10\%}$ ..... First instant at which the speed reaches 10% of the commanded change

t_0 ..... Starting instant of a reference transition

u ..... Voltage

u_a ..... Phase-a voltage

u_{a0} ..... Phase-a switch-node voltage

$\mathbf{u}_{abc}$ ..... Three-phase stator-voltage vector

u_α ..... Stationary α-axis voltage component

u_a^* ..... Phase-a voltage reference

u_b ..... Phase-b voltage

u_{b0} ..... Phase-b switch-node voltage

u_β ..... Stationary β-axis voltage component

u_b^* ..... Phase-b voltage reference

u_c ..... Phase-c voltage

u_{c0} ..... Phase-c switch-node voltage

u_c^* ..... Phase-c voltage reference

u_d ..... Direct-axis stator-voltage component

U_{DC} ..... DC-link voltage

$u_{d,\mathrm{comp}}^*$ ..... Compensated direct-axis voltage reference

$u_{d,\mathrm{ctrl}}$ ..... Direct-axis current-controller output voltage

$u_{d,\mathrm{ff}}$ ..... Direct-axis feedforward-decoupling voltage

u_{float} ..... Approximated voltage of the floating phase

u_i ..... Mean current-vector utilization

$U_{\mathrm{max,SPWM}}^*$ ..... Maximum phase-voltage reference amplitude for SPWM

$U_{\mathrm{max,SVPWM}}^*$ ..... Maximum fundamental phase-voltage reference amplitude for SVPWM

u_{n0} ..... Motor neutral-point potential

u_{pq} ..... Average line-to-line voltage between active phases p and q

u_q ..... Quadrature-axis stator-voltage component

$u_{q,\mathrm{comp}}^*$ ..... Compensated quadrature-axis voltage reference

$u_{q,\mathrm{ctrl}}$ ..... Quadrature-axis current-controller output voltage

$u_{q,\mathrm{ff}}$ ..... Quadrature-axis feedforward-decoupling voltage

u_{ref} ..... Current-sense amplifier reference voltage

U^* ..... Peak phase-voltage reference amplitude

u_x ..... Phase-to-neutral voltage of phase x

u_{x0} ..... Switch-node voltage of phase x

$u_{x,\mathrm{mod}}^*$ ..... Modified phase-voltage reference after common-mode injection

u_x^* ..... Voltage reference of phase x

u_0 ..... Injected common-mode voltage

V_5 ..... Fifth active inverter switching vector

V_4 ..... Fourth active inverter switching vector

$V_{\mathrm{LL}}(t)$ ..... Measured instantaneous line-to-line voltage

V_1 ..... First active inverter switching vector

V_7 ..... Second zero inverter switching vector

V_6 ..... Sixth active inverter switching vector

V_3 ..... Third active inverter switching vector

V_2 ..... Second active inverter switching vector

V_0 ..... First zero inverter switching vector

x ..... Generic sampled quantity or phase index

$\overline{x}$ ..... Arithmetic mean of a sampled quantity

x_I ..... Integrator state

x_{lim} ..... Quantity limiting maximum-speed operation

x_n ..... Value of a sampled quantity at sample n

y ..... Generic controller output

y_{lim} ..... Constrained controller output

y_{max} ..... Upper controller-output limit

y_{min} ..... Lower controller-output limit

y^* ..... Unconstrained controller output

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A. Note on the Use of Artificial Intelligence

Within the scope of this thesis, Artificial Intelligence (AI)-based tools were used. Table A.1 provides an overview of the tools used and their respective purposes.

Table A.1.: List of AI-based tools used

Tool	**Description of Use**
Perplexity	Preliminary research and identification of potentially relevant technical sources and terminology. Identified information and sources were subsequently reviewed before use.
Microsoft Copilot	Language revision, proofreading, improvement of clarity and academic style, restructuring and shortening of individual passages, assistance with LaTeX formulations, and review internal consistency.

B. Pilotix Motor Parameterization

Pilotix provides the motor data for the *Pilotix 6210 210KV* summarized in Table B.1. This information is used to determine the required model parameters either analytically or experimentally.

Table B.1.: Manufacturer-specified motor parameters.

Parameter	**Symbol**	**Value**
Motor speed constant	K_V	$210\,\mathrm{rpm/V}$
Number of pole pairs	p	14
Line-to-line resistance	R_{LL}	$42.5\,\mathrm{m\Omega}$
Stator radius	r_1	$31\,\mathrm{mm}$
Motor radius	r_2	$34.4\,\mathrm{mm}$
Total motor mass	m_{motor}	$331\,\mathrm{g}$

For a symmetrical, wye-connected machine, the measured line-to-line resistance corresponds to the series connection of two phase resistances. The stator phase resistance is therefore given by

$$R_s = \frac{R_{\mathrm{LL}}}{2} = \frac{42.5\,\mathrm{m\Omega}}{2} = 21.25\,\mathrm{m\Omega}. \tag{B.1}$$

The torque constant K_t is derived from the motor speed constant K_V. Under the adopted SI-unit convention, the torque constant is numerically equal to the back-EMF constant K_e. Converting K_V from rpm/V to rad/(s V) yields

$$K_t = K_e = \frac{60}{2\pi K_V} = \frac{60}{2\pi \cdot 210\,\mathrm{rpm/V}} = 0.04547\,\mathrm{N\,m/A}. \tag{B.2}$$

The permanent-magnet flux linkage is subsequently obtained by rearranging the torque equation, described in Equation 2.11:

$$\psi_{\mathrm{PM}} = \frac{K_t}{1.5p} = \frac{0.04547\,\mathrm{N\,m/A}}{1.5 \cdot 14} = 2.165\,\mathrm{mWb}. \tag{B.3}$$

The rotor moment of inertia J is estimated from the motor geometry and mass. Because the rotor mass is not specified by the manufacturer, it is assumed to constitute 36 % of the total motor mass, based on a conservative upper-bound assumption of similar heavy-lift drone outrunner motors [5]. This assumption results in

$$m = 0.36 m_{\text{motor}} \approx 0.36 \cdot 331\,\text{g} \approx 120\,\text{g}.$$

The rotor mass is concentrated primarily in the permanent magnets and the surrounding rotor structure. Accordingly, the rotor is approximated as a hollow cylinder with an inner radius equal to the stator radius r_1 and an outer radius equal to the motor radius r_2. Its moment of inertia is therefore estimated as

$$\begin{aligned} J &= \frac{1}{2} m \left(r_1^2 + r_2^2\right) \\ &= \frac{1}{2} \cdot 0.12\,\text{kg} \left[\left(31 \times 10^{-3}\,\text{m}\right)^2 + \left(34.4 \times 10^{-3}\,\text{m}\right)^2\right] \\ &= 1.29 \times 10^{-4}\,\text{kg}\,\text{m}^2. \end{aligned} \tag{B.4}$$

The stator inductance L_s is determined by measuring the line-to-line inductance L_{LL} using an LCR meter. For a symmetrical, wye-connected three-phase machine, a line-to-line measurement represents the series combination of two phase inductances. Neglecting mutual inductance, the stator phase inductance can therefore be approximated as

$$L_s = \frac{L_{\text{LL}}}{2}. \tag{B.5}$$

To assess variations among the phase pairs, the line-to-line inductance is measured for all three terminal combinations. The measured values are

$$\begin{aligned} L_{\text{LL},ab} &= 40.27\,\mu\text{H}, \\ L_{\text{LL},ac} &= 31.78\,\mu\text{H}, \\ L_{\text{LL},bc} &= 32.80\,\mu\text{H}. \end{aligned}$$

The measurements exhibit a pronounced variation, with one terminal pair yielding a higher inductance than the remaining two. When the rotor is manually rotated, the terminal pair associated with the highest measured inductance changes sequentially. This observation indicates that the measured inductance depends on the rotor position.

To obtain a single representative value for the motor model, the three measured line-to-line

inductances are averaged:

$$\begin{aligned}\overline{L}_{\mathrm{LL}} &= \frac{L_{\mathrm{LL},ab} + L_{\mathrm{LL},ac} + L_{\mathrm{LL},bc}}{3} \\ &= \frac{40.27 + 31.78 + 32.80}{3}\,\mu\mathrm{H} \\ &= 34.95\,\mu\mathrm{H}.\end{aligned} \tag{B.6}$$

The corresponding stator phase inductance is

$$L_s = \frac{\overline{L}_{\mathrm{LL}}}{2} = \frac{34.95\,\mu\mathrm{H}}{2} = 17.475\,\mu\mathrm{H}. \tag{B.7}$$

Accordingly, $L_s = 17.475\,\mu\mathrm{H}$ is adopted as the representative stator phase inductance in the machine model. The observed rotor-position-dependent inductance variation suggests that the machine may exhibit magnetic saliency, such that the direct- and quadrature-axis inductances differ, that is, $L_d \neq L_q$ [9]. Because the model uses a single stator inductance, it does not represent this distinction. This simplification may contribute to deviations between the simulated and measured motor behavior.

The viscous-friction coefficient B is estimated from the free-deceleration response of the motor. For this purpose, the motor is accelerated and subsequently allowed to coast freely until it comes to rest. During the coast-down phase, the line-to-line back-EMF voltage is measured using an oscilloscope, and the positive and negative voltage extrema are identified. Figure B.1 presents the measured voltage and the detected extrema. The red vertical line indicates the beginning of the deceleration interval used for the parameter estimation.

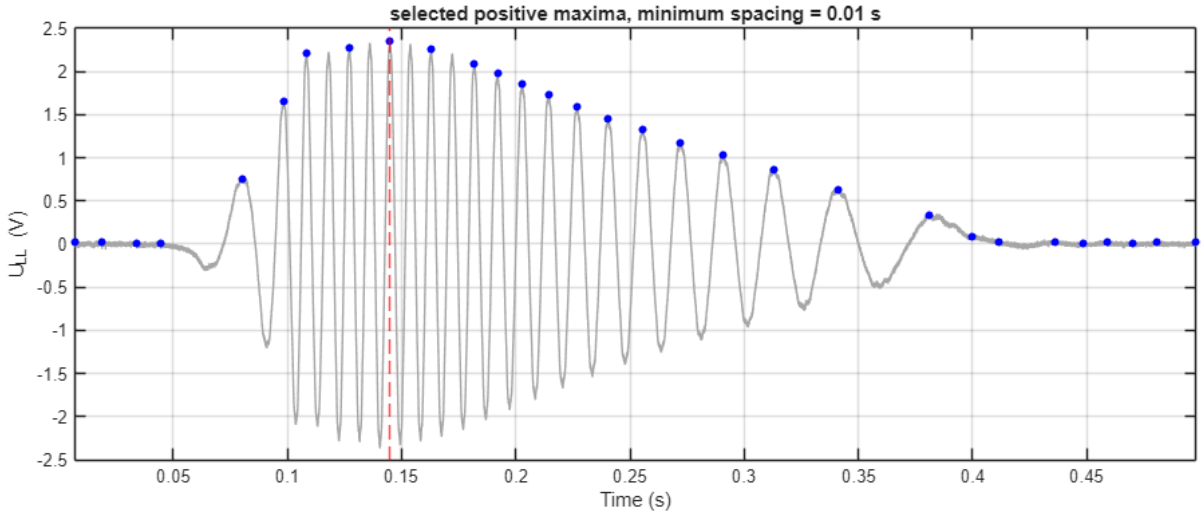


Figure B.1.: Measured line-to-line back-EMF voltage and detected extrema during motor acceleration and free deceleration.

The line-to-line voltage extrema recorded during the deceleration interval are converted into mechanical angular velocity using the motor speed constant:

$$\omega(t) = \frac{2\pi}{60} K_V V_{\mathrm{LL}}(t). \tag{B.8}$$

The natural logarithm of the resulting angular-velocity values is then evaluated. A linear least-squares fit is applied to the logarithmic data, as shown in Figure B.2.

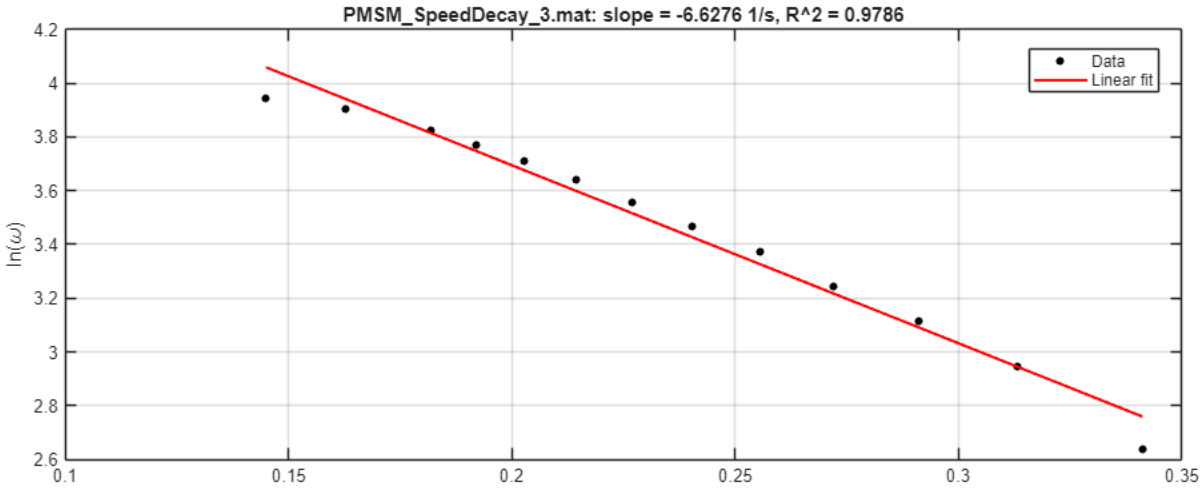


Figure B.2.: Natural logarithm of the angular-velocity estimates obtained from the back-EMF extrema during free deceleration.

Assuming that the mechanical losses can be represented exclusively by viscous friction, the equation of motion during free deceleration is

$$J\frac{\mathrm{d}\omega}{\mathrm{d}t} = -B\omega. \tag{B.9}$$

The corresponding solution is

$$\omega(t) = \omega_0 \exp\left(-\frac{B}{J}t\right). \tag{B.10}$$

Applying the natural logarithm yields

$$\ln\left(\omega(t)\right) = \ln\left(\omega_0\right) - \frac{B}{J}t. \tag{B.11}$$

Consequently, the slope s of the fitted line is

$$s = -\frac{B}{J}, \tag{B.12}$$

and the viscous-friction coefficient is calculated as

$$B = -sJ. \tag{B.13}$$

The coast-down experiment is repeated five times, and the individual estimates are averaged. Combining the mean fitted slope with the previously estimated rotor moment of inertia results in

$$B = 8.61444 \times 10^{-4}\,\mathrm{N\,m\,s/rad}.$$

C. Simulink Models

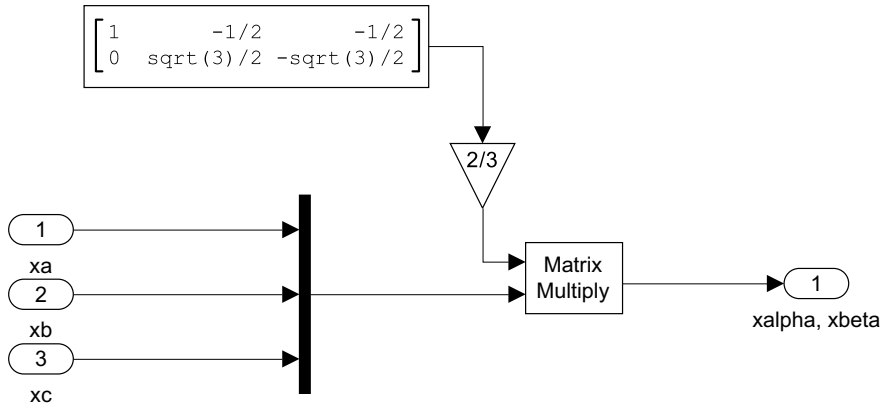


Figure C.1.: Clarke Transformation

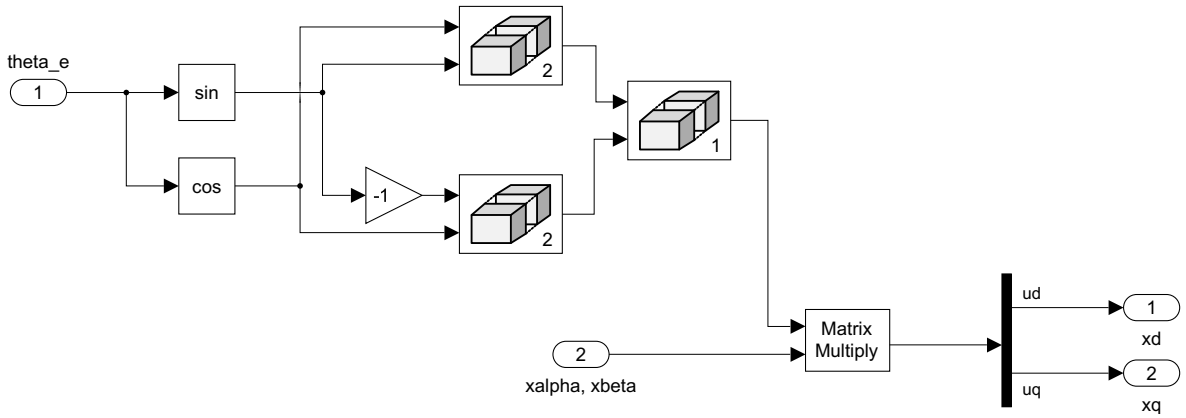


Figure C.2.: Park Transformation

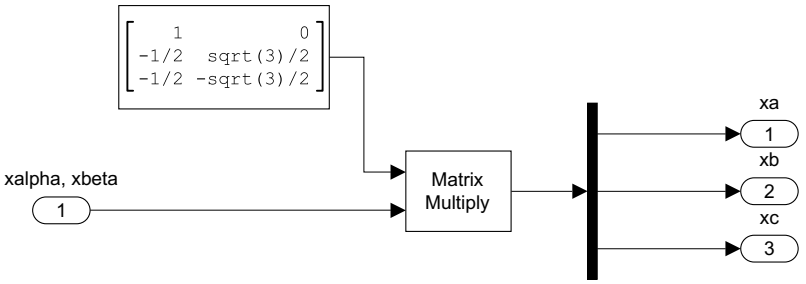


Figure C.3.: Inverse Clarke Transformation

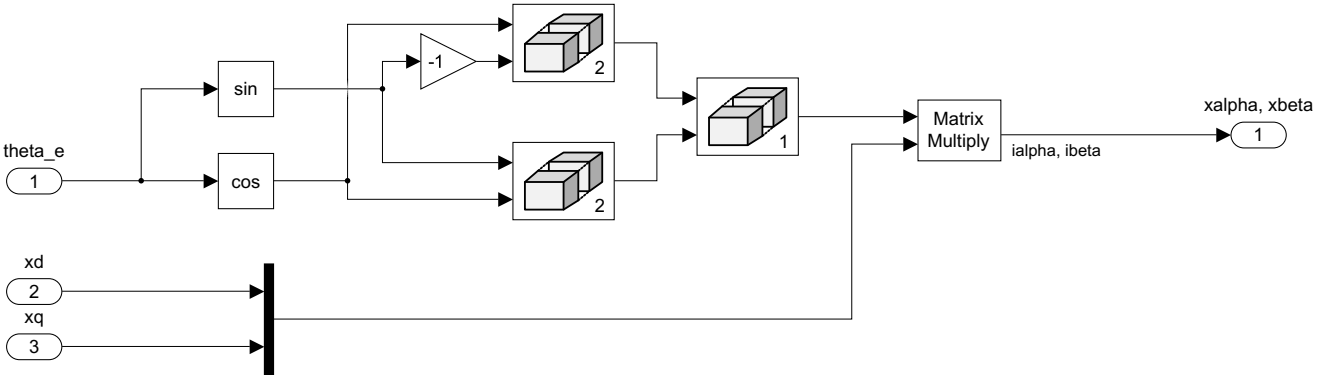


Figure C.4.: Inverse Park Transformation

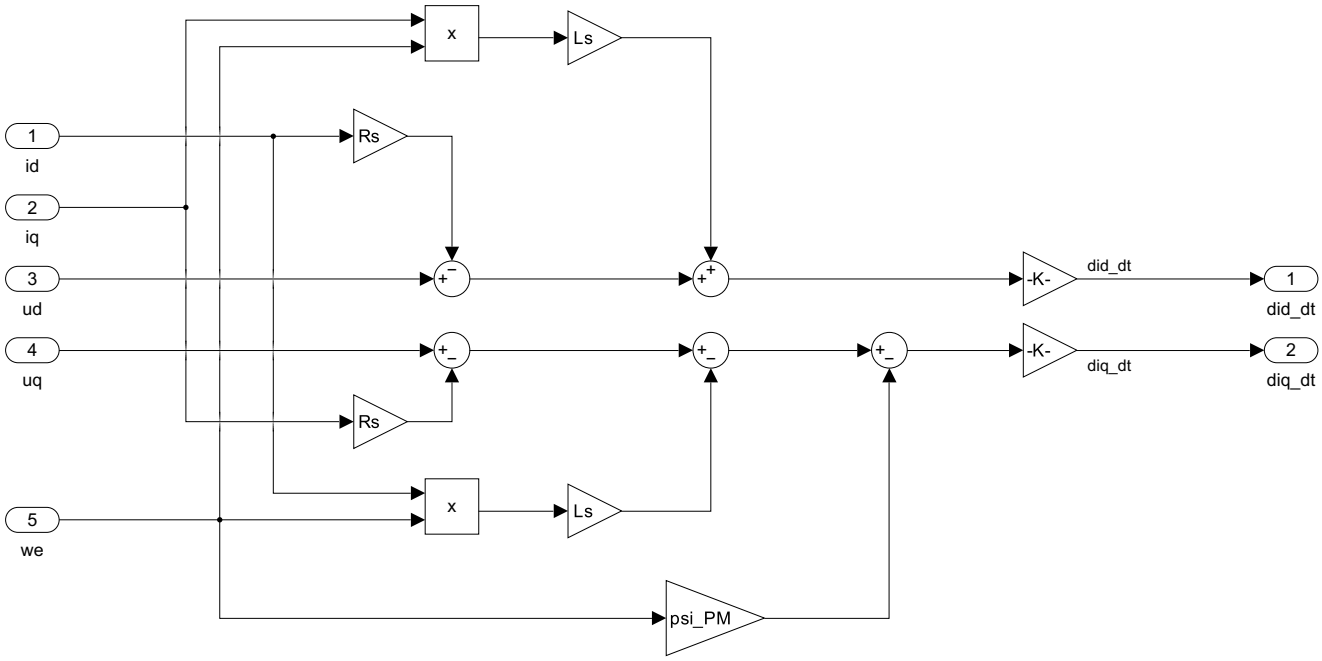


Figure C.5.: PMSM Electrical Dynamics

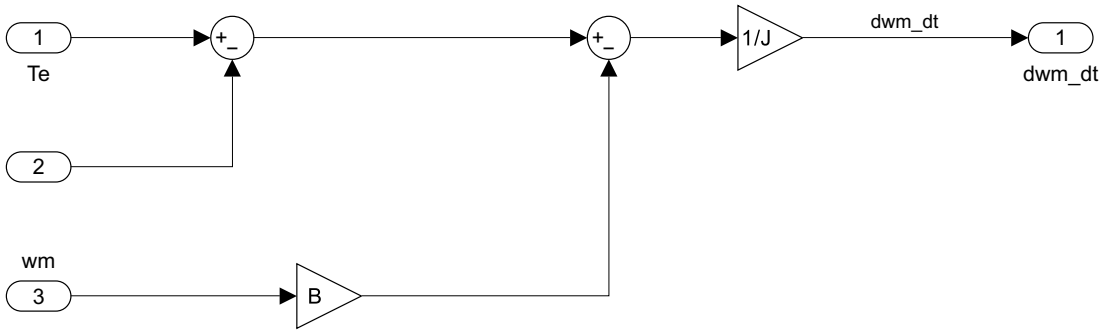


Figure C.6.: PMSM Mechanical Dynamics

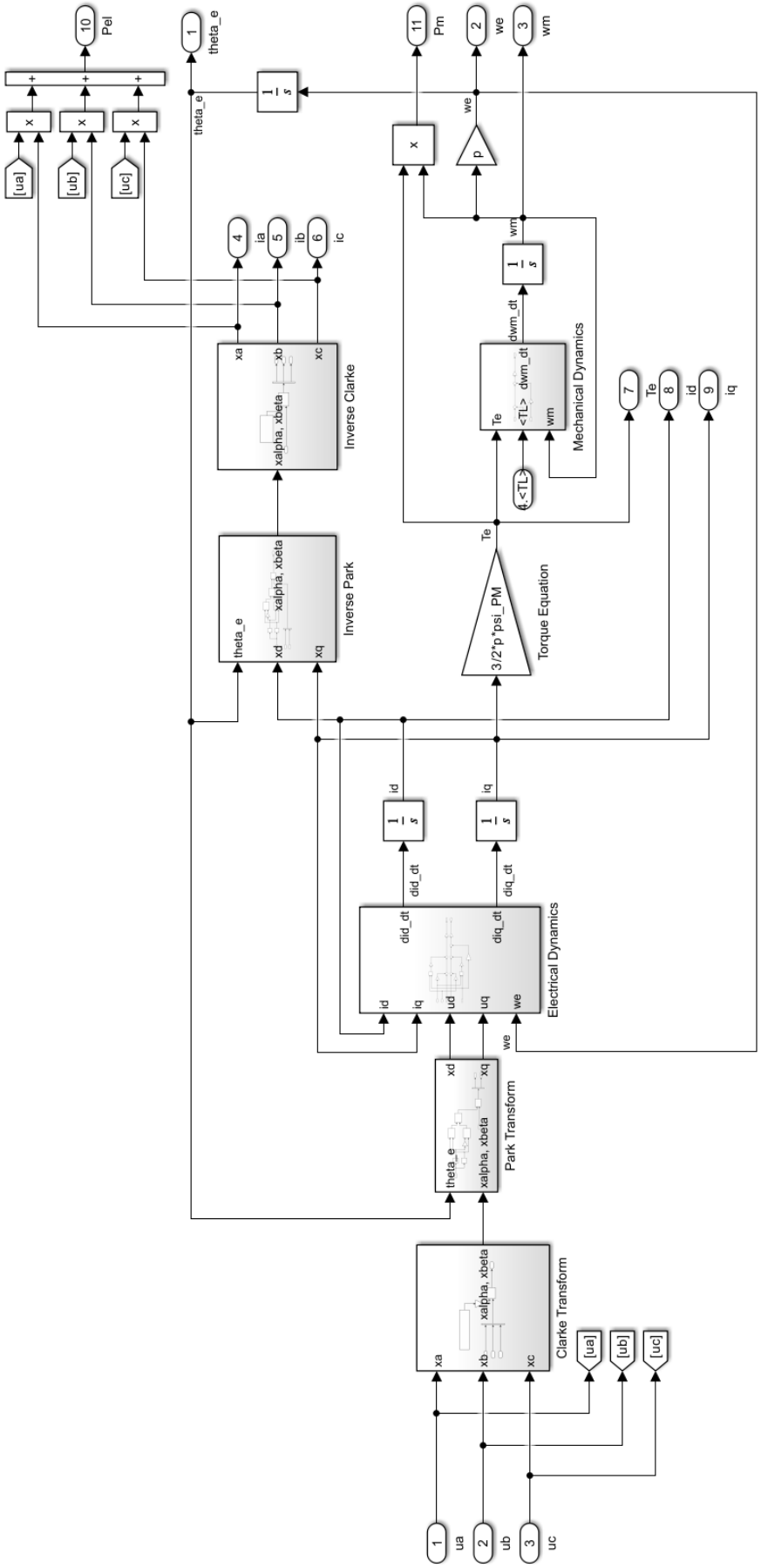


Figure C.7.: Complete PMSM Model

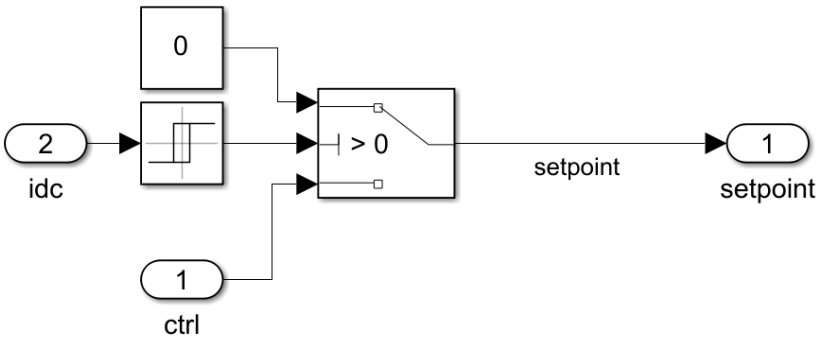


Figure C.8.: PWM Current Limiter

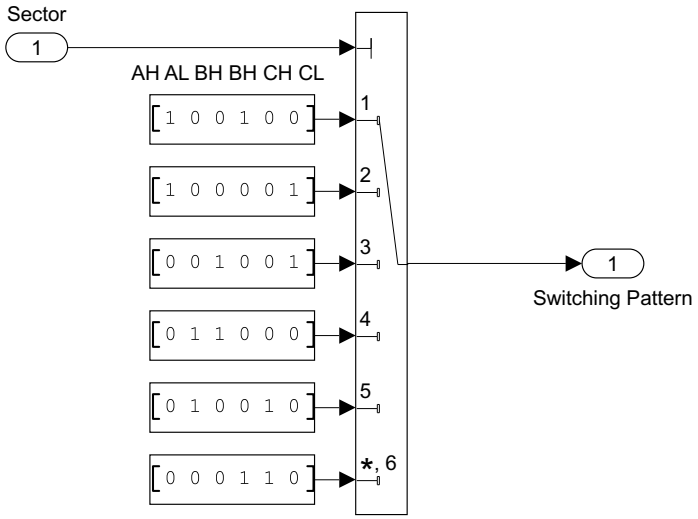


Figure C.9.: Six-Step Commutation Logic

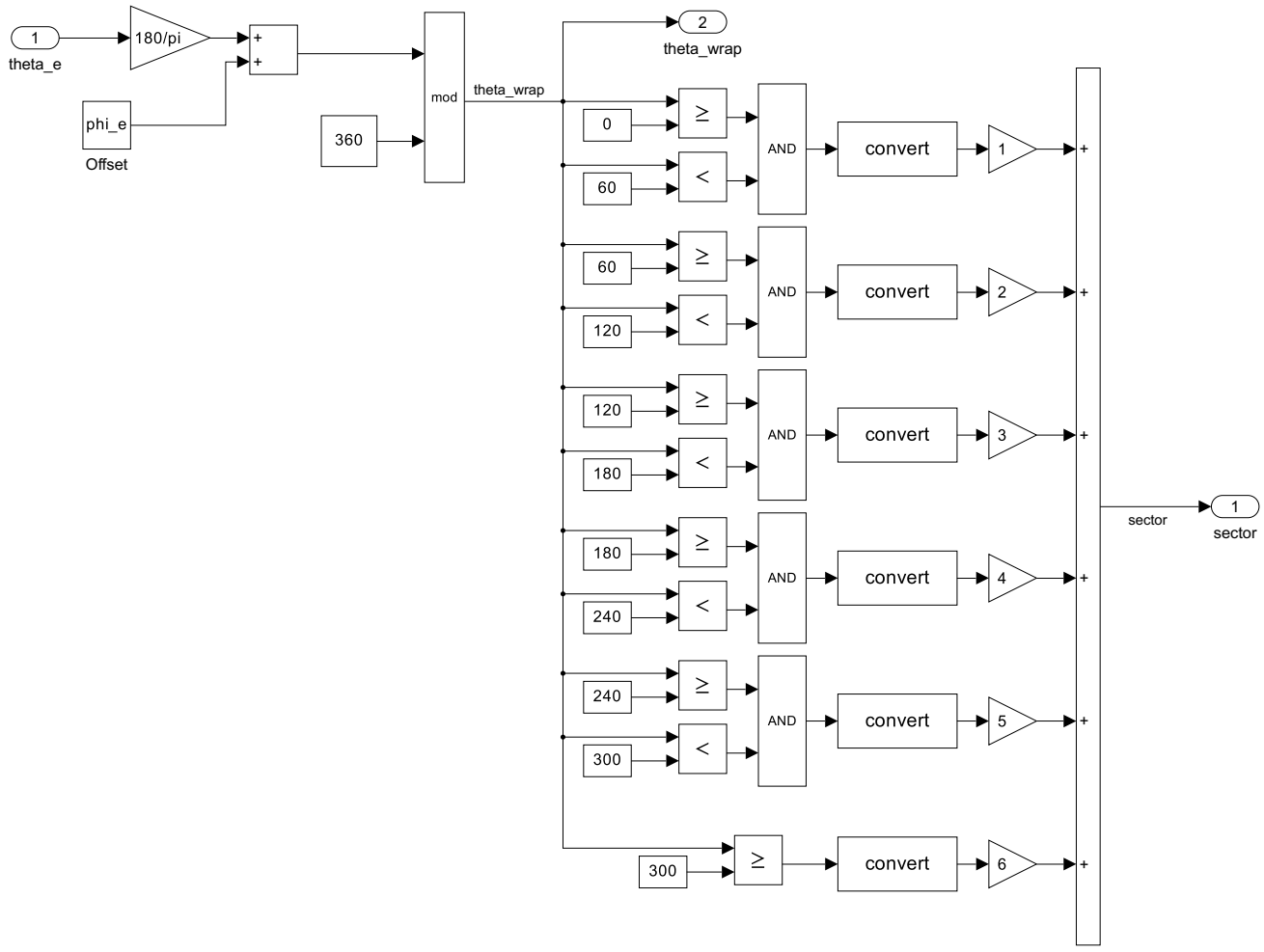


Figure C.10.: Six-Step Commutation Sensor

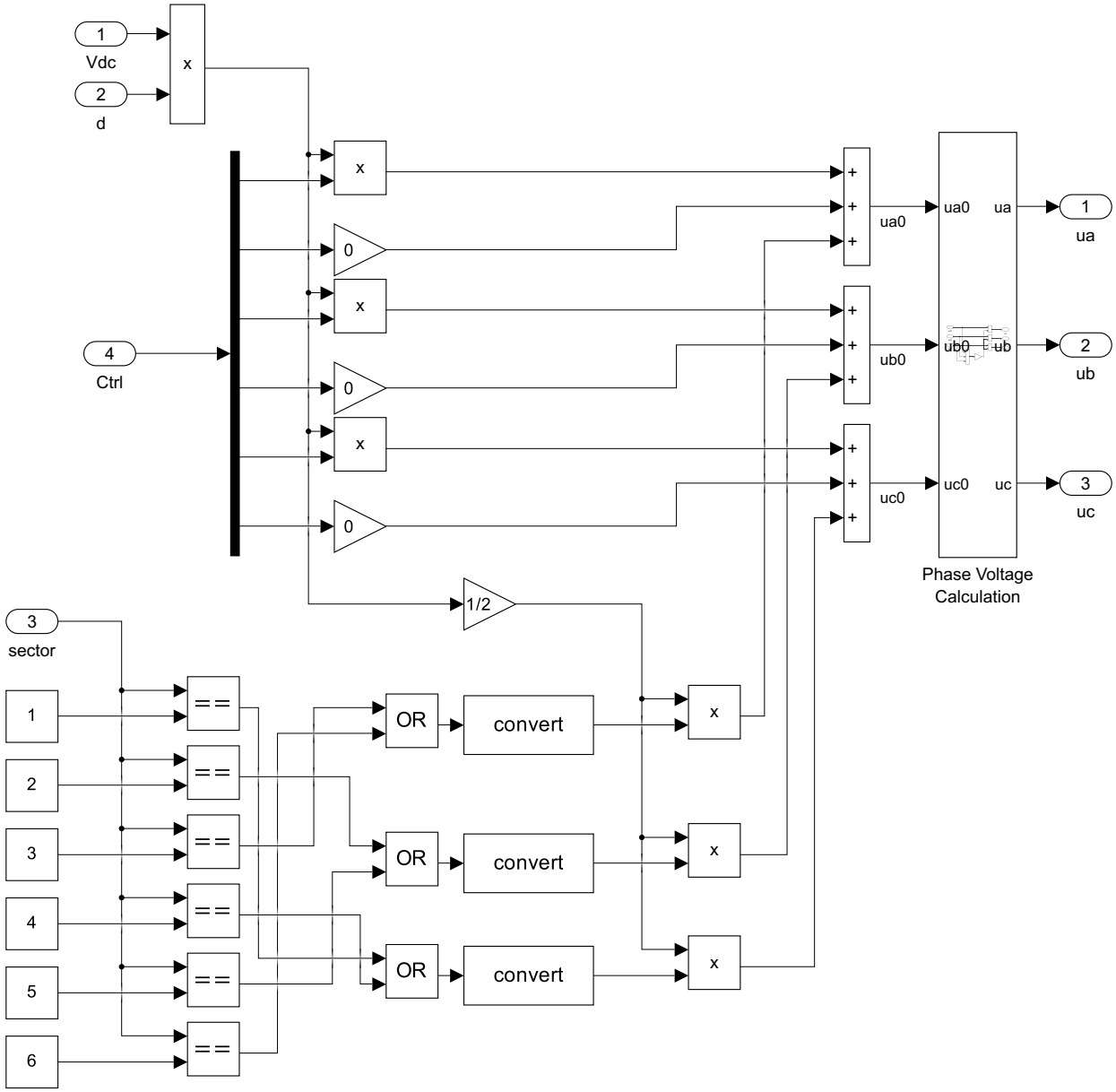


Figure C.11.: Six-Step Trapezoidal Reference Generation

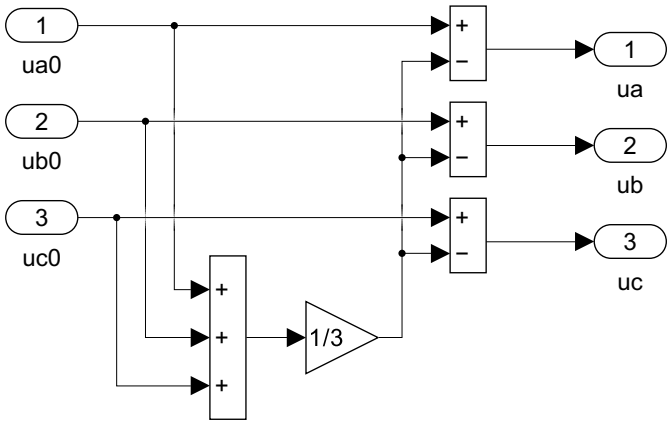


Figure C.12.: Six-Step Phase-Voltage Calculation

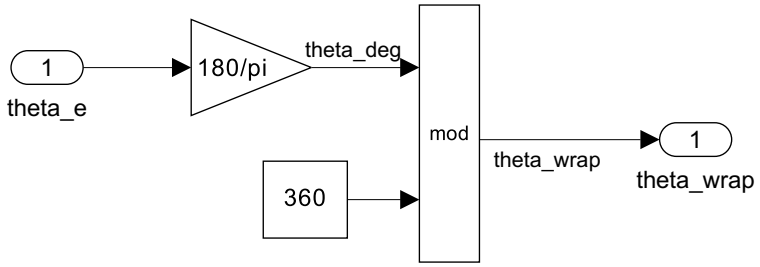


Figure C.13.: Angle Wrapper

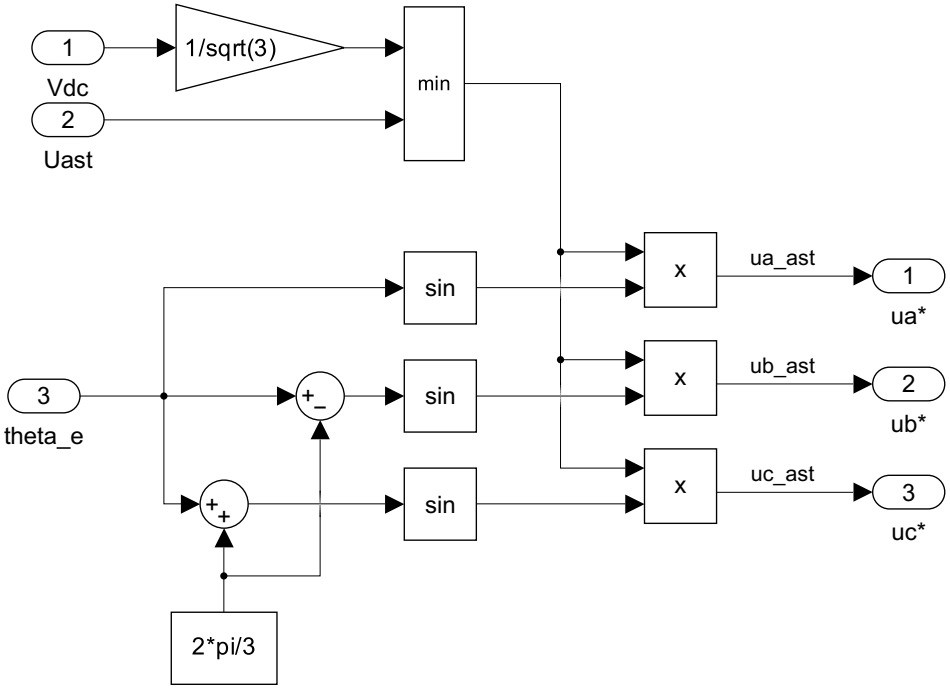


Figure C.14.: Sinusoidal Voltage-Reference Generator

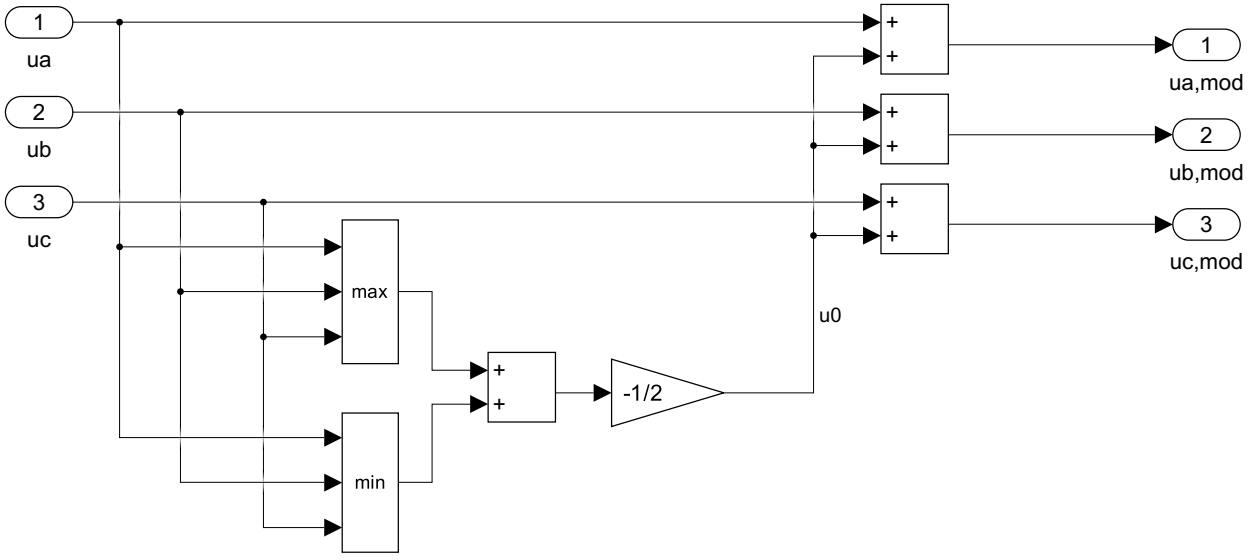


Figure C.15.: Common-Mode Injection

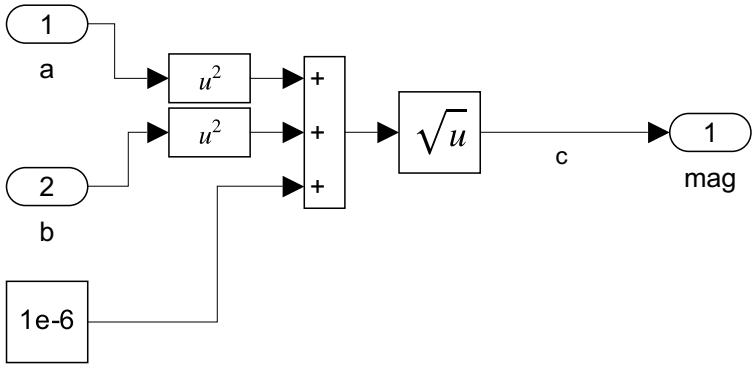


Figure C.16.: Vector-Magnitude Calculator

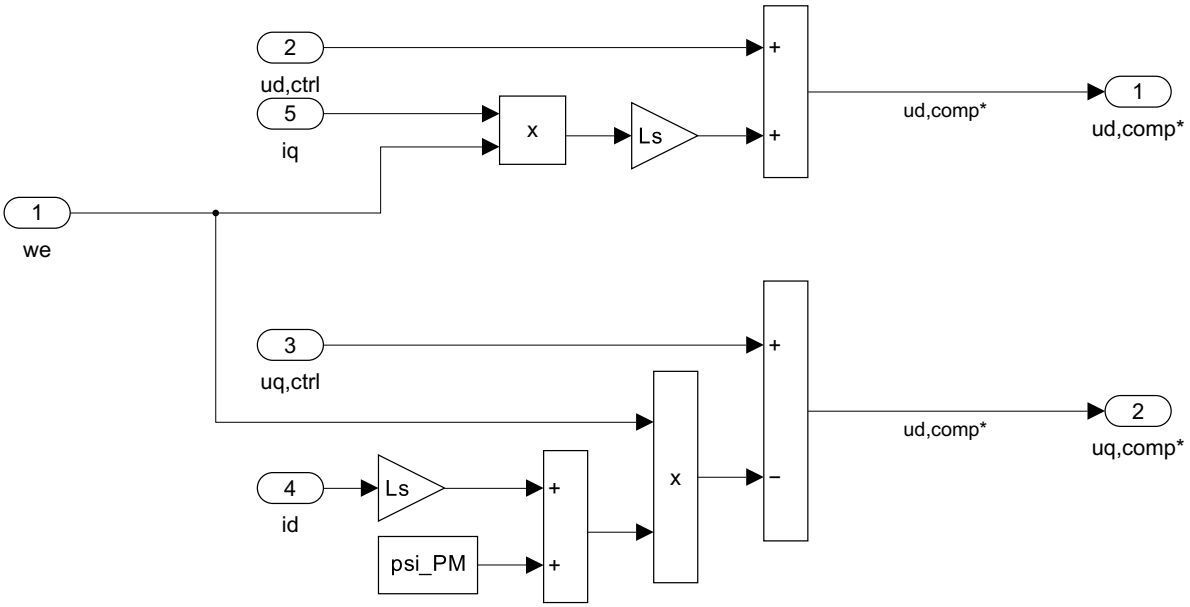


Figure C.17.: FOC Decoupling Network

D. Hardware Details

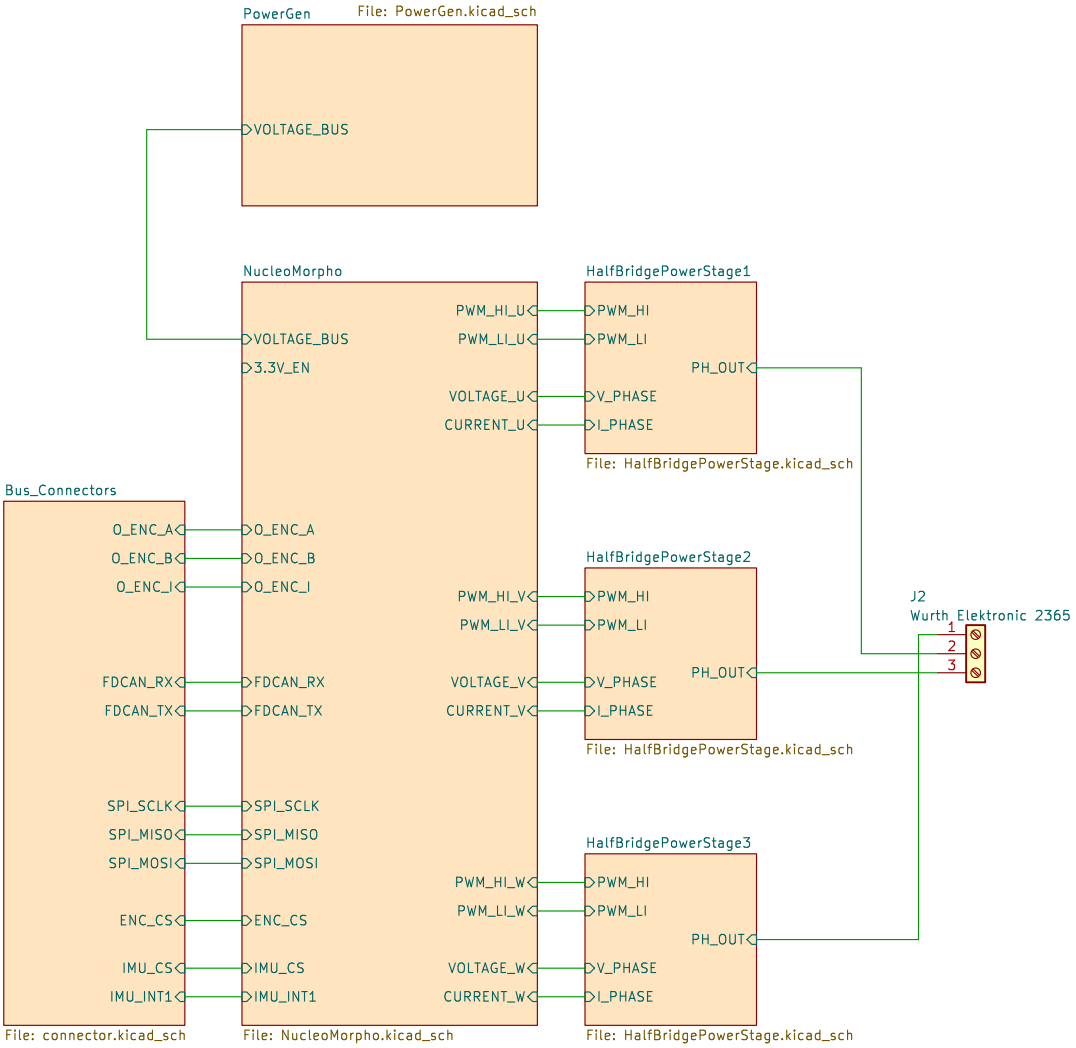


Figure D.1.: Control board circuit overview

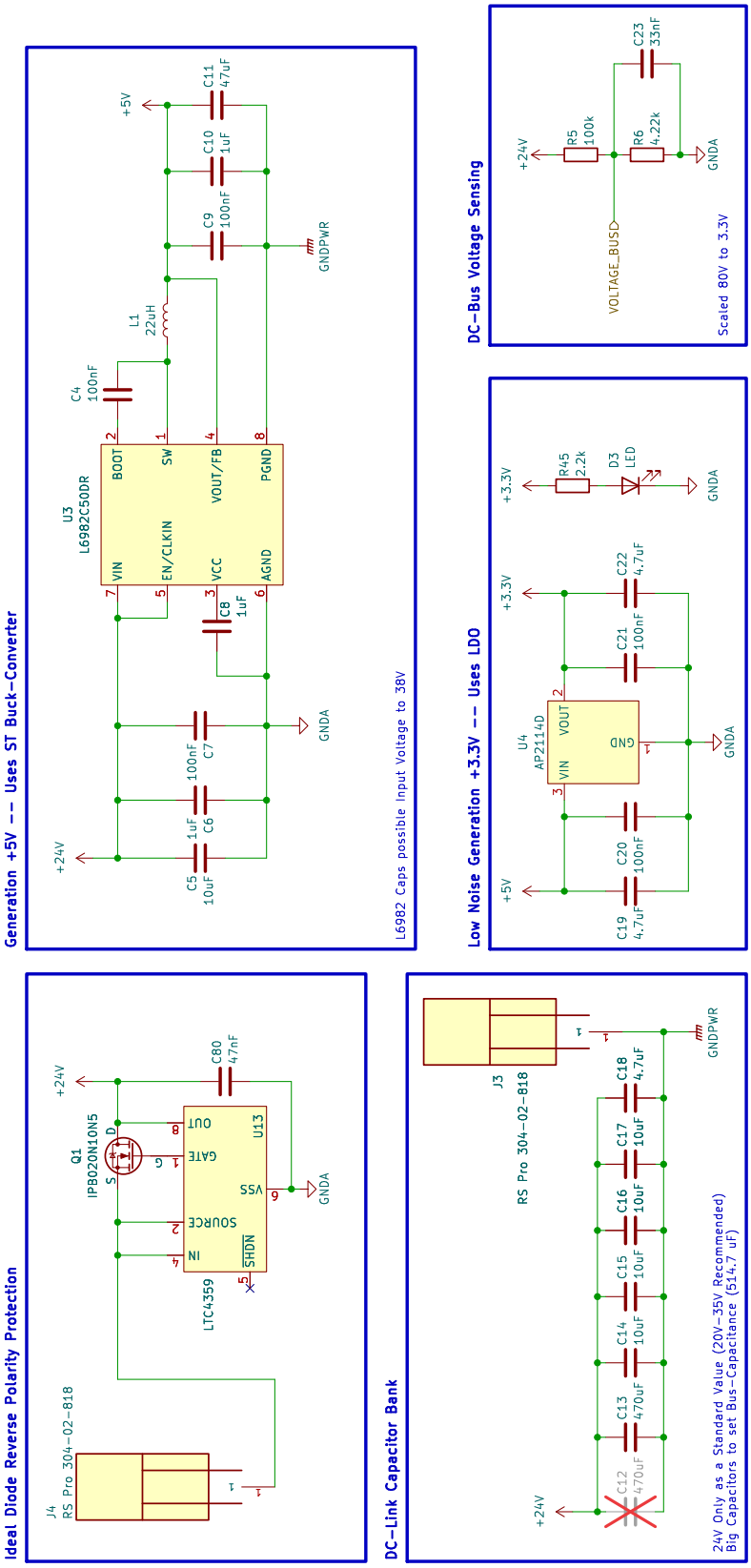


Figure D.2.: Power conditioning circuit

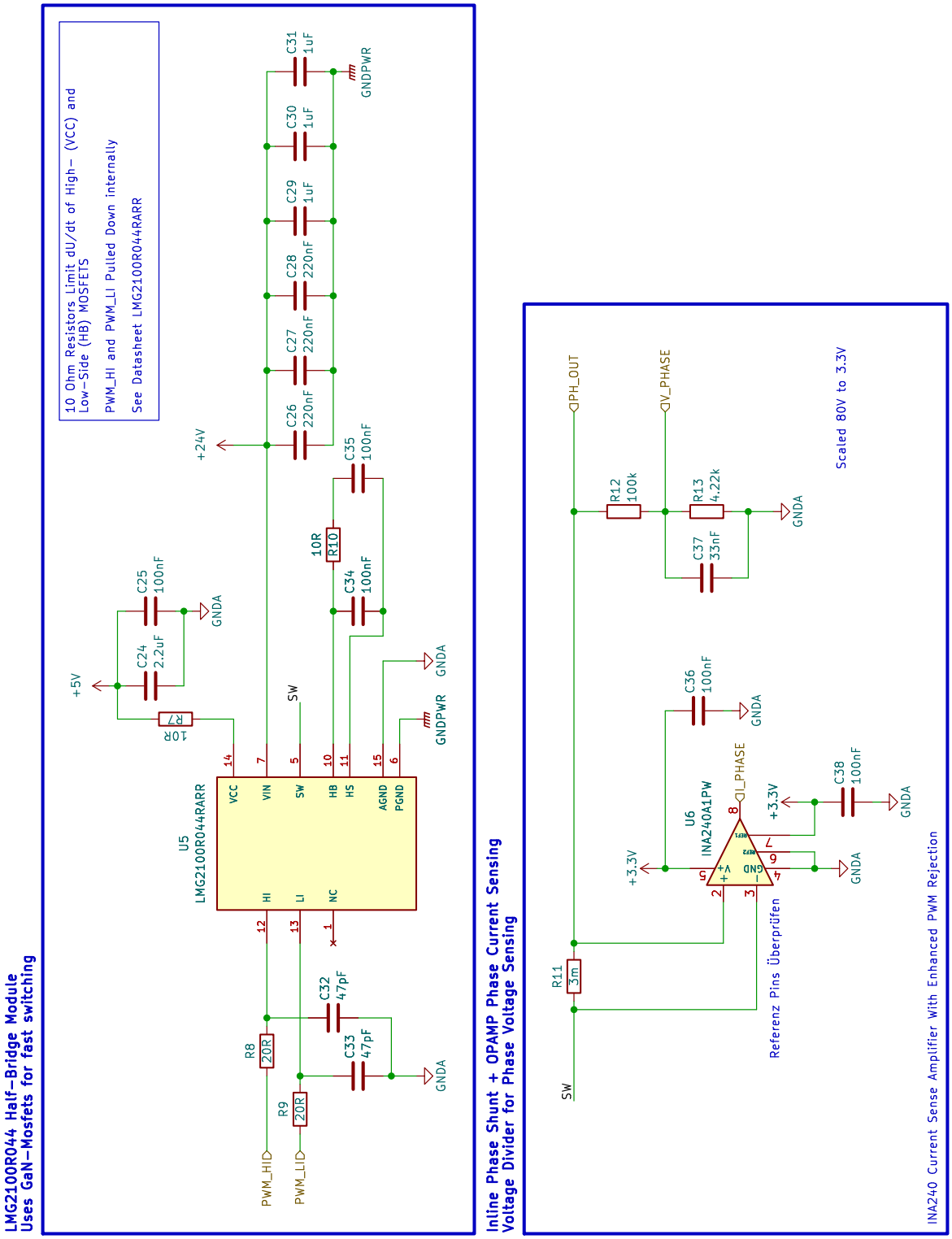


Figure D.3.: Half-bridge circuit

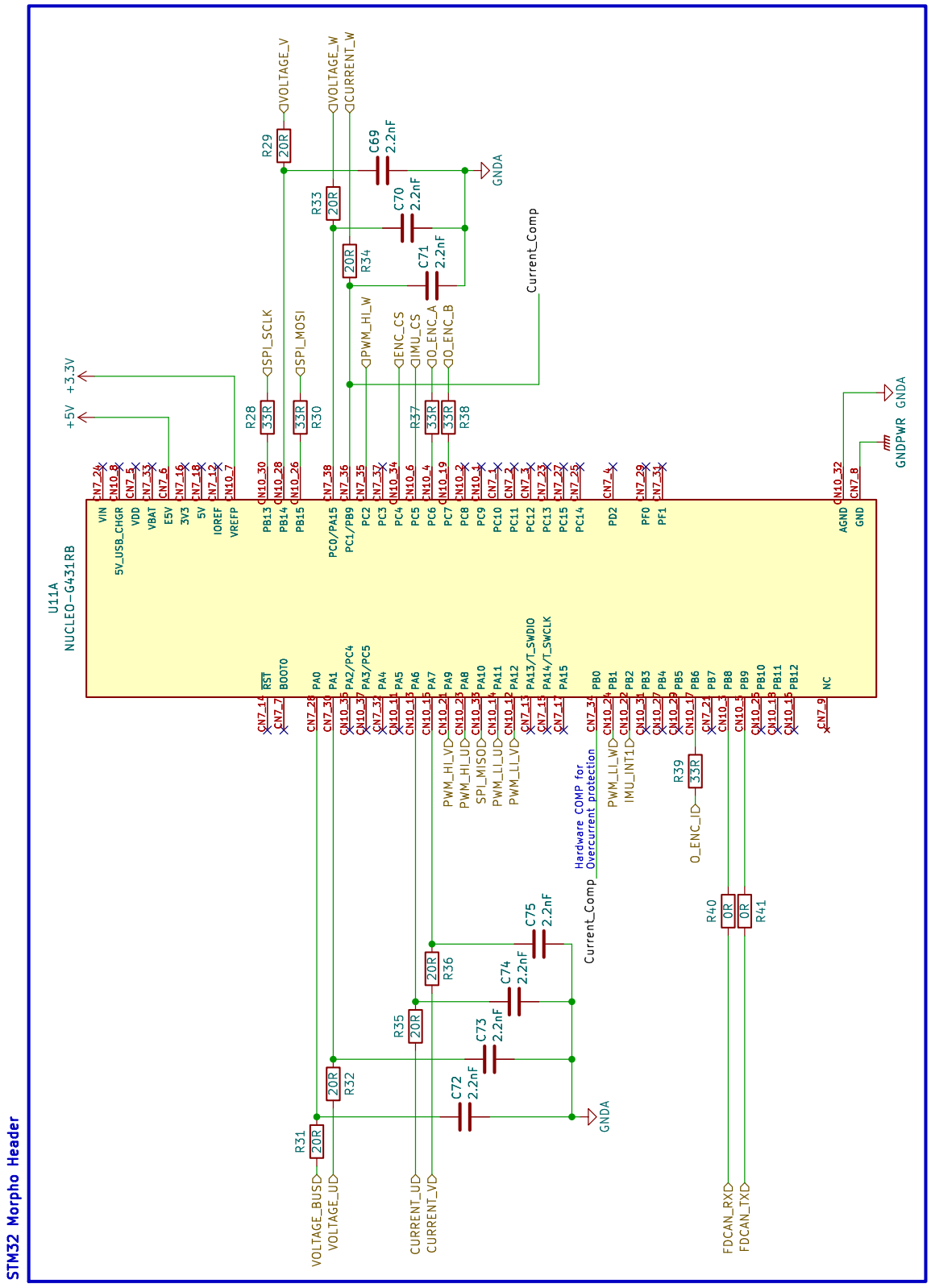


Figure D.4.: Nucleo pin header connections

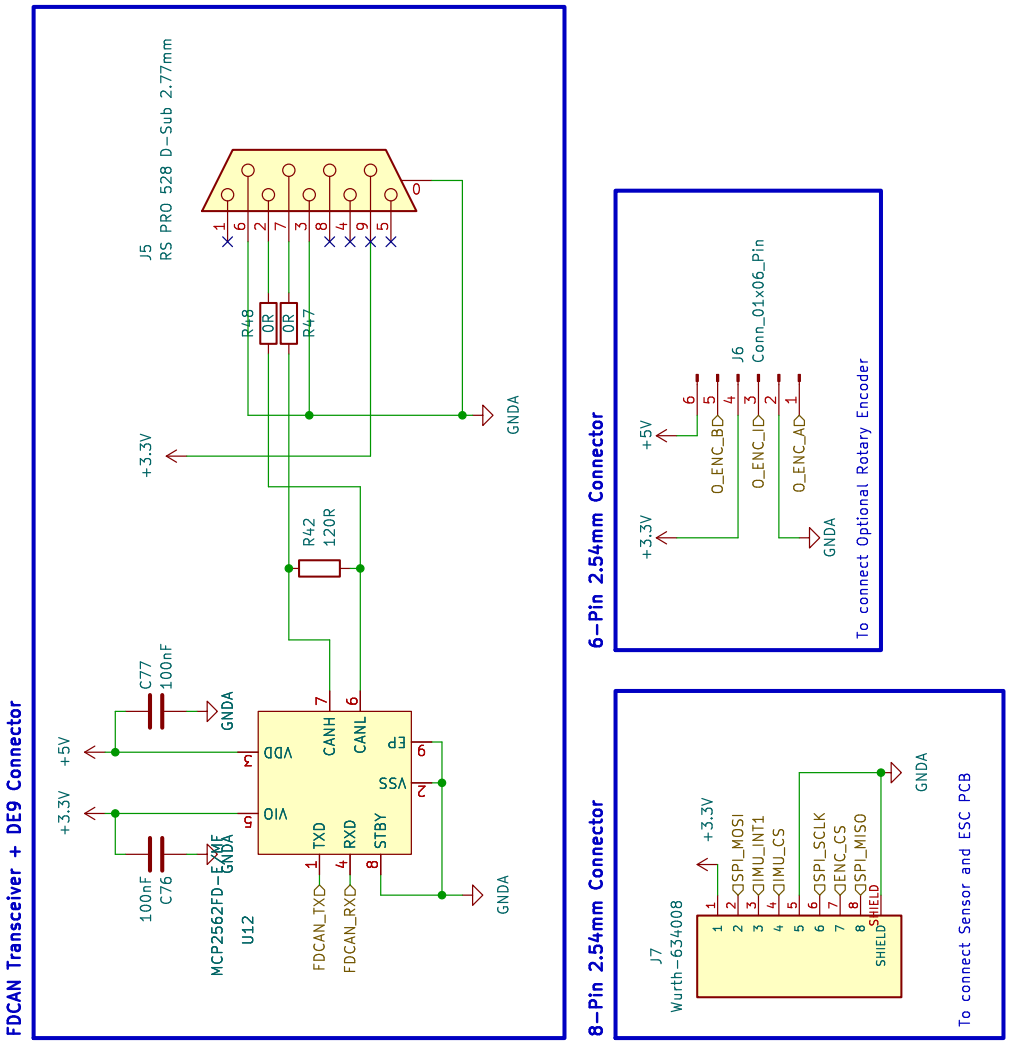


Figure D.5.: Connection circuit

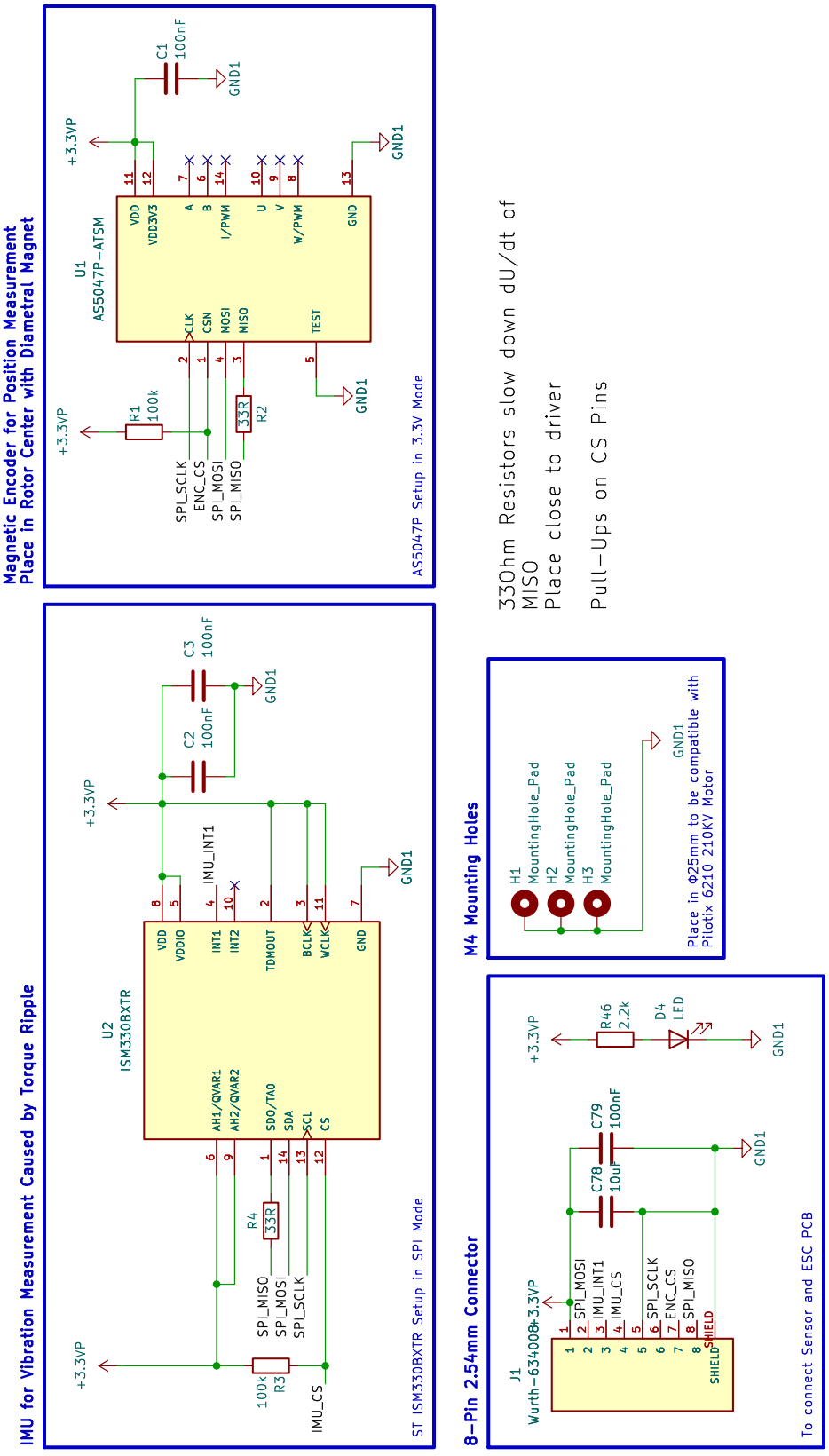


Figure D.6.: Sensor board circuit

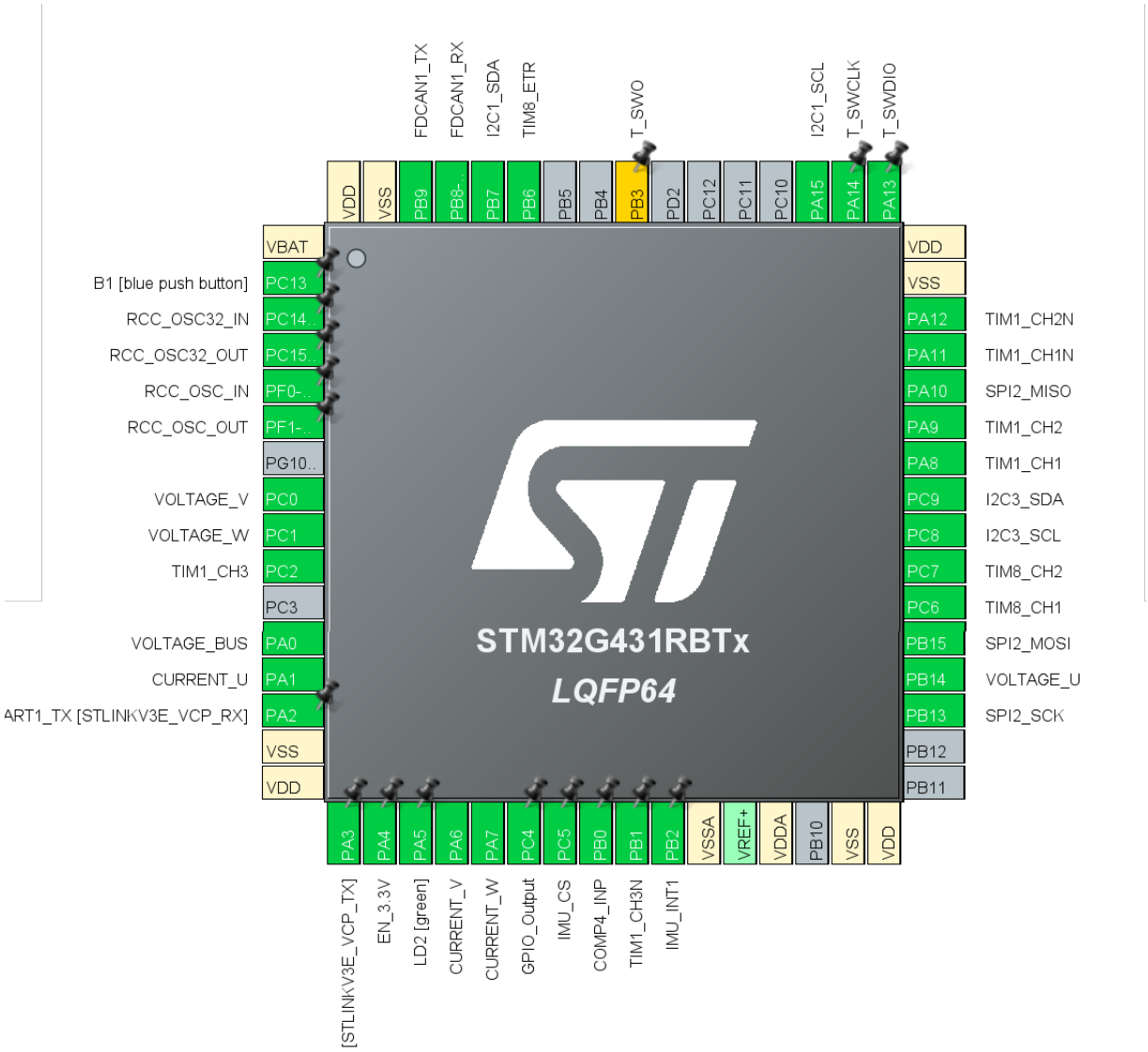


Figure D.7.: STM32CubeMX pin and peripheral configuration of the `NUCLEO-G431RB`.

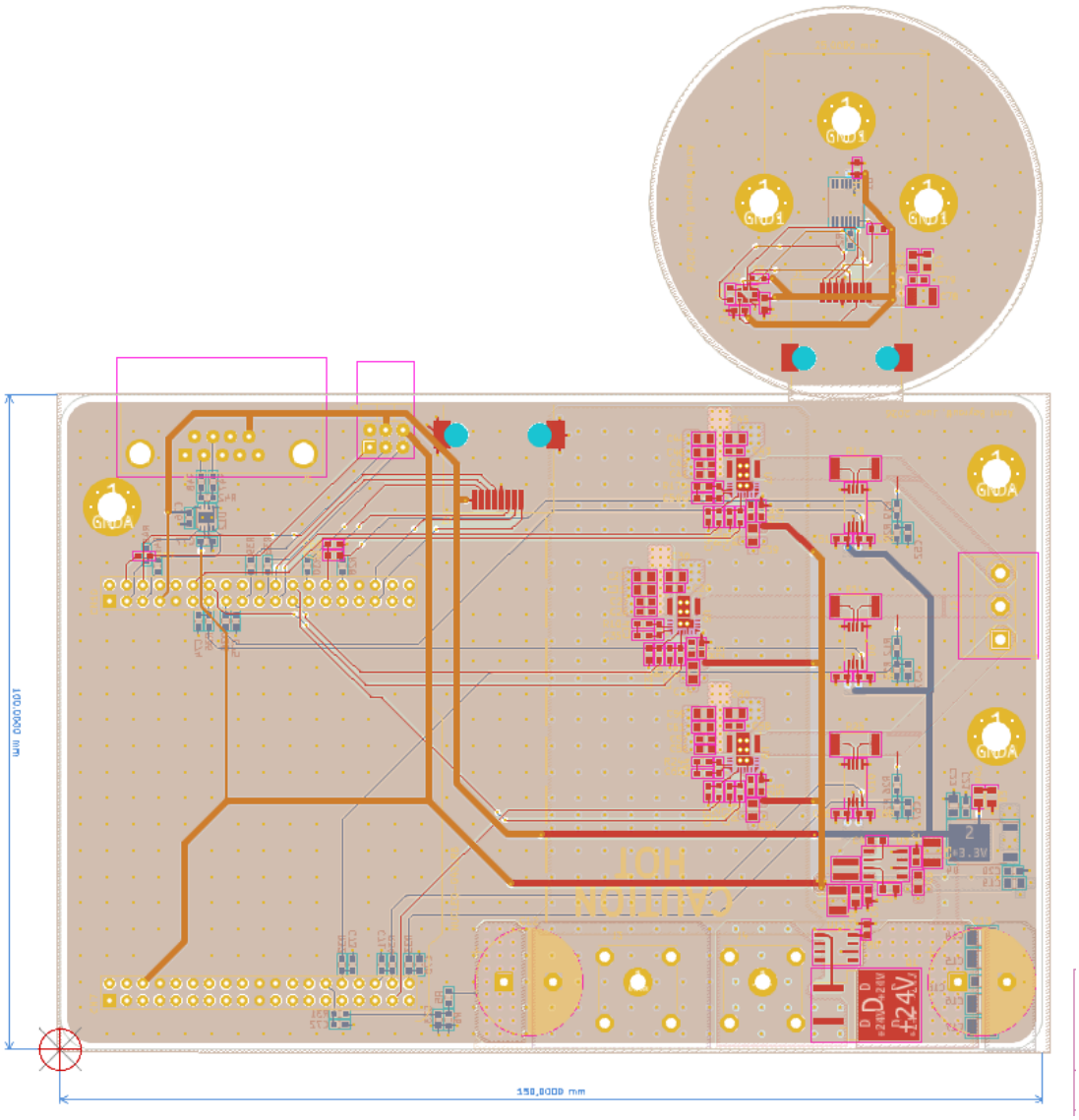


Figure D.8.: Layout of the motor-control extension board.